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The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations (2025)

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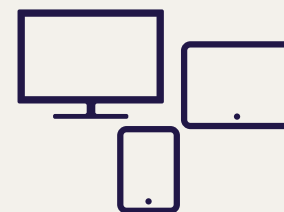
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The Age of AI in the Life Sciences

Benefits and Biosecurity Considerations

Committee on Assessing and Navigating Biosecurity
Concerns and Benefits of Artificial Intelligence Use in
the Life Sciences

Board on Life Sciences

Division on Earth and Life Studies

Computer Science and Telecommunications Board

Division on Engineering and Physical Sciences

Committee on International Security and Arms Control

Policy and Global Affairs

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ASSESSING AND NAVIGATING BIOSECURITY CONCERNS AND BENEFITS OF ARTIFICIAL INTELLIGENCE USE IN THE LIFE SCIENCES¹

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This Consensus Study Report was reviewed in draft form by individuals chosen for their diverse perspectives and technical expertise. The purpose of this independent review is to provide candid and critical comments that will assist the National Academies of Sciences, Engineering, and Medicine in making each published report as sound as possible and to ensure that it meets the institutional standards for quality, objectivity, evidence, and responsiveness to the study charge. The review comments and draft manuscript remain confidential to protect the integrity of the deliberative process.

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Acronyms and Abbreviations

AAV	adeno-associated virus
AI	artificial intelligence
AISI	U.S. AI Safety Institute
API	application programming interface
ARPA-H	Advanced Research Projects Agency for Health
BARDA	Biomedical Advanced Research and Development Authority
BW	biological warfare
BWC	Convention on the Prohibition of the Development, Production and Stockpiling of Biological and Toxin Weapons and on Their Destruction
CDC	U.S. Centers for Disease Control and Prevention
DARPA	Defense Advanced Research Projects Agency
dbGaP	Database of Genotypes and Phenotypes
DBTL	Design-Build-Test-Learn
DHS	U.S. Department of Homeland Security
DoD	U.S. Department of Defense
DOE	U.S. Department of Energy
FAIR	Findable, Accessible, Interoperable, and Reusable
GDS	genomic data sharing
GISAID	Global Initiative on Sharing All Influenza Data
GWAS	genome-wide association studies
HHS	U.S. Department of Health and Human Services
IDP	intrinsically disordered protein
IGSC	International Gene Synthesis Consortium
LLM	large language model
MCM	medical countermeasure
MERS-CoV	Middle East respiratory syndrome coronavirus
NAIRR	National Artificial Intelligence Research Resource
NIAID	National Institute of Allergy and Infectious Diseases
NIH	National Institutes of Health
NIST	National Institute of Standards and Technology
NSF	National Science Foundation
OSTP	Office of Science and Technology Policy

Acronyms and Abbreviations

PDB	Protein Data Bank
QA	quality assurance
R&D	research and development
SARS-CoV-2	severe acute respiratory syndrome coronavirus 2
SOC	sequence of concern
SRA	Sequence Read Archive
USDA	U.S. Department of Agriculture

Preface

At the turn of the 21st century, scientists predicted that biotechnology would play a major role in public health, medicine, agriculture, and the economy. Enabled by rapidly decreasing costs, particularly in DNA sequencing, synthesis, and engineering, many global benefits were envisioned.

Only one year into the millennium, however, we were reminded that biological agents could be used for nefarious purposes, as happened with the Amerithrax attacks in the United States. This led to heightened attention on biosecurity and, subsequently, biosafety. The term “dual-use” took on a new meaning. Whereas previously it signified technologies that had both a civilian and military purpose, the National Research Council report released in 2004, *Biotechnology Research in an Age of Terrorism*, redefined in reference to the life sciences to mean anything that had both a beneficial and harmful use. The U.S. government, and others worldwide, paid close attention to developments in the understanding of infectious agents and how other new and developing technologies might change the ability to engineer biology.

Perhaps the most recent addition to this toolbox of technologies is artificial intelligence (AI). Large amounts of data can be accessed and analyzed, and AI models can be applied to facilitate the design of biologics such as therapeutic proteins, including antibodies. Like biotechnology, however, AI is also a dual-use technology, with the potential to be harnessed by a malicious actor to do harm. One possibility would be to apply AI to bioengineering, facilitating the development of biological threats.

Recognizing the rapidity with which these technologies are progressing, the U.S. Department of Defense (DoD), as directed in Executive Order on Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence, tasked the National Academies of Sciences, Engineering, and Medicine to undertake a study to examine the benefits and risks presented by the intersection of AI and biotechnology. The National Academies assembled a committee of experts in AI, infectious agents, and biosecurity to advise DoD on how to frame the benefits and address and mitigate biosecurity concerns of this rapidly advancing technology, as applied to the life sciences. This report, the product of that study, presents an overview of the issue and several conclusions and recommendations about how the United States can best reap the benefits of the AI–biotech interface while minimizing any potential threats.

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Committee on Assessing and Navigating Biosecurity
Concerns and Benefits of Artificial Intelligence Use in the Life Sciences

Summary

Artificial intelligence¹ (AI) applications in the life sciences have the potential to enable biological discovery and design at a much-accelerated pace and volume when combined with classical experimental approaches. AI models in biology are currently trained on existing biological datasets built from experimental data. For example, AlphaFold, which predicts protein structures from sequence data with significant accuracy, was developed and trained on experimentally derived structures of about 200,000 unique protein sequences in the publicly accessible database Protein Data Bank (PDB). The convergence of AI and biology is a relatively new and emerging area of research—particularly in large neural network models—that spans several disciplines and affects much of the research workflow from hypothesis generation to data analysis. Current AI applications in the life sciences and biomedical research include analysis of genomics and other -omics, synthetic biology,² drug discovery, and healthcare delivery, among many others. The promise of these developments was recognized in 2024 with the Nobel Prize for Chemistry for the development of computational protein design and protein structure prediction.

However, some concerns have been raised that AI-enabled biological tools trained on biological data and intended for beneficial applications may be misused for harmful applications, particularly by facilitating the design of transmissible biological agents that could pose an epidemic- or pandemic-scale threat. Questions regarding the impact of AI on biosecurity risks are highlighted in Executive Order 14110 on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence. The creation of biological weapons is not a new concept or concern; however, whether AI-enabled biological tools could appreciably affect that risk is. The focus of the report is assessing how AI-enabled biological tools can *uniquely* enable biosecurity risks, or the capability uplift referred to as Δ AI and how AI in biology can be used to mitigate such biosecurity risks.

This report represents the outcome of a technical assessment by the committee of the capabilities of AI-enabled biological tools, which can be utilized in conjunction with the framework developed in the National Academies' 2018 report *Biodefense in the Age of Synthetic Biology* (hereafter referred to as “the 2018 framework”) for a full consideration of the threat landscape and risk assessment. The 2018 framework comprehensively identified the different risk factors associated with synthetic biology capabilities but can be applied more broadly to other biotechnologies.

AI-ENABLED BIOLOGICAL DESIGN

Synthetic biology is typified by ideation followed by the process of Design-Build-Test-Learn (DBTL). The most significant capability uplift of AI-enabled biological tools, Δ AI, is in the ideation and design phases. AI-enabled biological tools can accelerate and optimize research design and discovery by providing data-driven guidance for experimental validation, generating novel hypotheses and ideas, and discovering patterns in large datasets. In addition, AI may facilitate data analysis and enhance insights gained from experimental data in the “Learn” phase of the DBTL cycle, subsequently enabling continuous refinement of the DBTL cycle and knowledge generation in a feedback loop. Importantly, AI-

¹ Artificial intelligence refers to “a machine-based system that can, for a given set of human-defined objectives, make predictions, recommendations, or decisions influencing real or virtual environments.” See https://csrc.nist.gov/glossary/term/artificial_intelligence (accessed December 30, 2024).

² Synthetic biology, also known as engineering biology, refers to “the intentional design, manufacture and deployment of living cells, cell systems, and other biological processes to create materials and products that benefit us, like crops that resist drought and pests, renewable energy sources, advanced medicines, and more.” See <https://www.nist.gov/engineering-synthetic-biology> (accessed December 30, 2024).

enabled design does not replace the need for experimental validation; rather, it enhances the process by combining in silico modeling with experimental approaches.

As previously mentioned, there is concern that AI-enabled biological tools could be misused to design transmissible biological agents with epidemic or pandemic potential. The committee explored Δ AI in the context of three harmful applications (see Figure S-1): (1) design of biomolecules, such as proteins and toxins; (2) modification of existing pathogens for increased virulence; and (3) de novo design of a virus. These three applications exemplify both the biological nature of the problem and the capabilities of AI-enabled biological tools—ranging from simple to complex. The committee’s assessment of the Δ AI for each of the three applications is summarized below:

- **Design of biomolecules, such as toxins.** Available AI-enabled biological tools currently are capable of designing and redesigning toxins using different amino acid building blocks, but the scale of potential threats likely would be limited to the local level rather than elevated to an epidemic or pandemic level.
- **Modification of existing pathogens for increased virulence.** Available AI-enabled biological tools may be capable of modeling very specific features that may predict phenotypes related to virulence. However, certain limitations impact model performance, including insufficient datasets and the complex challenge of modeling networks of interactions and evolutionary pressures.
- **De novo design of a virus.** No available AI-enabled biological tool currently possesses the capability to design a virus de novo. Furthermore, no biological datasets that can be used to train such models are known to exist. However, developments that could lead to this capability would represent the most significant AI-enabled capability uplift risk of a potential epidemic- or pandemic-scale threat.

Importantly, AI-enabled biological tools do not necessarily reduce the bottlenecks and barriers to crossing the digital–physical divide (i.e., the physical production of AI-designed experiments or outputs). However, future improvements in automated laboratories may reduce the bottlenecks and barriers at the “Build” and “Test” phases of the DBTL cycle.

AI-ENABLED APPLICATIONS FOR BIOSECURITY

AI-enabled biological tools can improve biosecurity and mitigate biological threats by enhancing prediction, detection, prevention, and response. This includes improving biosurveillance and accelerating the development of medical countermeasures (e.g., vaccines, antibodies) in response to both intentional biological threats and naturally occurring infectious disease outbreaks. AI-driven acceleration of countermeasure design is especially critical in an emergency response to a biological threat. Research programs for understanding the biology of infectious agents are important to support the development of vaccines and therapeutics to prevent and treat diseases caused by biological agents, whether resulting from natural occurrences or intentional threats.

Recommendation 1-1: The U.S. Department of Defense, U.S. Department of Health and Human Services, U.S. Department of Agriculture (USDA), U.S. Department of Energy, U.S. Department of Homeland Security, and other agencies that support and conduct research should continue to invest in vigorous research programs to understand the biology of infectious agents. As a part of these programs, agencies should also invest in and implement biosurveillance networks through the U.S. Centers for Disease Control and Prevention (CDC) and USDA in cooperation with public health agencies globally.

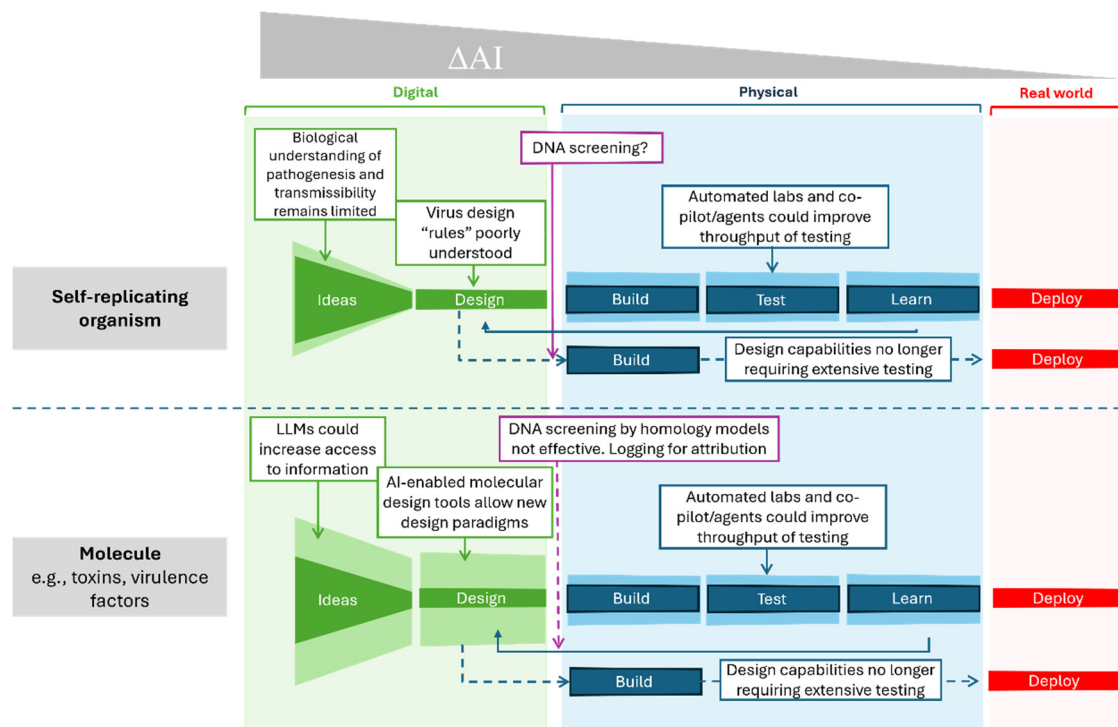


FIGURE S-1 Current capability uplift and limitations of AI-enabled biological tools in two biosecurity risk applications with different levels of design complexity. At present, AI biological models are limited in the capability to accurately model and predict pathogenesis and transmissibility that can be misused to design a virus de novo or modify an infectious agent with the potential for epidemic- or pandemic-scale consequences. Future improvements in AI-enabled biological tools and automated laboratories that may either reduce the intensive requirements of experimental validation in building and testing or eliminate the need for testing would represent a significant development in overcoming current bottlenecks and barriers. In contrast, available AI-enabled biological tools are currently capable of designing simple biomolecules such as toxins that may evade security checkpoints, such as nucleic acid synthesis screening.

NOTE: LLM = large language model.

Recommendation 1-2: Entities within the U.S. Department of Health and Human Services (e.g., National Institutes of Health, Biomedical Advanced Research and Development Authority, and Advanced Research Projects Agency for Health) and the U.S. Department of Defense (e.g., Defense Advanced Research Projects Agency and Chemical and Biological Defense) should fund approaches using AI-enabled design tools for application in medical countermeasure development, especially in the face of an epidemic, pandemic, or other biological threat.

Recommendation 1-3: As part of a national preparedness strategy, the Biomedical Advanced Research and Development Authority, U.S. Department of Energy, and U.S. Department of Defense should establish a public-private partnership, analogous to Operation Warp Speed and the COVID-19 High-Performance Computing Consortium, that can both leverage and provide access to AI-enabled tools and computational resources and be activated rapidly in an emergency response.

Associated biological datasets and insight from such research will be useful and necessary for training AI models to accelerate the development of such medical countermeasures, biosurveillance, and other biosecurity applications previously mentioned. A risk-benefit assessment may be useful to carefully

balance the need to invest in critical research intended to protect against harm from biological agents with the potential for misuse of the information or tools by malicious actors with harmful intent.

Unlike in other domains where AI in the digital world can directly generate nefarious products that can immediately be used for harm (e.g., misinformation, deep fakes, discriminatory practices), the risk of AI-related biological threats materializes in the physical world *only* when biological agents are produced. Given the production requirement of any AI-enabled design of transmissible biological threats, nucleic acid synthesis screening has been recognized as an important checkpoint for actors that uses synthesis services. AI-enabled biological tools also may be used to screen and thereby prevent the creation of biological threats by augmenting current nucleic acid screening methods. However, this specific application of AI is an emerging area and thus needs more research and development.

Conclusion: More research in new methodologies for nucleic acid synthesis screening, including how to leverage AI-enabled biological tools for screening, is needed for this process to be an efficient control point and possible strategy to mitigate potential biosecurity risks. Engaging practitioners and developers of AI-enabled biological tools to explore the most effective approaches for implementation is important as is engaging biologists and other developers to assess impacts of screening on public health preparedness and response.

PROMOTING AND PROTECTING AI-ENABLED INNOVATION FOR BIOSECURITY: GUIDING PRINCIPLES FOR CONTINUOUS EVALUATIONS

New and rapidly developing capabilities in AI biological models and design tools provide vast opportunities for discovery, ideation, and design of biomolecules and biological agents that have useful purposes and applications in health, agriculture, industry, and beyond. In addition, AI applications to mitigate biosecurity risks are promising. However, AI-enabled biological tools and models also may be misused for harmful applications. Here, it is important to distinguish between capability and intent when considering the potential biosecurity risks to avoid unnecessarily restricting or hindering biological research and development. At present, AI-enabled biological tools have limitations (see Chapter 4), whether for beneficial or harmful applications. Namely, biological knowledge and datasets are scarce to train AI biological models for the purpose of designing a novel virus or bacterium or modifying an existing biological entity broadly. Physical production of such an entity is also a significant bottleneck and barrier.

The committee recommends the adoption of an “if-then” strategy to account for the dynamic and rapid pace of development of AI-enabled biological tools that may be improved with training data. The if-then strategy accounts for these dynamics without being prescriptive or restrictive based on predicting future capabilities. Amassing significant datasets, sometimes generated through compute-intensive simulations, is a prerequisite for training AI models (e.g., PDB for AlphaFold); therefore, the availability of high-quality, robust data can be the leading indicator of an emerging or developing AI capability.

The if-then approach allows for the continual assessment and periodic reassessment of vulnerabilities and risks using a principles-based thought framework. These evaluations should be anchored in both data availability and the associated emerging AI capabilities as well as real-world threat models that incorporate all four factors articulated in the NASEM 2018 framework (i.e., usability of the technology, usability as a weapon, requirement of actors, and potential for mitigation). In addition, the development of an if-then strategy should incorporate benchmarks and metrics to track data availability and associated capability progress as well as a set of thresholds that, once surpassed by emerging capabilities, activate risk assessment and mitigation.

Recommendation 2: The U.S. Department of Defense and the U.S. Artificial Intelligence (AI) Safety Institute should develop an “if-then” strategy to evaluate continuously both the availability and quality of data and emerging AI-enabled capabilities to anticipate changes in the risk landscape (e.g., if dataset “x” is collected, then monitor for the emergence of

capability “y”; if capability “y” is developed, then watch for output “z”). Evaluation of AI-enabled biological tool capabilities may be conducted in a sandbox environment.

- Example datasets of interest:
 - If clear associations between viral sequences and virulence parameters become known, then evaluate the capability of AI models to predict or design pathogenicity and/or virulence.
 - If robust viral phylogenomic sequence datasets linked to epidemiological data become available, then assess for the development of new AI models of transmissibility that could be used to design new threats.
- Example AI models that warrant assessment:
 - If AI models are developed that infer mechanisms of pathogenicity and transmissibility from pathogen sequencing data, then watch for attempts to modify existing pathogens to increase their virulence or transmissibility.
 - If AI models are developed that could predictably generate a novel replication-competent virus, then assess risk for bioweapon development.

OPTIMIZING DATA RESOURCES AND INFRASTRUCTURE

Biological data are the leading indicators of the emergence of new AI applications in the life sciences. However, with the notable exception of nucleic acid sequences and structural data, data in the life sciences are fragmented, and robust, reliable, and well-curated data that are amenable to model training are scarce. Both top-down generation of high-quality datasets and bottom-up aggregation of diverse and smaller datasets alongside tools for data harmonization are valid approaches to address the paucity of biological data.

In addition, standardized contextual information is critical for data reuse. For example, the PDB is a valuable resource that provided the foundation for AlphaFold because it is standardized, curated, and available to the public. Lack of data aggregation hinders research and development, especially in the training of new AI-enabled biological models and capabilities. The generation, curation, and preservation of a data repository requires significant long-term investments in laboratory and computing infrastructure capacity as well as subject matter expertise. Building new national data resources and the strategic collection of AI-ready biological datasets should be a national research priority for the United States to maintain scientific competitiveness and innovation.

Recommendation 3-1: To maximize the benefits of artificial intelligence (AI) in biology, U.S. federal research funding agencies such as the National Institutes of Health, Advanced Research Projects Agency for Health, Defense Advanced Research Projects Agency, National Science Foundation, U.S. Department of Energy, and others should fund efforts and infrastructures to do the following:

- Create and steward AI-compatible data and training sets for biological applications that are made publicly available;
- Initiate and support public–private partnerships for the generation of training datasets; and
- Develop frontier AI-enabled biological tools that are specifically trained on biological data and focused on biological capabilities.

Recommendation 3-2: Federal research agencies that house or fund biological databases should consider these to be strategic assets and concordantly implement robust data provenance mechanisms to maintain the highest data quality. They should provide strong incentives and infrastructure for the standardization, curation, integration, and continuous maintenance of high-quality, publicly accessible biological data at scale.

Recommendation 3-3: While a public–private partnership is a reasonable approach to pilot the creation of an artificial intelligence (AI)–ready data resource on government-owned infrastructure, a dedicated organization is central to a sustainable national data stewardship resource. The National Artificial Intelligence Research Resource (NAIRR) pilot should explore approaches for establishing central repositories for training AI models.

1

Introduction

The maturation of biology into the field of biotechnology and the precise manipulation of biological form and function have accelerated considerably in recent years, largely driven by improvements in measurement, editing, and engineering techniques such as directed evolution and CRISPR/Cas9 genetic engineering. Computational tools such as cloud computing and, in recent years, artificial intelligence (AI) have added another dimension as drivers of scientific and technological progress.

These advances have also elevated biotechnology to a U.S. emerging technology priority that is critical to national security. In 2019, the U.S. Department of Defense established biotechnology as an enterprise modernization priority to place an emphasis on the importance of its responsible development (U.S. Department of Defense, 2018; Titus, van Opstal, and Rozo, 2020), and the White House has placed biotechnology on the U.S. Critical and Emerging Technology list (Fast Track Action Subcommittee on Critical and Emerging Technologies, 2024). In parallel, the COVID-19 pandemic caused devastating economic and health damage to countries around the world and placed an intense spotlight on the risks of emerging infectious diseases. The rapid development of vaccines and therapeutics with which to treat COVID-19 is a testament to the importance of biotechnology in the 21st century. The release of ChatGPT in 2022 launched an international race to advance the capabilities of modern AI across every field, leading to the Executive Order on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence in 2023.¹

The rapid parallel expansion of capabilities in AI and biotechnology, set against the backdrop of the COVID-19 pandemic, has renewed concerns that the convergence of these two technologies may pose significant biosecurity risks. However, at the same time, these technologies may present opportunities to mitigate these risks. As a result, the 2023 Executive Order on AI called for an in-depth exploration to assess the biosecurity concerns and mitigation opportunities of the use of AI in the life sciences. As this report outlines, consideration of both the potential benefits and risks of AI applications in the life sciences is determined by data, scientific underpinnings of the biology, and the technical capabilities of AI models. See Box 1-1 for the statement of task.

THE CONVERGENCE OF AI AND LIFE SCIENCES

Computational approaches to understanding and applying biology are not new, having been used now for the better part of four decades (Deshpande et al., 2024). AI applications in the life sciences promise to enhance and accelerate research capabilities to understand biological systems, given the increasing availability of biological data, technologies such as sequencing to generate multi-omics data, and advanced computing power. The convergence of AI and the life sciences is a relatively new area. The foundation of modern AI, or deep learning based on neural networks, emerged in 2012, and the application of AI to the life sciences lagged by a few years. Early work applying AI tools to study biology demonstrated promising results, but it was the 2021 release of AlphaFold and RoseTTAFold, which computationally met the 50-year-old challenge of predicting protein structure from an amino acid sequence, that captured the most attention. In 2024, the Nobel Prize in Chemistry was awarded to David Baker for computational protein design and to Demis Hassabis and John M. Jumper for protein structure

¹ Exec. Order No. 14110, Fed. Reg. 24283 (October 30, 2023). See <https://bidenwhitehouse.archives.gov/briefing-room/presidential-actions/2023/10/30/executive-order-on-the-safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence/> (accessed October 10, 2024).

prediction.² In recent years, new tools to study and manipulate biology as well as AI biological models (see Appendix A) have increased rapidly across diverse applications, albeit with variable performance.

The expansive and pervasive nature of AI in the life sciences has generated debate as to just how powerful these tools are for enabling a wider range of users to conduct biological experiments and intentionally engineer new forms and functions. The challenge posed in assessing the risks and benefits of AI in the life sciences is largely driven by the expansive definition of AI and the pace of progress in both fields. At the same time, a principal concern is whether bad actors can recreate existing biological agents or create new agents of previously unknown form and function and/or that have tailored properties. This biosecurity concern existed before AI arrived on the scene, and the question now is whether and how AI changes it.

An additional dimension of uncertainty is just how capable AI biological models are at designing and predicting novel functions based on currently available computational, data, and experimental resources as well as future capabilities. This uncertainty in the AI threat model has given rise to heightened concern from the biosecurity and national security communities. In addition, questions remain as to whether AI lowers the barriers for actors to create a biothreat, such as a novel functional and pathogenic virus, or enhance the harmful traits of an existing infectious agent. These are the types of questions that were addressed by the 2018 framework for synthetic biology (NASEM, 2018) and now need to be revisited to include recent advancements of AI tools.

At the same time, AI tools are being used by practitioners every day to develop new medical countermeasures (MCMs) and pharmaceutical interventions for a wide range of illnesses, from cancer to infectious diseases (Lee, Bubeck, and Petro, 2023; Wong et al., 2024; Xu et al., 2024). Given the emerging capabilities of AI tools, can researchers develop a new vaccine for an emerging infectious disease threat faster than ever before? Can these tools enable public health and biosecurity practitioners to identify and respond to biosecurity incidences more rapidly and efficiently?

With each new advance that aids in manipulating biological systems (e.g., genetic engineering techniques, AI-enabled tools), there has been a concomitant struggle to extract the benefits of the technology to human health, agriculture, and the economy as well as to prevent the misuse of those advances from causing harm. Owing to the dual-use nature of biological knowledge and tools, constant monitoring and evaluation of the potential risks and benefits of biotechnological advances will be needed, along with tailoring of current and development of new governance mechanisms to reduce the potential for misuse or accidents while still allowing beneficial advances that promote health and grow the economy (Titus and Russell, 2023).

Compounding the difficulty of preventing harm is the wide range of potential actors to deter—from nation-states to organized non-state actors to individuals—all with a wide range of motivations, resources, technical know-how, and capabilities. A wide range of potential pathogens and toxins already exists and could be deployed for harm anywhere in the world without any further engineering. While many pathogens that have been developed historically as weapons do not readily spread from person to person, such as *Bacillus anthracis* (the causative agent of anthrax), many regulated pathogens, designated as Biological Select Agents and Toxins in the United States,³ arguably have that potential, and infectious disease agents famously do not respect national boundaries. The experiments and biological tools that could be used to modify pathogens to make them more challenging to respond to (e.g., transmissibility or host range) are exquisitely dual-use in that the same tools and experiments can be used to help develop countermeasures and assist in the response. Often, basic biological research improves our understanding of biological systems in small increments, making it very difficult to decide a priori which pieces of information should be classified or protected. Moreover, most, if not all, basic research in the discipline falls under the fundamental research exclusion rules.

Finally, there are the significant known and unknown biological threats from nature, including those that stem from human-made perturbations of nature, such as the wildlife trade (Gómez and Aguirre,

² See <https://www.nobelprize.org/prizes/chemistry/2024/press-release/> (accessed October 9, 2024).

³ See <https://www.selectagents.gov/sat/list.htm> (accessed October 10, 2024).

2008); alteration of human–animal interactions due to urbanization and population growth (Gibb et al., 2020); and climate change–driven expansion of environments in which potential disease vectors can thrive (Ebi et al., 2017). The latter is an increasing concern especially regarding arboviruses (i.e., viruses that replicate in and are transmitted to humans by arthropods [insects and ticks], including West Nile virus, yellow fever virus, and Zika virus) but also regarding protozoans such as plasmodia, some of which cause malaria. The historical record is full of misery and changed fortunes as a result of communicable diseases, and changing climate is likely to affect modern experiences and frequency of spillovers as well. Some have argued that there are upward of 320,000 viruses in nature that have the potential to infect humans (Anthony et al., 2013), and this number does not consider threats to animals other than humans or to plants, which could be devastating to the economy. Thus, there are potentially many more threats from nature than biosafety and biosecurity concerns surrounding research laboratories that may impact animal (including humans) and plant health under the appropriate circumstances.

Countering these biological threats and benefiting economically while curtailing the inappropriate use of biotechnology for harm is no small feat. Overlapping governance structures aim to balance these at-times conflicting priorities, and often they are criticized as either being too open—lowering barriers to misuse or accident—or too stringent—hampering economic growth of new biotechnologies, generation of new knowledge, and the ability to counter emerging and re-emerging natural threats to health and safety. The balance is often determined by how these relative risks are perceived by the person or group empowered to make these decisions, as there can be no unassailably objective formula to predict the future and determine where resources are best placed. Natural disease events happen all the time. Deliberate and accidental events have been rare, but they remain theoretically possible and could have catastrophic consequences. It is the responsibility of policymakers and scientists to understand and prepare for these occurrences.

Some regulatory and oversight guidelines specifically related to AI and the life sciences⁴ have been generated, though this is a relatively new policy space. Executive Order 14110 on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence⁵ contains sections related to the use of AI in biological applications. About a year later, the Memorandum on Advancing the United States’ Leadership in Artificial Intelligence; Harnessing Artificial Intelligence to Fulfill National Security Objectives; and Fostering the Safety, Security, and Trustworthiness of Artificial Intelligence⁶ (hereafter referred to as the National Security Memorandum on AI) was released. The sections specifically relevant to AI and biology are listed in Table 1-1.

STUDY SCOPE AND APPROACH

AI-enabled biological tools are an emerging area of research and development in the life sciences. The increasing prevalence of AI tools utilized in the life sciences and their rapid development have brought into focus both the promising benefit of their uses and concern that they may increase biosecurity risks. The National Academies of Sciences, Engineering, and Medicine established the Committee on Assessing and Navigating Biosecurity Concerns and Benefits of Artificial Intelligence Use in the Life Sciences to conduct this study (see Appendix C for biographical information). The statement of task provided to the committee is in Box 1-1.

⁴ Policies and regulatory frameworks that govern research and development in AI and the life sciences, separately, will inevitably overlap with and impact how AI and the life sciences is governed and regulated. For the purposes of this report, we focus primarily on the policies that apply specifically to AI and the life sciences.

⁵ Exec. Order No. 14110, Fed. Reg. 24283 (October 30, 2023). See <https://bidenwhitehouse.archives.gov/briefing-room/presidential-actions/2023/10/30/executive-order-on-the-safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence/> (accessed October 10, 2024).

⁶ See <https://home.treasury.gov/system/files/216/Memorandum-on-Advancing-the-United-States-Leadership-in-Artificial-Intelligence-Harnessing-Artificial-Intelligence-to-Fulfill-National-Security-Objectives-and-Fostering-the-Safety-Security-and-Trustworthiness-of-AI.pdf> (accessed November 3, 2024).

TABLE 1-1 Summary of National Security Memo on AI Relevant to Biology

Task	Section	Agency
Facilitate testing of frontier AI models; establish capability to lead testing and risk assessment related to biosecurity	3.3(c)	U.S. AI Safety Institute (AISI) (National Institute of Standards and Technology [NIST]/Commerce)
Pursue voluntary preliminary testing of at least two frontier AI models that includes assessing models' capabilities to accelerate bioweapons development	3.3(e)i	AISI
Issue guidance on measuring capabilities relevant to the risk that AI models could enable the development of bioweapons; develop mitigation measures and testing efficacy of safety and security	3.3(e)ii, A-E	AISI
Develop a roadmap for future classified evaluations of advanced AI models' capacity to generate or exacerbate deliberate chemical and biological threats	3.3(g)i, A-B	U.S. Department of Energy (DOE), U.S. Department of Homeland Security (DHS), and AISI, in consultation with the U.S. Department of Defense (DoD) and other relevant agencies
Establish a pilot project to provide expertise, infrastructure, and facilities capable of conducting classified tests in this area	3.3(g)i, C	DOE
Support efforts to utilize high-performance computing resources and AI systems to enhance biosafety and biosecurity in development of AI systems trained on biological and chemical data:	3.3(g)iii, A-E	DoD, U.S. Department of Health and Human Services (HHS), DOE, DHS, National Science Foundation (NSF), and other relevant agencies
(A) The development of tools for screening in silico chemical and biological research and technology;		
(B) The creation of algorithms for nucleic acid synthesis screening;		
(C) The construction of high-assurance software foundations for novel biotechnologies;		
(D) The screening of complete orders or data streams from cloud labs and biofoundries; and		
(E) The development of risk mitigation strategies such as medical countermeasures.		
Incorporate guidance by AISI outlined in subsection 3.3(e) for agencies that develop publicly available dual-use foundation AI biological and chemical models	3.3(g)iii	Relevant agencies
Convene academic research institutions and scientific publishers to develop voluntary best practices and standards for publishing computational biological and chemical models, datasets, and approaches, including those that use AI	3.3(g)iv	NSF, DoD, Commerce (AISI within NIST), HHS, DOE, other relevant agencies, Office of Science and Technology Policy (OSTP)

continued

TABLE 1-1 *continued*

Task	Section	Agency
Develop guidance promoting the benefits of and mitigating the risks associated with in silico biological and chemical research	3.3(g)v	OSTP, National Security Council, Office of Pandemic Preparedness and Response Policy, in consultation with relevant agencies

SOURCE: Adapted from “Memorandum on Advancing the United States’ Leadership in Artificial Intelligence; Harnessing Artificial Intelligence to Fulfill National Security Objectives; and Fostering the Safety, Security, and Trustworthiness of Artificial Intelligence,” The White House, October 24, 2024, <https://home.treasury.gov/system/files/216/Memorandum-on-Advancing-the-United-States-Leadership-in-Artificial-Intelligence-Harnessing-Artificial-Intelligence-to-Fulfill-National-Security-Objectives-and-Fostering-the-Safety-Security-and-Trustworthiness-of-AI.pdf>.

BOX 1-1
Statement of Task

The National Academies of Sciences, Engineering, and Medicine will, as directed by the President in Executive Order 14110, conduct a consensus study that will:

- (A) Assess the ways in which artificial intelligence (AI) can increase biosecurity risks, including risks from generative AI trained on biological data, and make recommendations on how to mitigate those risks;
- (B) Consider the national security implications of the use of data and datasets, especially those associated with pathogens and omics studies, that the United States Government hosts, generates, funds the creation of, or otherwise owns, for the training of generative AI models, and make recommendations on how to mitigate the risks related to the use of these data and datasets; and
- (C) Assess the ways in which AI applied to biology can be used to reduce biosecurity risks, including recommendations on opportunities to coordinate data and high-performance computing resources.

This study will focus specifically on concerns of transmissible biological threats that could pose significant epidemic and pandemic-scale consequences. This study will build on the 2018 study *Biodefense in an Age of Synthetic Biology* but would not duplicate that work. An unclassified consensus report will provide evidence-supported analysis of AI model capabilities, including trends and actionable recommendations for near- and mid-term implementation.

The charge to the committee is focused on the ways in which AI-enabled biological tools trained on biological data may impact the design of transmissible biological agents that could lead to epidemic- or pandemic-scale consequences or how they could be leveraged for mitigation strategies and other beneficial applications. In a public briefing held on June 21, 2024, the study sponsors stated their interest in the study providing a balanced understanding of both the benefits and risks.⁷ In interpreting the statement of task, the committee considered it worthwhile also to examine current AI-enabled biological tools’ capabilities with respect to designing transmissible biological threats that may pose a more limited or local scale of impact than that of an epidemic or pandemic. General-purpose AI trained on data beyond biological data, such as chatbots or general large language models (LLMs; e.g., ChatGPT, Claude, Gemini, Llama), is considered outside the scope of this study. Scientific LLMs that are trained or fine-tuned on biological data are, however, within the scope. The committee also considered future developments in utilizing general-purpose LLMs as interfaces or assistants with other biological

⁷ Presentation to the committee by representatives from the U.S. Department of Defense, National Security Council, and Office of Science and Technology Policy, June 21, 2024. See Appendix B.

computational tools. The committee also acknowledged that while nonbiological AI-driven technologies may increase biosecurity risks that are not focused on the design aspect, they are also beyond the scope of this report.

The committee reviewed relevant literature and heard from experts with a broad range of expertise and perspectives as part of the information-gathering process. Three public meetings were held from August 2024 to October 2024 and covered topics such as the current state of AI-enabled biological tools, understanding the science of pathogenesis of transmissible biological agents, AI-driven MCM development, biological data resources, and high-performance computing (see Appendix B for agendas of public sessions).

The committee did not access classified information in consideration of the questions related to the statement of task, and the resulting report is unclassified. As such, the report focuses on the capabilities of AI biological models and the state of science and is not intended to be a threat assessment. Instead, the study utilized a framework for assessing potential biodefense concerns developed in a 2018 National Academies report, entitled *Biodefense in the Age of Synthetic Biology*.

In the current report, the committee adopts the term “AI-enabled biological tools” to refer broadly to applications used to interact with AI models that are trained on biological data. The committee also distinguishes between biology-specific tools and general AI tools that have broad domain applications. References to biological data in this report are limited to include the types of biomolecules such as DNA, RNA, protein, or large sets of biomolecules known collectively as omics data (e.g., genomics, transcriptomics, proteomics, metabolomics) that are publicly available. Unless stated otherwise, biological data as used in the report do not include health or medical data.

REPORT ORGANIZATION

This report is divided into five chapters that examine, from different angles, the biosecurity risks and benefits of AI applied to biological sciences and potential mitigations. Chapter 2 examines the capability uplift of AI-enabled biological tools on the Design-Build-Test-Learn cycle of the synthetic biology process and discusses future developments that may help increase DAI. It also provides a detailed discussion of the considerations for AI-enabled design with respect to differing biological complexities, from simple molecules to full replicative infectious agents. Because AI model innovation will rely heavily on access to data, Chapter 3 discusses how AI-enabled biological tools can uniquely enable biological design at different levels of biological complexity in harmful applications. AI-enabled biological tools have vast potential to enhance biosecurity, and Chapter 4 covers the ways AI can be leveraged to bolster biosurveillance and accelerate the development of MCMs in response to biological threats. Finally, Chapter 5 explores the importance of high-quality biological data to train AI models and optimizing and coordinating data infrastructure and access to resources. The chapter also discusses the potential vulnerabilities of biological datasets used for training, and various approaches for model evaluations and mitigation.

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2

Design-Build-Test-Learn: Impact of AI on the Synthetic Biology Process

Artificial intelligence (AI)-enabled biological tools can enhance research and design across the various disciplines in the life sciences. Appendix A provides a more detailed discussion of current capabilities and limitations of AI models in biological applications. This chapter presents a broad overview of the impact of AI in synthetic biology and three areas of future development—scientific large language models (LLMs), automated laboratories, and synthetic datasets—that have the potential to accelerate biological design and discovery by improving biological models.

In synthetic biology, the Design-Build-Test-Learn (DBTL) cycle is an iterative approach for designing, constructing, and evaluating biological systems with a built-in feedback loop for refining designs based on experimental data (see Figure 2-1). Ideation or hypothesis generation typically precedes design. The design phase defines the conceptual plan, describes how to achieve the desired output, and can utilize computational modeling tools. AI-enabled biological tools can accelerate ideation and design significantly by providing data-driven insights, discovering patterns in large datasets, and generating novel ideas and design. For example, in drug discovery applications such tools can generate thousands of molecules as potential candidates in days (Chan et al., 2019; Melo, Maasch, and de la Fuente-Nunez, 2021; Food and Drug Administration, 2023; Ren et al., 2024), a feat that would probably take years for researchers.

The build phase is the key point in the DBTL cycle, where digital outputs transition to physical implementation. During the test phase, designed agents are evaluated for whether they behave as expected. During these phases, automated laboratories can improve efficiency and throughput significantly, but there are challenges, as will be discussed below. Extracting data from build and test to refine and optimize design in the learn phase allows for continuous improvement and knowledge generation. The iterative process of DBTL accounts for the inherent variability of biological systems and highlights the necessity of coupling AI-driven design or modeling with experimental validation.

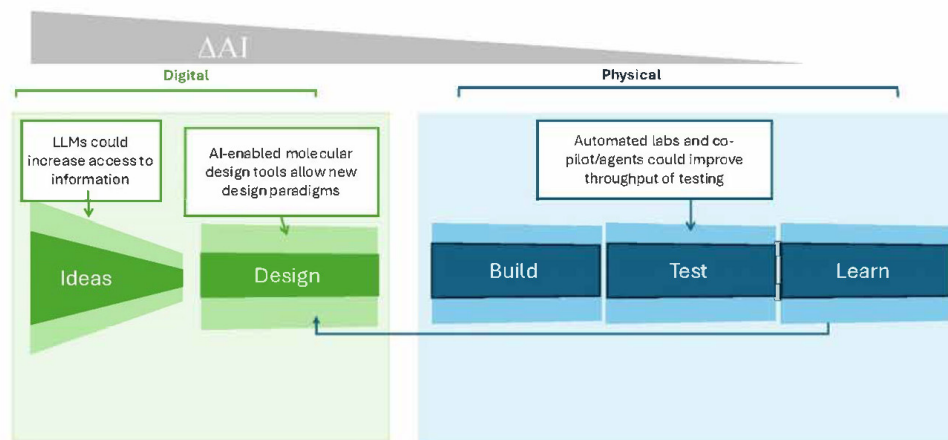


FIGURE 2-1 The impact of AI-enabled biological tools across digital and physical tasks in synthetic biology at present. For example, LLMs can facilitate hypothesis generation, and other AI-enabled biological tools allow for the design of novel biomolecules for experimental validation in build and test. Automated laboratories can enhance the efficiency and scalability during the build and test processes.

IDEATION AND DESIGN: LARGE LANGUAGE MODELS AND FOUNDATION MODELS

Thanks to their scaling properties (Hoffmann et al., 2022) and in-context learning (Garg et al., 2024), LLMs are now approaching trillions of learnable parameters. LLMs have made significant advances in natural language understanding and generation, as demonstrated by their increasing performance on benchmark tasks such as language modeling, machine translation, and answering questions (Kumar et al., 2021; Xie et al., 2022; Yue, Yao, and Sun, 2022). Despite existing and well-publicized challenges on staying true to a corpus of knowledge and reducing hallucinations, LLMs are powering valuable tools for knowledge acquisition and idea generation. While most widely known commercial LLM-enabled platforms, such as chatbots, focus on natural language understanding and generation, researchers are building interactive platforms to advance discovery in specific scientific disciplines or domains.

One category, commonly referred to as “LLMs for science,” is being developed at an increasing pace to help students and researchers quickly grasp key concepts and findings over vast amounts of literature in a particular discipline or across disciplines; interpret complex scientific data to support the discovery process; and, more importantly, assist and guide the generation of scientific hypotheses (Birhane et al., 2023). LLM-assisted ideation goes beyond simple question-answering grounded in a particular domain such as Alphabet’s Med-PaLM 2 for medicine¹. An exemplar of a scientific LLM is CRISPR-GPT, an LLM for the automated design of gene-editing experiments (Huang et al., 2024). LLM-enabled ideation assistants/agents are growing. A non-exhaustive list includes ChemCrow, assisting with planning and executing chemical synthesis procedures (Bran et al., 2024); BioGPT, trained on biomedical literature to assist in biomedical research and ideation (Luo et al., 2022); MatSciBERT, trained on materials science literature to aid in materials discovery and research (Gupta et al., 2022); AstronomyGPT, trained on astronomical data and literature to assist in astrophysics research and hypothesis generation (Ciucă et al., 2023); and ClimateGPT, helping researchers analyze climate data and generate research ideas (Thulke et al., 2024).

While building powerful LLMs currently remains within the reach of only a few companies and countries, two core capabilities may continue to fuel the rise of LLM-enabled ideation assistants: (1) many companies, such as Meta, are releasing their LLMs and sharing model functional characteristics but not the details of the training process or training data; and (2) LLMs are examples of *foundation models* that learn representations of the input data in a task-agnostic manner through a process referred to as pretraining. Although LLMs are pretrained over large nonspecific datasets as foundation models, LLMs can be further refined in capabilities and performance in a specialized task or domain through continued training, fine-tuning, and instruction tuning.

How well do current ideation assistants perform? Evaluating LLMs as ideation assistants is challenging as it is an emerging area of research with no clear benchmarks. Anecdotal evidence tends to be positive. For instance, Si, Yang, and Hashimoto (2024) demonstrated that LLMs can generate research ideas consistently rated as more novel than those produced by 100 experts, but the ideas lacked depth.

On the other hand, there are concerns that an actor may exploit general LLMs for harmful intentions, such as providing guidance to create a bioweapon, and in turn lower the level of expertise needed. A recent study investigated the dual-use potential of chatbots by having students prompt chatbots for information related to creating a pathogen (Soice et al., 2023). However, prompts to the chatbot were highly guided and detailed, and they potentially contained the very information requested.

Rigorous analysis over planning problems indicates that current LLMs cannot reason well (Kambhampati, 2024). Reasoning is key to planning out a recipe or scientific protocol and so is central to effective ideation. However, reasoning capabilities of LLMs are growing (see GPT-4o²), and LLM-

¹ Big tech and start-ups, such as Alphabet, Kili Technology, and others, are counting on an increasing need by researchers for domain-specific LLMs for quick knowledge acquisition and ideation.

² See <https://openai.com/index/hello-gpt-4o/> and <https://openai.com/index/learning-to-reason-with-llms/> (accessed November 12, 2024).

modulo frameworks that includes external verifiers may provide a path forward (Kambhampati et al., 2024). In the future, LLM-enabled assistants may be capable of being real reasoning partners, representing a transformative point for scientific discovery.

Yet, developing general LLMs remains a challenge owing to the computational cost to power them. While this cost can be circumvented by researchers fine-tuning already pretrained models, there is another computational cost in deployment. An active area of research seeks alternative non-transformer-based architectures to lower both costs. State space models, such as S4 (Gu, Goel, and Ré, 2022), S5 (Smith, Warrington, and Linderman, 2023), Mamba (Gu and Dao, 2024), Hawk (De et al., 2024), and others, constitute a particular category of efficient recurrent networks that promise linear-time pretraining and constant-cost inference, with capabilities approaching those of transformer-based models (Blouir et al., 2024).

Confining LLMs to text or ideation, however, misses a big part of the picture and their true potential. Growing research in bioinformatics and cheminformatics leverages sequential representations of biological entities in a way that can be used by computational tools and LLMs to generate new proteins and new small molecules. For example, the Simplified Molecular Input Line Entry System is used to represent chemical structures (Hunter, Culver, and Fitzgerald, 1987). It is worth noting that the research advances foundation models, which are architecture-agnostic; that is, the models are not restricted to transformer-based architectures but rather are inclusive of other generative architectures, such as those based on variational autoencoders, generative adversarial networks, diffusion models, and others.

In addition, foundation models can be trained to learn information from multiple types of data. DALL-E, for instance, is an example of a foundation model that is trained over two data modalities, text and images.³ Sora is another example, trained over text and video data.⁴ Indeed, foundation models trained over multiple data modalities promise to revolutionize scientific discovery and present an emerging area of research (see Appendix A). A non-exhaustive list illustrates this point:

- IsoFormer is a multimodal foundation model that learns over DNA, RNA, and proteins to predict how multiple RNA transcript isoforms originate from the same gene and map to different transcription expression levels across various human tissues (Garau-Luis et al., 2024).
- IBM's Biomedical Foundation Models⁵ include foundation models for identifying novel diagnostic and therapeutic target discovery (trained over DNA, bulk RNA, single-cell RNA expression data, and other cell-level signaling information); foundation models for biologics discovery, which further integrate protein sequences, protein complex structures, and protein–protein complex binding free energies; and various small molecule models trained over multiple representations of small molecules to advance drug discovery.
- Many foundation models are being advanced for medical imaging and pathology, including multimodal AI systems for outcomes prediction in precision oncology, trained over textual and image data (Truhn et al., 2024).

In principle, foundation models can be trained over diverse sets of multi-omics⁶ data and combine ideation and biological design, but many challenges remain for developing multimodal models, including data size, data sparsity across modalities, and data fidelity. For natural language processing, the scaling

³ See <https://openai.com/index/dall-e-2/> (accessed November 12, 2024).

⁴ See <https://openai.com/index/sora/> (accessed November 12, 2024).

⁵ See <https://research.ibm.com/projects/biomedical-foundation-models> (accessed November 12, 2024).

⁶ Multi-omics refers to “combined information derived from data, analysis, and interpretation of multiple omics measurement technologies to identify or analyze the roles, relationships, and functions of biomolecules (including nucleic acids, proteins, and metabolites) that make up a cell or cellular system. Omics are disciplines in biology that include genomics, transcriptomics, proteomics, and metabolomics.” See <https://www.nist.gov/bioscience/nist-bioeconomy-lexicon#multiomic-information>.

laws demand very large training datasets (Hoffman et al., 2022⁷). It is possible that advancements in architectures and training procedures may lower data demands to facilitate the development of multimodal foundation models that can incubate both ideation and design internally.

Provided that these challenges are surmountable, multimodal foundation models have the potential to be a one-stop shop, such that one interacts with a master AI agent rather than patching single-task AI agents. Advancing this analogy, multimodal foundation models hold great potential to operationalize ideation to actual biological design. Currently, however, one-stop-shop models are out of reach owing to computational cost and data requirements as well as the sophistication needed to build them.

BUILD AND TEST: AUTOMATED LABORATORIES

Today, applications of bioengineering require many successive DBTL cycles to yield a biological design that meets the desired performance target. To increase the probability of success and reduce the total amount of time and cost it takes to develop a successful design, automation is often used to perform build and test activities. Facilities that automate design-build-test cycles for synthetic biology are commonly referred to as foundries, in reference to foundries that fabricate silicon chips (Hillson et al., 2019).

While the automation used in foundries has become significantly more advanced, foundries vary in their capabilities and are often designed for specific purposes (e.g., assembly of large DNA molecules, microbial engineering to produce a useful product, or mammalian cell engineering). Foundries are now being leveraged to generate datasets for AI training, the results of which can be used to curate new datasets that improve model performance. This “lab in the loop” approach promises to accelerate AI model development, which in turn could reduce the number of DBTL cycles needed to realize a successful design. Additionally, this combination of automated laboratories with AI may address the need to generate vast biological datasets broadly by using the generated data to further refine AI models in the iterative DBTL cycle.

The concept of lab in the loop involves AI-driven experimental design and automation of the build and test work. This raises the concern that fully automated foundries could lower the intensive requirements of build and test and be co-opted by malicious actors to develop harmful products. However, the full suite of activities needed to design, build, and test in automated or fully “self-driving laboratories” is a future capability that at present has significant challenges. Currently, foundries are typically application-specific and resource-intensive, and they require a large amount of human input and capital investment (Martin et al., 2023).

LEARN: SYNTHETIC DATA

Training AI models on synthetic data is an increasingly appealing proposition in several application domains where data scarcity and potential privacy and ethical considerations are key concerns. Synthetic data refers to artificial data that is generated to mimic real-world data. In biology, future developments may lead to improvements of biological models from incorporating synthetic data as training data, but more work is needed. Biological processes are physics-driven dynamic processes, and no amount of wet-lab data will sufficiently capture all aspects of a physics-based process as it unfolds across disparate spatiotemporal scales. Synthetic data, such as those obtained from Monte Carlo or molecular dynamics physics-based simulations, may fill in the gaps of experimentally derived data. However, model collapse, defined as the decrease in performance from using synthetic data generated

⁷ The Chinchilla scaling laws in Hoffman and colleagues (2022) suggest that for compute-optimal training, the number of model parameters and the number of training tokens should scale at approximately equal rates. This is typically operationalized as a ratio of about 20 tokens per parameter.

from one machine learning model in another machine learning model, may occur (Shumailov et al., 2024).

Future developments in LLMs, automated laboratories, and the use of synthetic data to train models represent a few areas to monitor for increases in AI-enabled biological capabilities.

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3

AI-Enabled Biological Design and the Risks of Synthetic Biology

Artificial intelligence (AI) integration in the biological sciences is a relatively nascent area of research that creates a complex landscape for risk assessment. To date, there is a distinct lack of empirical data and research studies focused on evaluating the potential biosecurity risks of AI-enabled biological tools. Most risk assessments and subsequent critiques of studies at the interface of AI and biology have focused on general chatbots (Soice et al., 2023; Mouton, Lucas, and Guest, 2024; Patwardhan et al., 2024) and are outside the scope of this study. Urbina and colleagues (2022) published a study reporting the use of a predictive machine learning model to generate *de novo* chemicals predicted to be toxic. However, as will be discussed in this chapter, designing transmissible biological agents is more complicated than designing toxic biochemical molecules using current AI predictive models. This chapter will explore the complexities of biological design and engineering and subsequent biosecurity risk implications, focusing on the current capabilities of AI-enabled biological tools combined with scientific understanding of the biology. Although this report does not represent a formal risk assessment, which requires a broader assessment beyond technical capability to include intent and other factors, the 2018 framework (NASSEM, 2018) may be utilized to consider the biosecurity risks in context of a given capability.

ADDRESSING BIOLOGICAL COMPLEXITY IN DESIGN AND ENGINEERING

The committee examined the following three representative problems in biological design and engineering in consideration of the concern stated in the statement of task—transmissible biological threats that could pose significant epidemic- and pandemic-scale consequences: (1) design of biomolecules, such as proteins and toxins; (2) modification of existing pathogens for increased virulence; and (3) *de novo* design of a virus.

AI-Enabled Design of Biomolecules

Current AI-enabled biological tools are capable of various modeling tasks, such as design and prediction with biological applications. For example, RoseTTAFold (Baek et al., 2021) and AlphaFold (Jumper et al., 2021) are accurate predictive protein models trained on structures in the Protein Data Bank (PDB). RFdiffusion is a generative model that can generate or design novel proteins by combining structure prediction from RoseTTAFold with a diffusion model (Watson et al., 2023). Recently developed generative AI-based approaches (see Appendix A) perform more sophisticated tasks, such as designing entirely new molecules with new functions and specificities to a specific ligand (Bennett et al., 2023; Vázquez Torres et al., 2024). How can these capabilities be misused for harmful applications?

One possibility is to *redesign* known toxins using different amino acid building blocks and, in so doing, potentially bypass simple homology-based DNA screening (Ekins et al., 2023; Baker and Church, 2024; Whitmann et al., 2024). For example, ProteinMPNN can be utilized in protein design to assign amino acid building blocks to create a specific structure. Importantly, ProteinMPNN (Daupras et al., 2022) can be used not only when building new molecules but also when redesigning an existing molecule to maintain the same structure with different amino acids. This capability can be used to generate a structurally similar molecule with limited sequence homology to the parent (Sumida et al., 2024) and may bypass typical homology-based DNA screening methodologies.

Likewise, intrinsically disordered proteins (IDPs) comprise 30–50 percent of proteomes and perform multiple essential functions in cells as well as play key roles in disease. These proteins do not form stable conformations and are refractile to traditional protein folding tools. Prions, which are proteins whose misfolding leads to accumulation of abnormal protein in the central nervous system, are IDPs that in specific conformations are able to self-template and “replicate.” AI-enabled prediction tools are beginning to decipher these conformational transitions (Erdős and Dosztányi, 2024). While it is theoretically possible that, as with toxin simulants above, a prion simulant could be created using IDP analytical tools, wet-lab testing is still necessary to assess the functionality of the designed protein or molecule.

Furthermore, limitations with current tools, such as fine-grained control (i.e., generating outputs that satisfy desired requirements), remain for any AI-enabled biological design. For AI-designed small molecules, a useful molecule must be synthetically accessible, able to permeate specific physical barriers, and interact with specific intended molecules in the cell (and potentially no others). The current AI-enabled design methodologies focus on satisfying one or more phenotypical properties. The ability to satisfy an arbitrary number of chemical and phenotypical constraints remains an outstanding challenge in generative AI models. Similarly, controllable generation task challenges manifest themselves in macromolecular protein design. For instance, we are currently able to control for structure, but AI models cannot fully capture dynamics at disparate spatiotemporal scales and the full equilibrium conformational ensemble expected of a designed molecule in the cell. While not insurmountable, these are key challenges in the future capabilities of AI biological models. Quantum computing platforms may help address some of these challenges in the future.

The capabilities and limitations of current AI-enabled biological tools for protein design demonstrate the numerous factors that must be accounted for in designing a protein, a relatively simple biological problem. These variables significantly increase when the objective is to design *complex* biological systems that replicate, such as a bacterium or virus (see Figure 3-1).

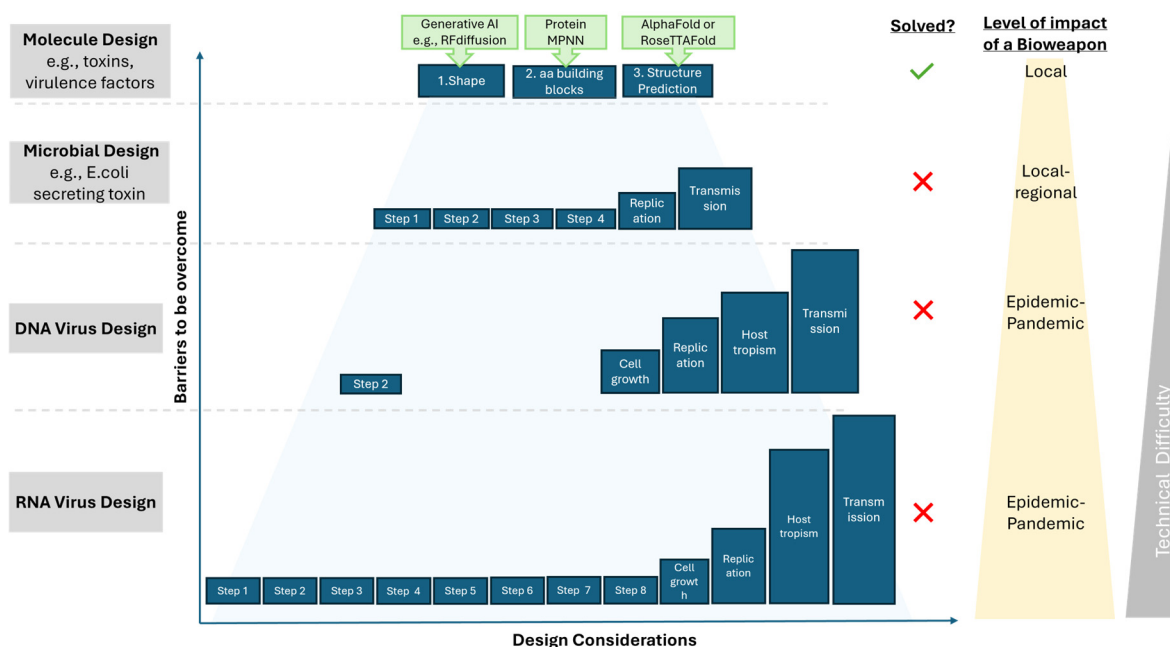


FIGURE 3-1 Levels of complexity for molecular and microbial design. Current state-of-the art AI-enabled tools can design a new molecule in three steps (top row). In contrast, there is increasing complexity of designing replicating pathogens (from top to bottom rows), represented by a number of factors and steps to consider in designing a biological transmissible agent. The number of design considerations illustrates the barriers needed to be overcome. To date, AI has not been able to address hurdles other than designing molecules.

AI-Enabled Design of Self-Replicating Agents with Epidemic or Pandemic Potential

The potential use of AI-enabled biological tools to design self-replicating and transmissible biological agents that do not already exist or are modified in some manner to be rendered more dangerous is challenging at every phase—from design to build and test. Specifically, to design new viral pathogens, generative AI models need to predict structural, virulence, and transmissibility determinants of a virus accurately, as discussed below. The de novo design of a virus would represent a significant capability uplift enabled by AI biological tools. Furthermore, this capability could have the highest impact in terms of consequences, as the potential for mitigation would be hampered and relatively delayed absent of information. However, it is unlikely that currently available viral sequence data are sufficient to train such a model, which would require not just sequence data but also the knowledge of sequences that are linked to specific phenotypes. Furthermore, virulence and transmissibility phenotypes are often associated with multiple molecular determinants, along with host interaction components, and therefore require an understanding of multiple network interactions (Eisfeld et al., 2024; Waters, 2024). Human viruses have had hundreds of thousands of years to evolve these traits, which are often best suited for a particular host with which the virus co-evolved or closely related organisms. In addition, viruses with RNA genomes are already functioning near their limit in terms of mutational tolerance, meaning that most changes are likely to affect viral fitness (Goldhill et al., 2018; Li et al., 2023).

Modification of Existing Organisms

AI biological models have demonstrated promise in predicting phenotypes from genotypic, proteomic, and other types of molecular data but have several limitations at present. It should be noted that several existing non-AI-based models and methodologies can be used to predict individual characteristics of pathogens, such as transmission rates and antibody escape (Wagh et al., 2023; Dadonaite et al., 2024; May and Rannala, 2024; Meijers et al., 2024). Appendix A includes a brief overview of AI biological models that predict phenotypes of viruses (see Box 3-1). The unique capability uplift (Δ AI) of utilizing AI-based methods stems from the predictive modeling that in turn may provide guidance for experimental validation that can be further refined.

BOX 3-1

AI Biological Models for Predicting Viral Fitness

Unsupervised protein language models have shown promise in predicting viral infectivity and escape, utilizing high-throughput deep mutational scanning data that measure how amino acid changes affect these properties (Haddox, Dingens, and Bloom, 2016; Haddox et al., 2018; Lee et al., 2018; Starr et al., 2020; Wu et al., 2020; Starr et al., 2022). Viral escape refers to mutations allowing viruses to evade immune responses, while infection is mediated by specific proteins interacting with host cell receptors. Initial work demonstrated that statistical properties learned by protein sequence models correspond to viral escape characteristics (Riesselman, Ingraham, and Marks, 2018; Hie et al., 2021; Thadani et al., 2023). Subsequent work showed that larger protein language models like the Evolutionary Scale Modeling (ESM), either zero-shot or when fine-tuned on viral proteins, can competitively model various aspects of viral evolution (Hie, Yang, and Kim, 2022; Beguir et al., 2023; Ito et al., 2024; Lamb et al., 2024). These diverse methods offer promising tools for studying mutational effects on near-term infectivity or escape, but their ability to predict long-term viral evolution remains limited. Viral evolution across multiple generations and long evolutionary trajectories inherently has substantial levels of uncertainty. Better models are needed to forecast viral evolution accurately over the long term, highlighting the ongoing challenges in translating these insights into reliable vaccine design and pandemic preparedness strategies.

SOURCE: Appendix A.

Understanding the molecular determinants of fitness characteristics of pathogens is important for developing vaccines, medical countermeasures, and other public health responses, as noted in the referenced studies and discussed in more detail in Chapter 4. However, can AI-enabled biological tools be misused to modify existing pathogens by designing a single or a few specific changes to viral or bacterial sequences to increase virulence? The committee notes several key limitations of current AI-enabled biological tools in this regard and also for designing novel pathogens. These include limitations in the performance and capabilities of current models, while a second type of limitation involves the downstream processes of building, testing, and production. For accuracy in modeling, existing datasets that can be used to train the models are limited and noisy. Many datasets lack the coverage to capture the full range of molecular data influencing phenotypes. Biological datasets are not all standardized, often contain errors, and may contain biases that will affect model performance. Biological systems are also complex, and phenotypes result from intricate interactions at the molecular level, between environmental factors and host-pathogen interactions, and under evolutionary pressures that are difficult to model accurately. Phenotypes can change over time, requiring dynamic modeling that current models and tools are not yet capable of with great accuracy. Finally, regardless of the capabilities of AI models, the production of a bioweapon requires experimental validation, which is resource-intensive and therefore remains an outstanding bottleneck. The bottleneck in physical production is not affected by the capabilities of AI biological models.

The chaotic nature of viral evolution also challenges accurate prediction of the intended results of virus design, including modification. The “butterfly effect” (Lorenz, 1972) is a chaos theoretical term defined as “the sensitive dependence on initial conditions in which a small change in one state of a deterministic nonlinear system can result in large differences in a later state”¹. In public parlance, the butterfly effect refers to the possibility that a butterfly flapping its wings could theoretically lead to a tornado but that the time and place of the tornado formation would be unpredictable mathematically due to the vast number of variables that would have to be taken into consideration. Chaos theory (Hasselblatt and Katok, 2003) is related to Stephen Wolfram’s concept of “computational irreducibility,” which postulates that many systems in nature are so complex that the system’s future state cannot be predicted without simulating each step in the process. Although both chaos theory and computation irreducibility are typically applied to physics, both concepts clearly should be taken into consideration when contemplating evolutionary pathways (Bennett, 2010; Rego-Costa, Débarre, and Chevin, 2018; Hoggard, 2024). In other words, it is possible that the various evolutionary pressures on pathogens, and, by extrapolation, even more so the evolution of newly created viruses, remain unpredictable even with the best AI tools and the fastest super computers.

Understanding Data Limitations in Viral Design

The lack of relevant and well-curated data is currently a limiting factor to training AI biological models that allow the design of novel pathogens with epidemic or pandemic potential. However, collecting large-scale data on virulence, pathogenicity, immune evasion, and other properties of viruses to use for training models is challenging. By comparison, protein folding models such as AlphaFold were trained on several hundred thousand experimentally determined protein structures, the vast majority of which were contributed over decades by individual research laboratories. This was possible in large part because the genotype-to-phenotype relationship is measurable: the three-dimensional structure of the protein is the phenotype itself, and scientists have been studying protein structure for decades. In addition to the lack of data, the quality of existing datasets may not be suitable for training accurate AI models (see Box 3-2).

The COVID-19 pandemic drove an unprecedented collection of high-quality sequences of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) genomes as the global research and public health communities tracked the spread of new variants. To date, the GISAID (Global Initiative on Sharing

¹ See https://en.wikipedia.org/wiki/Butterfly_effect (accessed October 20, 2024)

All Influenza Data) repository, which began as an influenza virus resource but expanded to include SARS-CoV-2 during the pandemic, has more than 15 million deposited SARS-CoV-2 variant sequences.² If these sequence data were paired with comprehensive phenotypic data, such as evasion of neutralizing antibodies raised against vaccine sequences, these data could be used to design sequences that enhance immune evasion. However, these types of phenotypic data have only been collected for a small fraction of the sequenced SARS-CoV-2 genomes, making the training of an immune evasion model difficult. Yet, the sheer number of SARS-CoV-2 genome sequences available, even without phenotypic data, does offer insight into the evolutionary space that has been effectively explored over the course of the pandemic. As a comparison, vaccine researchers who engineer prefusion stabilized spike proteins for SARS-CoV-2 report that AI has sped up their design process but that high-throughput screening and testing remains essential to their process.³

BOX 3-2

Current Limitations of Using Virus Sequences as Training Data

At present, numerous viruses discovered in the human host are not associated with any meaningful, useful sequence information in databases as only fragments are sequenced. Many human virus genome sequences may appear complete according to the sequence database descriptors but are incomplete due to missing difficult-to-sequence segments such as genomic termini. This is particularly true for negative-sense RNA viruses such as filoviruses, which consist of hairpin structures that are hard to resolve; highly repetitive or highly GC-rich (guanine-cytosine) regions in orthopoxviruses (Monzón et al., 2024); or missing genomic segments (Liu et al., 2024). Furthermore, many human viruses are represented in genetic databases by just one to a few representative sequences and may contain errors. A large number of representative sequences would increase confidence. Many virus genome sequences in the database are so-called consensus sequences, or the average of all genome sequences in a sample. The consensus sequence may not necessarily be the correct sequence to allow virus rescue. Viruses can produce copious amounts of shorter “defective interfering” genomes, which may tilt the consensus sequence into the wrong direction. Some viruses produce quasi-species—that is, numerous genomes that individually are defective or attenuated but in combination become functional through reciprocal compensation of defects. There is currently no straightforward way to determine whether an individual sequence in a database is fully functional without trial-and-error experimentation, even when high-quality standards in sequence determination have been met (Adriaenssens et al., 2023).

Another example is the family of replication-incompetent adeno-associated viruses (AAVs), one of the few groups of viruses with datasets linking viral sequences to human cell types infected (i.e., tropism) (Wang et al., 2024). These datasets have enabled AAVs to be engineered to deliver gene therapies, including Food and Drug Administration–approved treatments for hemophilia A (FDA, 2023) and neuromuscular diseases in children (FDA, 2019, 2024). However, even with the comparatively large amounts of data, predictively altering the tropism of AAVs remains challenging. Currently, the state of the art in AI-assisted AAV design depends on the use of AI to generate diverse libraries of AAV designs followed by successive design-build-test cycles to narrow the libraries down to functional sequences (Bryant et al., 2021; Ghauri and Ou, 2023). In other words, zero- or one-shot design of retargeted AAVs remains beyond the state of the art.

Conversely, attempting to design replication-competent viruses in a bottom-up fashion using protein design models is far beyond the capabilities of existing tools. For example, the authors of AlphaFold3 report that current protein design models are still poor predictors of protein dynamics (e.g., different conformational states), stemming from the fact that state-of-the-art protein design models use mainly static protein structure data from sources such as PDB (Abramson et al., 2024). See Appendix A

² See <https://gisaid.org/lineage-comparison/> (accessed October 14, 2024).

³ Presentation to the committee by Jason McLellan and Emanuele Andreano, October 1, 2024. See Appendix B.

for a more detailed discussion on the types of data that at scale may be used to develop new AI models for biology.

Finally, beyond the challenges associated with the design or modification of viruses, there are challenges associated with the physical production of any digital outputs to test and subsequently deploy, as discussed in the following sections.

IMPLICATIONS FOR BIOSECURITY

In assessing the impact of AI-enabled biological tools on lowering barriers or eliminating bottlenecks to creating or modifying existing pathogens, the committee considered how the capabilities made possible by these tools change the specific aspects of the Design-Build-Test-Learn (DBTL) cycle for a given threat. Presently, AI-enabled biological tools can facilitate the design of simple biomolecules. This capability has the potential to be exploited for harmful applications such as the redesign of toxins. However, this capability does not remove the need for building and testing that requires an extensive footprint. In contrast, current AI-enabled biological tools are not capable of the *de novo* design of a self-replicating organism such as a virus. Figure 3-2 summarizes the current capabilities and limitations of AI-enabled biological tools with respect to biological design complexity.

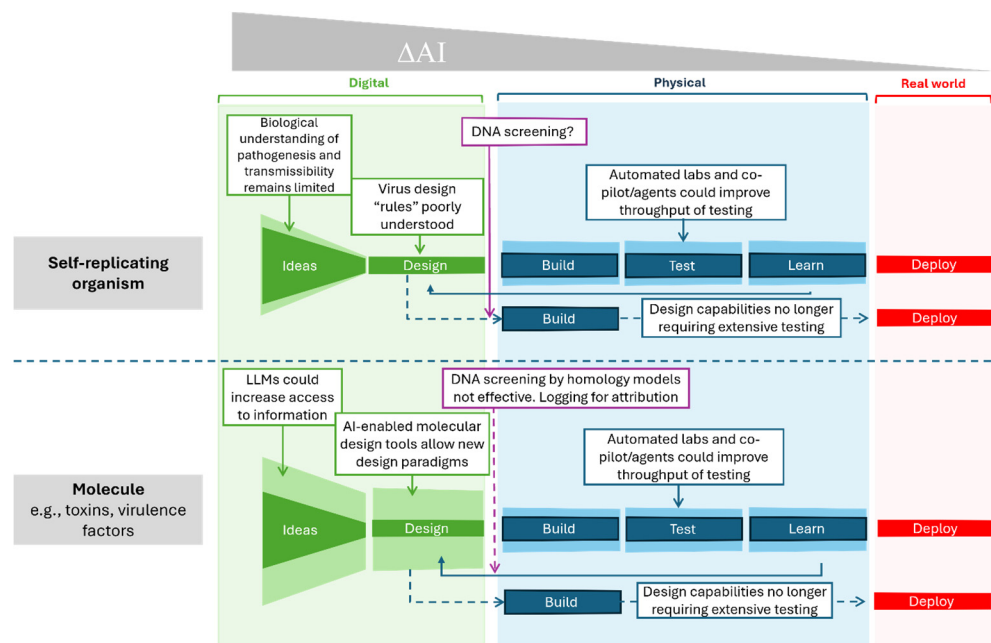


FIGURE 3-2 Current capability uplift and limitations of AI-enabled biological tools in two biosecurity risk applications with different levels of design complexity. At present, AI biological models are limited in the capability to accurately model and predict pathogenesis and transmissibility that can be misused to design a *de novo* or modify an infectious agent with the potential for epidemic- or pandemic-scale consequences. Future improvements in AI-enabled biological tools and automated laboratories that may either reduce the burden of the intensive requirements of experimental validation in building and testing or eliminate the need for testing would represent a significant development in overcoming current bottlenecks and barriers. In contrast, current AI-enabled biological tools are capable of designing simple biomolecules such as toxins that may evade synthesis screening.

NOTE: LLM = large language model.

Overall, AI-enabled biological tools provide data-driven guidance to filter and select designs that reduce the amount of wet-lab testing and iterations to a potentially manageable number. The ultimate application of AI-enabled biological design would be to obviate the need for DBTL altogether, so called

zero-shot (de novo design without a preexisting example of a design to work from) or one-shot (design from a single reference design) learning. Significant capability uplift of AI-enabled biological tools that may be concerning and should be monitored include the following: (1) models able to predict transmissibility and pathogenesis with high accuracy, (2) AI-enabled design of fully replicating infectious agents, (3) design of models for molecules or pathogens that no longer require minimal wet-lab testing, or (4) improved AI-driven automated laboratories.

Conclusion: The relative lack of biological and mechanistic understanding about virulent phenotypes and the paucity of high-fidelity biological data mean that AI-enabled biological tools currently cannot be used to de novo design and subsequently build complex biological systems that can successfully replicate as transmissible biological agents with epidemic or pandemic potential.

BROADER CONTEXT FOR ASSESSING BIOSECURITY RISKS: THE 2018 FRAMEWORK

In assessing biosecurity risks, intent cannot be separated from capability. While this report provides an analysis of the technical capabilities of AI-enabled biological tools that may be misused for harmful applications, the committee acknowledges the broader context of the risk landscape that can be assessed using the 2018 framework developed in the report *Biodefense in the Age of Synthetic Biology* (NASEM, 2018).

The framework allows various aspects of the threat development and use process to be considered, such that an analyst can properly weigh how technical advances may change one part of the process, for example, but leave others unchanged. The framework involves the analysis of four main attributes as they relate to the development and use of a synthetic biology-generated threat:

1. Usability of the technology,
2. Usability as a weapon,
3. Requirements of actors, and
4. Potential for mitigation.

Each of these categories was further divided into a series of subcategories, as shown below (see Figure 3-3).

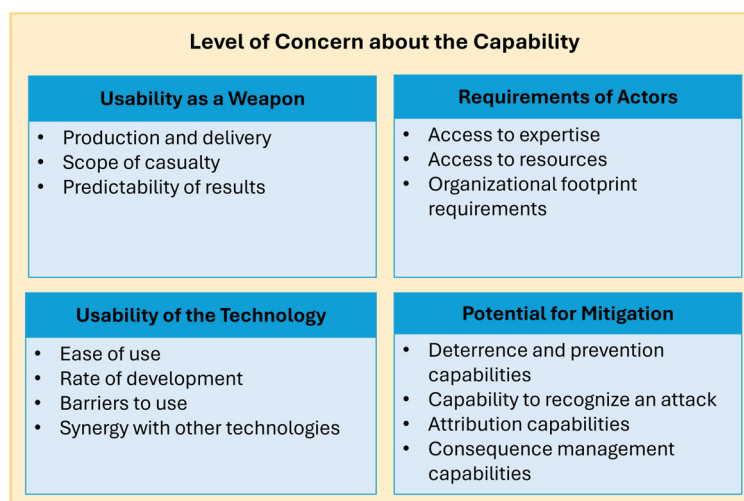


FIGURE 3-3 Framework for assessing concern. The framework consists of four factors, along with descriptive elements within each factor. The factors are usability of the technology, usability as a weapon, requirements of actors, and potential for mitigation. These factors delineate the information used to assess the level of concern for particular synthetic biology-enabled capabilities.

SOURCE: Adapted from NASEM (2018).

As shown above, the framework considers the requirement of actors with intent to deploy a biological agent in an attack, including access to resources and expertise. Malicious actors may range from an individual to a state-sponsored bioweapons program (see Box 3-3). With respect to current AI-enabled biological tools, some degree of expertise is required to be able to exploit them for harmful applications. In looking toward the future, the committee considered the possibility of the development of AI agents that would lower the tacit knowledge required for an actor with malicious intent (see Box 3-4).

BOX 3-3

State Sponsored Biological Weapons Programs

Section 403 of the Arms Control and Disarmament Act requires a report by the President on Adherence to and Compliance with Arms Control, Nonproliferation, and Disarmament Agreements and Commitments (U.S. Department of State, 2024). The report presents findings related to compliance with the terms of the Convention on the Prohibition of the Development, Production and Stockpiling of Biological and Toxin Weapons and on Their Destruction (BWC), opened in 1972 and entered into force in 1975. The U.S. definition of a state biological warfare (BW) program includes activities such as research and development, possessing facilities for production, training or plans for use, and use or attempted use of BW agents. The report concludes that Russia and North Korea have active offensive BW programs and that China and Iran engage in activities that raise concerns about compliance with the BWC. Determining whether a state's biological research program is offensive or defensive is a complex undertaking (Regan and Dubin, 2023). The role of biotechnology in the relationship between a country's research program and its military strategy is equally difficult to analyze. The International Security Advisory Board in the U.S. Department of State recently reported on China's "military civil fusion strategy" (International Security Advisory Board, 2024) and observed that China has prioritized investment in the life sciences and biotechnology for decades. The committee discussed the advantages of increased resources provided by state level biological research programs and concluded that the barriers to de novo agent production are universal. While the United States maintains its biotechnological lead, the report calls urgently for significant commitment to scientific training and workforce development.

BOX 3-4

Future Impact of AI Agents on Expertise and Resources Required by Actor

The discovery-to-deployment process could be streamlined with AI-assisted ideation, in which a bespoke AI agent assists with specific inquiries and helps the user generate viable hypotheses. Some of these ideas, determined to be feasible or interesting by the user, are then reformulated as tasks for AI-enabled biological design, the latter producing a sufficient number of viable and property-specific designs that are then pushed into the wet laboratory for synthesis and characterization. Operationalization and deployment of viable pathogens are the last steps.

Research on AI agents for ideation is currently quite active. Using large language model (LLM)-enabled AI co-pilots/agents for ideation, such as CRISPR-GPT, or LLMs for biomedical discoveries or material science discovery, AI researchers are focused on introducing automation in the synthesis of scientific literature. The goal is often humble, as in lowering the entry barrier for graduate students in a specific fast-evolving scientific domain (Wang et al., 2023). In other cases, it is more ambitious. Borrowing from Gao and colleagues (2024, p.1) in their collaborative aspiration article, researchers "envision 'AI scientists' as systems capable of skeptical learning and reasoning that empower biomedical research through collaborative agents that integrate machine learning tools with experimental platforms. Rather than taking humans out of the discovery process, biomedical AI agents combine human creativity and expertise with AI's ability to analyze large datasets, navigate hypothesis spaces, and execute repetitive tasks." As described in another article by Lu and colleagues (2024), the goal is to remove all human assistance. There are substantial challenges with any of the aspirations laid out in the aforementioned articles. Realization of these agents in the future may substantially lower the entry barrier for ideation and design for organizations or individuals, including those with no to little background in the life sciences but with sufficient motivation.

continued

BOX 3-4 continued

The next steps, from ideation to design, currently require more expertise, as the AI models target separate tasks and are currently not deeply integrated to take the human expert out of the loop in a meaningful way. The current paradigm requires instead that the user first discern which criteria to employ to narrow down hypotheses and then formulate them into precise “tasks” for AI-enabled biological design tools. Lone actors are less likely to possess the resources and expertise needed to move beyond ideation and design than organizations or dedicated nation states. It is also worth noting that computational resources and cost are often major hurdles for lone actors or small organizations when considering whether general-purpose AI lowers the barrier to discovery. However, certain AI biological tools make use of various computational architectures, from variational autoencoders to graph adversarial neural networks (see Appendix A for more detailed explanations) to diffusion-based models and others, which have much lower computational and storage demands than LLMs. Provided with precise hypotheses and sufficiently granular and precise data, dedicated organizations can easily build over existing open-source AI models and customize them for tasks of interest. However, as one proceeds further down the paradigm of attempting to cross the barrier from the digital to the physical world, the challenges described above present a higher threshold for expertise and resources for even dedicated organizations and nation states.

Finally, weaponization of the transmissible biological agent is necessary, or the usability as a weapon, as described in the 2018 framework, which accounts for the “implications for production and delivery of a weapon, expected scope of casualty, and the predictability of the intended results” (NASEM, 2018, p. 27).

Large-scale virus production, weaponization, dispersal, and dissemination are primarily informed through biotechnological, engineering, or meteorological knowledge and largely unencumbered by classical biological challenges or dependent on biological datasets. The committee acknowledges that nonbiological AI tools may accelerate development and evaluation for weaponization and dissemination by predicting ideal meteorological or environmental conditions for dispersal or tracking population movement, but these aspects are beyond the scope of this report. However, the fundamental problems associated with biological weapons production and delivery have not changed in recent years (Roffey, Tegnell, and Elgh, 2002; Guillemin, 2006).

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4

Promoting and Protecting AI-Enabled Innovation for Biosecurity

Executive Order 14110 on the Safe, Secure, and Trustworthy Development and Use of Artificial Intelligence and the National Security Memorandum on AI both articulate the promise and potential vulnerabilities of artificial intelligence (AI) applications. To this end, the first two guiding principles and priorities listed in the Executive Order state, (1) “artificial intelligence must be safe and secure”; and (2) “promoting responsible innovation, competition, and collaboration will allow the United States to lead in AI and unlock the technology’s potential to solve some of society’s most difficult challenges.” This chapter highlights beneficial applications of AI-enabled biological tools to enhance biosecurity and mitigate biological threats through improved prediction, detection, prevention, and response.

AI-ENABLED INFECTIOUS DISEASE BIOSURVEILLANCE

AI-enabled biological tools have long contributed to multiple aspects of infectious disease surveillance, such as epidemiological tracking and forecasting, early-warning systems, hotspot detection, and resource allocation. Figure 4-1 lists several significant roles of AI-based methods for infectious disease surveillance. Rapid advances were seen during the COVID-19 pandemic as multiple AI models were built and used to support diverse aspects of the response to the pandemic (Arora et al., 2021; Chen et al., 2021; Syrowatka et al., 2021; Ahmed, Boopathy, and Sudhagara Rajan, 2022; Brownstein et al., 2023; Malhotra and Sodhi, 2023; Sarmiento Varón et al., 2023).

AI-based approaches to disease surveillance complement traditional approaches. Natural language processing is used to analyze global communication sources by parsing, filtering, and classifying the data to remove noise and focus on unusual infections. For example, HealthMap identified early cases of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) infections in China as a “cluster of pneumonia cases of unknown etiology” (Freifeld et al., 2008). Following an early-warning system’s identification of an outbreak, the pathogen in question can be classified by diagnostic algorithms, differentiating pathologic and epidemiologic characteristics of an outbreak and pointing to a potential pathogen. AI-enabled platforms are additionally promising to detect and investigate infectious disease outbreaks (Sintchenko and Holmes, 2015). For instance, phylodynamic approaches can incorporate epidemiological data to infer host-related events such as disease introductions into discrete geographic regions or to constrain transmission hypotheses with contact data (Volz, Koelle, and Bedford, 2013; Ingle et al., 2021). Likelihood-free deep learning models are lowering the computational cost of these methods to build phylodynamic models in a fraction of the computational time (Kupperman, Leitner, and Ke, 2022; Voznica et al., 2022; Thompson et al., 2023). To stop an outbreak, infections can be traced by spatiotemporal clustering and contact tracing, aiding in the analysis of complex transmission patterns. Transmission during a pandemic can be traced using genetic epidemiological methods, which were greatly expanded and refined during the COVID-19 pandemic (Li et al., 2021; Niemi, Daly, and Ganna, 2022).

AI models that utilize deep learning can now analyze genomic data rapidly to identify known pathogens (Singh, 2022; Hou et al., 2024). By comparing both genetic and structural features of biological agents to known organisms, it may be possible for future AI models to provide insights into potential functions, virulence, pathogenicity, and other properties (Suster et al., 2024). However, as emphasized earlier in this report, such functionality will depend on the development of larger, more robust datasets on which the models can train.

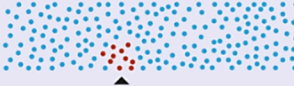
Function	Examples
Early warning 	- Natural-language processing of news sources to identify outbreaks (Freifeld et al., <i>JAMIA</i> 2008) - Unsupervised machine learning of social media data to detect unknown infections (Lim, Tucker, and Kumara, <i>J Biomed Inform</i> 2017)
Pathogen classification 	- Convolutional neural network model for reading antibiograms (Pascucci et al., <i>Nat Commun</i> 2021) - Convolutional neural network model to automate malaria microscopy and diagnosis (Liang et al., <i>IEEE</i> 2016)
Risk assessment 	- Reinforcement learning of Covid-19 positivity rates to target limited testing in Greece (Bastani et al., <i>Nature</i> 2021) - Machine-learning models including random forest and extreme gradient boosting to use syndromic surveillance for Covid-19 risk prediction (Dantas, <i>PLoS One</i> 2021)
Source identification 	- Automated data mining of electronic medical records to uncover hidden routes of infection transmission (Sundermann et al., <i>Clin Infect Dis</i> 2021) - Supervised machine learning in combination with digital signal processing for genomic tracing of Covid-19 (Randhawa et al., <i>PLoS One</i> 2020)
Hotspot detection 	- Neural computing engine to correlate sound from hospital waiting rooms with influenza spikes (Al Hossain et al., <i>Proc ACM Interact Mob Wearable Ubiquitous Technol</i> 2020) - Multilayer perceptron artificial neural network model to detect spatial clustering of tuberculosis (Mollalo et al., <i>Int J Environ Res Public Health</i> 2019)
Tracking and forecasting 	- Real-time stacking of multiple models to improve forecasts of seasonal influenza (Reich et al., <i>PLoS Comput Biol</i> 2019) - Machine learning to combine new data sources for monitoring Covid-19 (Liu et al., <i>J Med Internet Res</i> 2020)

FIGURE 4-1 Various functions of AI for infectious-disease surveillance. Shown is a non-exhaustive list of functions of AI-aided infectious-disease surveillance and representative examples from the published literature. Each example includes the type of AI algorithm, a brief description of its purpose, and the associated citation. “Covid-19” denotes coronavirus disease 2019. SOURCE: Brownstein et al. (2023).

AI-enabled real-time biosurveillance is indeed within reach. AI models can monitor genomic data continuously from various sources, amplify such data with predicted protein structure data, and detect emerging threats or unusual patterns that potentially indicate a new circulating pathogen. These same models can integrate genomic data with epidemiological information to predict outbreak patterns and inform public health responses in near real time (Brownstein et al., 2023; Suster et al., 2024).

AI models also can guide treatment decisions. In addition to serving as early-warning systems and identifiers of both potential biological agents and risks for outbreaks and epidemics, these same systems can be used to guide the design of effective treatments to inform public health strategies (Wong, de la Fuente-Nunez, and Collins, 2023; Wong et al., 2024).

AI-ENABLED BIODESIGN FOR COUNTERMEASURE DEVELOPMENT AND EMERGENCY RESPONSE

The rapid development of medical countermeasures (MCMs) (e.g., vaccines, biologics/therapeutics, and diagnostics) to address an emerging biological threat is essential to secure the world against such threats, whether they are natural or human-made. The past decade has seen a

significant shift in the approach to vaccine research and development (R&D). Building on the foundational work done by many vaccinologists working on HIV/AIDS, the field has shifted from empirical development of vaccines to an approach referred to as “structure-based” vaccine design (McLellan, Chen, Joyce, et al., 2013; McLellan, Chen, Leung, et al., 2013; Gilman et al., 2016; Crank et al., 2019; Graham, Gilman, and McLellan, 2019). In this approach neutralizing antibodies are used to identify and understand the best epitopes on the viral particle to target with a vaccine-elicited immune response or as a therapeutic monoclonal antibody. Once those antibodies have been identified, the structure of the target antigen is used to understand how the antibodies bind. Moreover, structural biology is used to guide stabilizing amino acids in the viral antigen that helped lock the target in a conformation that can best elicit protective immune responses. Such amino acid changes include stabilizing proline residues, discovered by Jason McLellan and Barney Graham working on Middle East respiratory syndrome coronavirus (MERS-CoV)—work that took years of wet-lab testing and optimization (Pallesen et al., 2017). When SARS-CoV-2 emerged, this foundational work fortunately had already been done; these mutations were ported into the viral spike protein to stabilize the antigen in many of the vaccine candidates, including both the Moderna and Pfizer/BioNTech mRNA vaccines, which ultimately were authorized for emergency use within a year (Corbett et al., 2020; Polack et al., 2020; Baden et al., 2021).

How will AI change this approach? Precedent that establishes a role for computational approaches to change how we mount an emergency response has already been set. During the COVID-19 pandemic, the Institute for Protein Design at the University of Washington used computational methods to generate a nanoparticle vaccine (SKYCovione) that was the first computationally designed protein medicine to complete a Phase III trial; SKYCovione was approved for emergency use during the COVID-19 pandemic (SK bioscience, 2023; Miranda et al., 2024). Subsequently, AI has further transformed the world of protein sciences and structural biology and, in so doing, is poised to affect discovery and design of MCMs in many positive ways. As an example, during the first weeks of COVID-19, researchers were “flying blind,” working with presumptive structures of what the SARS-CoV-2 spike might look like based only on prior knowledge of other betacoronaviruses. Six weeks after the sequence of SARS-CoV-2 was released, the solved SARS-CoV-2 structure was published online on February 15, 2020 (Wrapp et al., 2020). Now, only four years later, AI-enabled protein structure prediction tools have dramatically changed these timelines—AlphaFold, which has already been used to predict the structure of millions of proteins (Tunyasuvunakool et al., 2021; Varadi et al., 2022; Varadi et al., 2024), or other structure prediction models, would be able to predict the structure of a similar new viral glycoprotein accurately in a matter of hours, and AI could now be used to do more detailed molecular dynamic modeling and glycan profiling (Abramson et al., 2024; Krishna et al., 2024). Moreover, instead of relying on the proline residues that had been used to stabilize the MERS-CoV spike, generative AI can now be used to design new vaccines rapidly and explore a vast array of additional stabilizing strategies such as cavity filling, scaffolds, and particulate display,¹ thereby greatly reducing the time needed to design a vaccine against a new viral threat.

Similar acceleration can also be expected in the development of monoclonal antibodies for diagnostics and therapeutic use. In the pre-AI era, therapeutic monoclonal antibodies were produced by expression from antibody gene sequences identified in B cells isolated from infected humans that were shown to have viral neutralization capacity (Gilman et al., 2016). This process is time consuming, as it requires identification of a convalescent donor as well as selection of the appropriate B cells and isolation or expression of the monoclonal antibodies. During COVID-19, the first antibodies in the United States were isolated from the B cells of some of the first infected patients in North America (Zost et al., 2020). In the future, improved biodesign tools likely will make it possible for monoclonal antibodies or other protein biologics to be generated rapidly in silico (He et al., 2024). These designed biologics will be able to bind to predicted or actual structures on known viral antigens, circumventing the need to isolate antibodies from convalescent patients and expediting the development of both diagnostics and therapeutics.

¹ Presentation to the committee by Jimmy Gollihar, August 14, 2024. See Appendix B.

In the case of both vaccines and biologics, AI-enabled biological tools make it possible for much of the work to be performed *in silico*, allowing a global outbreak response to be initiated anywhere in the world based initially on viral sequence alone, which holds the promise of dramatically reducing the time from detecting an outbreak to initiating countermeasure development. The committee notes that while going from an AI-designed sequence for a potential pathogen to having the pathogen will be met with significant bottlenecks, going from an AI-designed sequence to a therapeutic antibody is more akin to the earlier example of toxin design—that is, there is much less of a bottleneck to make a single protein than a complex system such as a virus.

AI AND BIODESIGN SECURITY

Although AI-enabled biological tools could be misused for harmful applications, as described in Chapter 3, the tools may also be used to screen and prevent the creation of biological threats by augmenting current nucleic acid screening methods. However, this specific application of AI is an emerging area and would need more research. Nucleic acid synthesis is a critical chokepoint between the digital and physical divide; therefore, measures to mitigate the misuse of commercial services for nucleic acid synthesis have been developed and implemented, including screening of both sequences and customers. The International Gene Synthesis Consortium (IGSC) is an industry-led group of gene synthesis companies formed in 2009 that sets voluntary standards for screening nucleic acid synthesis orders based on U.S. and international biosecurity standards. Most major nucleic acid synthesis providers are part of the IGSC and use a common protocol to screen the sequences of synthetic gene orders and the legitimacy of customers. Screening nucleic acid sequences (and modified sequences that retain homology to known amino acid sequences) against “sequences of concern” (SOCs), such as those on the Select Agents and Toxins List, is routine and standard practice.

Executive Order 14110² includes directives for the U.S. federal government to reduce risks of misuse of synthetic nucleic acids. Subsequently, the Framework for Nucleic Acid Synthesis Screening was released in April 2024, requiring all federal agencies that fund life sciences research to mandate the use of nucleic acid synthesis providers that (1) attest they are implementing screening of synthetic nucleic acids; (2) screen purchase orders for synthetic nucleic acids to identify SOCs; (3) screen customers to verify their legitimacy; (4) report potentially illegitimate purchase orders of synthetic nucleic acids involving SOCs or of benchtop nucleic acid synthesis equipment; (5) retain records relating to purchase orders for synthetic nucleic acids and benchtop nucleic acid synthesis equipment; and (6) take steps to ensure cybersecurity and information security (National Science and Technology Council, 2024). This builds upon the longstanding National Institutes of Health (NIH) Guidelines for Research Involving Recombinant or Synthetic Nucleic Acid Molecules, most recently updated in April 2024 (National Institutes of Health, 2024).

At this time, an SOC is defined as a “best match” to a sequence of already federally regulated agents (such as the Biological Select Agents and Toxins List,³ which is codified in the USA PATRIOT Act [2001]⁴ and the Bioterrorism Act [2002],⁵ or the Commerce Control List)⁶ except when the sequence is also found in an unregulated organism or toxin. The Framework for Nucleic Acid Synthesis Screening has updated the SOC definition to also include those “sequences known to contribute to pathogenicity or toxicity, even when not derived from or encoding regulated biological agents” (Fast Track Action Committee on Synthetic Nucleic Acid Procurement Screening of the National Science and Technology

² Exec. Order No. 14110, Fed. Reg. 24283 (October 30, 2023). See <https://bidenwhitehouse.archives.gov/briefing-room/presidential-actions/2023/10/30/executive-order-on-the-safe-secure-and-trustworthy-development-and-use-of-artificial-intelligence/> (accessed October 12, 2024).

³ See <https://www.selectagents.gov/sat/list.htm> (accessed October 10, 2024).

⁴ USA PATRIOT Act of 2001, Pub. L. No. 107-56, 115 Stat. 272.

⁵ Public Health Security and Bioterrorism Preparedness and Response Act of 2002, Pub. L. No. 107-188, 116 Stat. 594.

⁶ See <https://www.bis.doc.gov/index.php/regulations/commerce-control-list-ccl> (accessed October 14, 2024).

Council, 2024). Determining these sequences is a challenging, highly contextual process that has been upended by generative AI tools (Godbold, 2024). This area will require more research to be useful for regulatory purposes.

Any new security measures would build on the existing practice recommended by the IGSC for logging synthesized nucleic acids to establish provenance. Logging designs and orders does not itself prevent misuse, but this practice can assist with attribution. While developing measures to diminish misuse of commercial services for nucleic acid synthesis is important, such an effort is an imperfect control point because it will not deter all potential malicious actors and there are technological means to bypass commercial synthesis. For example, future development of benchtop synthesis devices would bypass commercial synthesis screening processes. Customer screening of these devices and embedded security screening software, similar to the processes in nucleic acid synthesis, will remain important. Monitoring developments in this area is important as is engaging companies with biosecurity expertise in advance of any commercial launch of a benchtop device.

AI-enabled biological tools may pose new challenges for nucleic acid synthesis screening. It is difficult to infer the function of a novel design without context on its intended use, and current screening methods may not detect every possible sequence that may be concerning. Nucleic acid synthesis providers may need to rely increasingly on metadata to evaluate incoming orders. Today, there are no standards for metadata to be included with nucleic acid synthesis orders, beyond “know your customer” guidance. “Know your order” guidance could assist nucleic acid synthesis providers in screening orders more effectively.

Beyond nucleic acid synthesis screening, the development of a layered approach that might include “intent-based screening,” guardrails embedded within design tools and logging and tracking of who is using design tools (i.e., “know your design screening”), has been suggested. While these could complement existing screening approaches to reduce the chance of misuse of design tools and serve as deterrents, such approaches may be both impractical to implement and ineffective to deter more sophisticated actors.

Moreover, screening how users access and leverage AI tools is challenging and prone to false positives that can hinder or prevent legitimate research that may itself be beneficial. Closed-source AI tools are often accessible via an application programming interface (API), which can effectively put a “black box” around proprietary models and any security features that have been implemented. For example, the committee learned from Jason McLellan that expert vaccine researchers seeking to access the AlphaFold3 API to develop a monkeypox vaccine were initially barred from using the API; the AlphaFold3 developers had planned to prevent the use of their API for the design of certain viral sequences.⁷ While the researchers were eventually permitted to use AlphaFold3 for monkeypox vaccine design after several weeks’ delay, the event underscores the challenges of screening for misuse via design tool usage. Access restriction may present challenges to end users who have legitimate designs as an unintended consequence of security features, and further research to optimize screening methodologies is warranted. Here, AI-enabled biological tools themselves may be integrated in screening and help improve screening over traditional methods.

The implementation of security screening for AI-enabled biological tools has evolved rapidly with the release of newer generative models such as those based on large language models and represents an emerging area of research. Currently, AI-based approaches have a few limitations. Screening can occur prior to training a generative model by leaving data on harmful biological agents out of the training dataset completely. This approach, however, may impact the performance of a model intended for beneficial uses such as vaccine design (Kaplan et al., 2020; Hoffman et al., 2022). Agent-based models instructed to avoid designing harmful sequences or, alternatively, template-based design models can be prohibited from using particular structures or sequences as templates for generative design—as seen in the monkeypox example. In the near future, optimizing and increasing efficiency of AI-enabled biological tools in screening may provide an additional mitigation strategy.

⁷ Presentation to the committee by Jason McLellan and Emanuele Andreano, October 1, 2024. See Appendix B.

Conclusion: More research in new methodologies for nucleic acid synthesis screening, including how to leverage AI-enabled biological tools for screening, is needed for this process to be an efficient control point and possible strategy to mitigate potential biosecurity risks. Engaging practitioners and developers of AI-enabled biological tools to explore the most effective approaches for implementation is important as is engaging biologists and other developers to assess impacts of screening on public health preparedness and response.

BALANCING AI-ENABLED RESPONSES TO BIOLOGICAL THREATS AND POTENTIAL RISKS

AI applications for biosecurity have great potential to enhance biosurveillance, bolster early-warning systems, improve infectious disease monitoring, and accelerate the development of MCMs for a variety of diseases—from infectious to noncommunicable. In general, research programs for understanding the biology of infectious agents are important to support the development of vaccines and therapeutics to prevent and treat diseases caused by such agents, whether from natural occurrences or intentional threats. Along these lines, associated biological datasets will be useful and necessary for training AI models to accelerate the development of such MCMs, bolster biosurveillance, and support other previously mentioned biosecurity applications.

Various institutes and offices housed within the U.S. Department of Health and Human Services (HHS), such as the U.S. Centers for Disease Control and Prevention (CDC) and NIH, have existing programs that support infectious disease research. Specifically, CDC houses the National Center for Emerging and Zoonotic Infectious Diseases,⁸ which has the mission of preventing and controlling infectious disease. Within NIH, the National Institute of Allergy and Infectious Diseases (NIAID) is the primary funder of both intramural and extramural research to better understand the biology of infectious diseases affecting humans,⁹ including those that are emerging.¹⁰ Within NIAID, the Integrated Research Facility at Fort Detrick is a “collaborative resource that facilitates multidisciplinary research to understand, treat, prevent, and eradicate diseases caused by novel, emerging, and highly virulent viruses.”¹¹ Other institutes at NIH centers investigate how environmental conditions affect immunological defenses to infectious disease (National Institute of Environmental Health Sciences¹²) and which viruses cause human cancer (National Cancer Institute¹³).

The U.S. Department of Agriculture (USDA) also has research programs that study infectious diseases affecting plants and animals. Relevant USDA agencies include Animal and Plant Health Inspection Services,¹⁴ the National Institute of Food and Agriculture,¹⁵ and the National Bio and Agro-Defense Facility.¹⁶ The U.S. Army Medical Research Institute of Infectious Diseases,¹⁷ under the U.S. Department of Defense (DoD), also conducts MCM research “to deter and defend against current and emerging biological threats.” The National Biodefense Analysis and Countermeasures Center¹⁸ is situated within the U.S. Department of Homeland Security and plays an important role in filling knowledge gaps

⁸ See <https://www.cdc.gov/ncepid/about/index.html> (accessed November 17, 2024).

⁹ See <https://www.niaid.nih.gov/research/infectious-diseases> (accessed November 17, 2024).

¹⁰ See <https://www.niaid.nih.gov/research/centers-research-emerging-infectious-diseases> (accessed November 17, 2024).

¹¹ See <https://www.niaid.nih.gov/research/frederick-integrated-research-facility> (accessed November 17, 2024).

¹² See <https://www.niehs.nih.gov/research/supported/health/autoimmune> (accessed November 18, 2024).

¹³ See <https://ccr.cancer.gov/staff-directory/principal-investigators/research-areas/cancer-and-viruses> (accessed November 18, 2024).

¹⁴ See <https://www.aphis.usda.gov/> (accessed November 18, 2024).

¹⁵ See <https://www.nifa.usda.gov/about-nifa> (accessed November 18, 2024).

¹⁶ See <https://www.usda.gov/nbaf> (accessed November 18, 2024).

¹⁷ See <https://usamriid.health.mil/> (accessed November 18, 2024).

¹⁸ See <https://www.dhs.gov/science-and-technology/national-biodefense-analysis-and-countermeasures-center> (accessed November 18, 2024).

about biological agents and R&D related to MCMs. Other relevant programs that support research on infectious disease biology include the National Science Foundation's program on Ecology and Evolution of Infectious Diseases¹⁹ and CDC's One Health Initiative,²⁰ which involves collaborative efforts of several agencies including NIH and USDA.²¹ Though not directly related to infectious disease research, the U.S. Department of Energy (DOE) is home to many biological research programs (including those at DOE's national laboratories)—for example, the Joint Genome Institute²² is a user facility that provides a resource for genomic research. These research programs and investments will need to continue, and ideally expand, in support of understanding the biology of infectious agents toward enhanced biosecurity enabled by AI.

Recommendation: The U.S. Department of Health and Human Services (including the U.S. Centers for Disease Control and Prevention [CDC]), U.S. Department of Agriculture (USDA), U.S. Department of Defense, U.S. Department of Energy, U.S. Department of Homeland Security, and other agencies that support and conduct research should continue to invest in vigorous research programs to understand the biology of infectious agents. As a part of these programs, agencies should also invest in and implement biosurveillance networks through CDC and USDA, in cooperation with public health agencies globally.

In addition to this enhanced understanding through research and data collection, research supporting the development and testing of new models for MCM development and biosurveillance is also needed. Several programs are currently in place that focus on the development of MCMs. For instance, the Biomedical Advanced Research and Development Authority (BARDA), which is situated within the Administration for Strategic Preparedness and Response in HHS, “promotes the advanced development of medical countermeasures to protect Americans and respond to 21st century health security threats.”²³ As a part of its 2022–2026 strategic plan, BARDA committed to investing in technologies that could make MCMs “faster, safer, and more accessible” as a part of its preparedness goal.²⁴ BARDA's Division of Research, Innovation, and Ventures²⁵ specifically invests in a broad portfolio and would be well suited to invest in approaches using AI-enabled design tools for development of MCMs. Funders like the Advanced Research Projects Agency for Health (ARPA-H) and the Defense Advanced Research Projects Agency (DARPA) invest in high-risk projects and also can consider investing in AI-enabled design tools for MCM development. Both ARPA-H²⁶ and DARPA²⁷ already have plans to apply AI to biology in innovative ways. For instance, the new ARPA-H CATALYST program plans to develop predictive drug safety and efficacy models to develop *in silico* models of human physiology for predictive drug safety and efficacy testing, reducing the time from drug discovery to delivery. The DoD Chemical and Biological Defense²⁸ group, Defense Threat Reduction Agency,²⁹ and Defense Health Agency³⁰ all play a role in

¹⁹ See <https://new.nsf.gov/funding/opportunities/eeid-ecology-evolution-infectious-diseases>.

²⁰ See <https://www.cdc.gov/one-health/about/> (accessed November 18, 2024).

²¹ See <https://www.aphis.usda.gov/news/agency-announcements/usda-announces-awards-support-one-health-initiatives> (accessed November 18, 2024).

²² See <https://jgi.doe.gov/> (accessed November 18, 2024).

²³ See <https://aspr.hhs.gov/AboutASPR/ProgramOffices/BARDA/Pages/default.aspx> (accessed November 18, 2024).

²⁴ See <https://medicalcountermeasures.gov/barda/strategic-plan/> (accessed November 18, 2024).

²⁵ See <https://drive.hhs.gov/ventures.html> (accessed November 18, 2024).

²⁶ See <https://www.synbiobeta.com/read/arpa-h-unveils-program-to-revolutionize-drug-safety-and-efficacy-testing> (accessed November 18, 2024).

²⁷ See <https://www.synbiobeta.com/read/darparas-aixbto-initiative-pushing-bioinnovation-boundaries> (accessed November 18, 2024).

²⁸ See <https://www.acq.osd.mil/ncbdp/cbd/> (accessed November 18, 2024).

²⁹ See <https://www.dtra.mil/> (accessed November 18, 2024).

³⁰ See <https://www.health.mil/About-MHS/OASDHA/Defense-Health-Agency> (accessed November 18, 2024).

biosurveillance. In addition, innovations in AI tools are largely driven by the commercial sector, making public–private partnerships important in this endeavor.

Recommendation: Entities within the U.S. Department of Health and Human Services (e.g., National Institutes of Health, Biomedical Advanced Research and Development Authority, and Advanced Research Projects Agency for Health) and the U.S. Department of Defense (e.g., Defense Advanced Research Projects Agency and Chemical and Biological Defense) should fund approaches using artificial intelligence–enabled design tools for ongoing application in medical countermeasure development, especially in the face of an epidemic, pandemic, or other biological threat.

Recommendation: As part of a national preparedness strategy, the Biomedical Advanced Research and Development Authority, U.S. Department of Energy, and U.S. Department of Defense should establish a public–private partnership, analogous to Operation Warp Speed and the COVID-19 High-Performance Computing Consortium, that can both leverage and provide continuous access to artificial intelligence–enabled tools and computational resources and be activated rapidly in an emergency response.

The potential misuse of biological data and knowledge for harmful applications by an actor with malicious intent presents a security concern. When considering biosecurity risks, it is important to distinguish between technical capability and intent to avoid unnecessarily stifling R&D advances that are critical in response strategies for natural infectious disease threats and intentional threats. Efforts are ongoing to address these issues in various sectors including the government, the scientific community and its broader enterprise, and commercial developers of the relevant technologies. Both the Executive Order and National Security Memorandum on AI outline such initiatives, including the development of risk evaluations and stakeholder engagement (see Table 1-1), similar to recommendations articulated in this report. Other policies are in place that oversee pathogen research in general but do not include AI, such as the Policy for Oversight of Dual Use Research of Concern and Pathogens with Enhanced Pandemic Potential.³¹

The committee summarizes the following considerations discussed in this report. Application of AI-enabled biological tools is an emerging area of research that is developing rapidly. At present, the capabilities of AI tools have certain limitations that are described in previous chapters and in Appendix A. The present limitations, as well as uplift in capabilities, hold true whether for beneficial or harmful applications. AI-enabled biological tools provide data-driven guidance for experimental validation, and the resulting experimental data may then be used to improve the models in an iterative Design-Build-Test-Learn cycle, as discussed in Chapter 2. However, physical production of digital outputs remains a significant barrier. In the future, significant improvements in automated laboratories may assist with experimental throughput and scalability. Furthermore, AI-enabled biological tools have promising applications to enhance and augment biosecurity responses in naturally occurring outbreaks or intentional events.

Amassing significant datasets, sometimes generated through compute-intensive simulations, is a prerequisite for training AI models (e.g., Protein Data Bank [PDB] for AlphaFold); therefore, the availability of high-quality, robust data can be the leading indicator of an emerging or developing AI model capability. Currently, no approach exists to determine the absolute amount of data needed for training, but, as previously noted, AlphaFold was trained on a few hundred thousand protein structures in PDB. It is important to note that dataset size is comparatively significant for biology because the process to generate biological data is often laborious. Moreover, more datasets are necessary to capture the

³¹ See <https://bidenwhitehouse.archives.gov/ostp/news-updates/2024/05/06/united-states-government-policy-for-oversight-of-dual-use-research-of-concern-and-pathogens-with-enhanced-pandemic-potential/> (accessed November 18, 2024).

diversity and complexity of biology and subsequently account for them in models. The scaling hypothesis indicates that model performance improves as the size of the model, dataset, and computational resources increases (Kaplan et al., 2020; Hoffmann et al., 2022).

In light of the rapid pace of development of AI-enabled biological tools and the continued collection of training data that could lead to new or improved capabilities of such models, the committee suggests the adoption of an “if-then” approach to guide future vulnerability and risk assessment and mitigation. The if-then strategy accounts for these dynamics without being prescriptive or restrictive based on predicting future capabilities.

An if-then approach would allow for the continual assessment and periodic reassessment of vulnerabilities and risks using a principles-based thought framework. These evaluations should be anchored in both data availability and the associated emerging AI capabilities as well as real-world threat models that incorporate all four factors articulated in the 2018 framework and in Figure 3-3 (i.e., usability of the technology, usability as a weapon, requirement of actors, and potential for mitigation) (NASEM, 2018). In addition, the if-then strategy should establish benchmarks and metrics to track data availability and associated capability progress as well as a set of thresholds that, once surpassed by emerging capabilities, activate risk assessment and mitigation.

Recommendation: The U.S. Department of Defense and the U.S. Artificial Intelligence (AI) Safety Institute should develop an “if-then” strategy to evaluate continuously both the availability and quality of data and emerging AI-enabled capabilities to anticipate changes in the risk landscape (e.g., if dataset “x” is collected, then monitor for the emergence of capability “y”; if capability “y” is developed, then watch for output “z”). Evaluation of AI-enabled biological tool capabilities may be conducted in a sandbox environment.

- Example datasets of interest:
 - If clear associations between viral sequences and virulence parameters become known, then evaluate the capability of AI models to predict or design pathogenicity and/or virulence.
 - If robust viral phylogenomic sequence datasets linked to epidemiological data become available, then assess for the development of new AI models of transmissibility that could be used to design new threats.
- Example AI models that warrant assessment:
 - If AI models are developed that infer mechanisms of pathogenicity and transmissibility from pathogen sequencing data, then watch for attempts to modify existing pathogens to increase their virulence or transmissibility.
 - If AI models are developed that could predictably generate a novel replication-competent virus, then assess risk for bioweapon development.

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5

Importance of Data in AI-Enabled Biological Models

Artificial intelligence (AI)–enabled biological tools are likely to continue to require data-driven AI methods that learn patterns from large biological datasets, because the underlying rules of the types of biology being discussed in this report are generally unknown. This contrasts with AI systems that play and win difficult games with superhuman performance, such as chess (Campbell, Hoane, and Hsu, 2002) and Go (Silver et al., 2017), where the rules of the games are fully known and can be used to improve AI systems without additional data. Our recommendations on the importance of generating accessible high-quality biological data with robust data provenance are derived from three key properties commonly found in data-driven AI systems:

1. Risky data are hard to define. It is difficult to determine which aspects of training data (i.e., features) cause deep neural networks to make predictions (Ras et al., 2022). As a result, it is not feasible to a priori identify targeted biological data points that, if specifically excluded from training, would prevent AI from making certain kinds of risky predictions (e.g., increased viral virulence) while still enabling related beneficial predictions (e.g., lack of pathogenicity of viral vector vaccines). The main option is to exclude large swaths of data from training.
2. State-of-the-art AI uses all available data. The most competitive AIs, including those for clinical decision support (Singhal et al., 2023) and code generation (Du et al., 2024), are built using foundation models trained on enormous volumes of general data (e.g., text, images, genomic) and then specialized to certain domains using techniques such as fine-tuning, prompt engineering, and model distillation. A biosecurity approach based on restricting large swaths of data from AI training is more likely to hinder research and is impractical given that model capabilities are reliant on the data.
3. High-quality data are a strategic national asset. Data-driven AI requires trusted high-quality data to be effective, as witnessed by recent breakthroughs ranging from protein folding AI trained on Protein Data Bank (PDB) data to digital pathology AI for identifying infectious diseases trained on curated clinical datasets (Smith and Kirby, 2020; Jiao et al., 2021; Xu, Liu, et al., 2024). The availability of high-quality datasets with ensured data integrity will help maximize the benefits of AI-driven biosurveillance and faster medical countermeasure (MCM) design.

BIOLOGICAL DATASETS FOR TRAINING AI MODELS

As previously noted, publicly available and accessible data are necessary to train AI models. The accessibility of data follows a continuum from freely available and in the public domain to highly classified information—with several options in between. Guidance for federally funded research data advocates for data deposition in repositories that meet the criteria established in the report, *Desirable Characteristics of Data Repositories for Federally Funded Research*. These criteria include openness “consistent with legal and policy requirements related to maintaining privacy and confidentiality, Tribal and national data sovereignty, and protection of sensitive data,” as well as appropriate cybersecurity policies to prevent “unauthorized access to, modification of, or release of data, with levels of security that are appropriate to the sensitivity of data (e.g., the NIST [National Institute of Standards and Technology]

Cybersecurity Framework¹)” (Subcommittee on Open Science of the National Science and Technology Council, 2022). The guidance is meant to provide a balance of accessibility and security in recognition of the importance of data as a public good with dual-use potential. Data generated and funded by private entities have no requirement for submission to national repositories. The security of these data is the responsibility of the private entity.

The Findable, Accessible, Interoperable, and Reusable (FAIR) principles established by Wilkinson and colleagues (2016) serve as aspirational guidelines to the data management community for creating and maintaining data resources that are machine-accessible. Data can be FAIR while implementing access controls and use restrictions in accordance with federal guidelines. The data management community has improved data FAIRness through increased adoption of persistent identifiers, standardized authentication protocols, and metadata standardization. As repositories converge on common standards and ontologies, the interoperability of data from different studies improves, making data more machine-accessible and AI-ready. However, FAIR does not mean that all data are open or immediately accessible to humans or machines. Furthermore, as our understanding of biological data improves and new measurements are possible, the data models and standards as well as the policies must evolve to accommodate this new understanding.

Evolution of Biological Data Access Policy at the National Institutes of Health

Many factors influence the extent to which information is available—some are driven by organizational policy and others are driven by legal constraints, but all are consistent with the FAIR principles. In 2003, the National Institutes of Health (NIH) established a data-sharing policy that required a data-sharing or management plan for any researcher seeking more than \$500,000 in direct costs. However, researchers could seek exemptions for concerns over biosafety or national security. Also, any identifiable information had to be stripped from data related to research involving human subjects. After the final data-sharing policy, as high-throughput genome sequencing became a reality, NIH published the Policy for Sharing of Data Obtained in NIH Supported or Conducted Genome-Wide Association Studies (GWAS) in 2007,² which lowered the funding threshold from \$500,000 to \$0. In other words, NIH indicated that as much data as possible created through public tax dollars should be put into the public domain, in large part due to the rise of fraudulent publications that were accepted to high-impact journals but not reproducible. Thus, while it could be argued that NIH has pushed for public data sharing to speed up scientific investigation, it could also be argued that it has done this to address the reproducibility crisis in science. If investigators did not have a clear plan for sharing genomic data, their data can be sent to the Database of Genotypes and Phenotypes (dbGaP), which would curate and redistribute the data on behalf of the investigators.

After the GWAS policy, NIH released the Genomic Data Sharing (GDS) Policy in 2014.³ This policy was due to new advances in sequence generation capabilities. This time, whole exome sequencing and whole genome sequencing became (relatively) cost-effective. NIH recognized that the GWAS policy was a bit myopic because it focused mainly on basic variants and their associations with phenotypic expression as opposed to simply the entirety of the genome. Thus, the GDS policy aimed to resolve this issue. However, it should be recognized that both the GWAS and GDS policies indicated that data should be made available in a manner that was de-identified.

Many of the policies that have been set forth by NIH and the infrastructure that has been set up to manage biomedical data have focused on data derived from humans. The GWAS and GDS policies, for instance, are concerned mainly about human privacy and not about biosafety issues that could arise from the sharing of bacterial or viral data and their impact on human cell lines. That said, NIH has recognized

¹ See <https://www.nist.gov/cyberframework> (accessed January 6, 2025).

² See <https://grants.nih.gov/grants/guide/notice-files/not-od-07-088.html> (accessed January 2, 2025).

³ See <https://grants.nih.gov/grants/guide/notice-files/NOT-OD-14-124.html> and <https://sharing.nih.gov/genomic-data-sharing-policy/about-genomic-data-sharing/gds-policy-overview> (accessed January 2, 2025).

that there are many ethical questions regarding the sharing of infectious disease data, and one of the core components of NIH's soon-to-be-initiated Human Virome Project relates to the ethical, legal, and social implications regarding the collection and sharing of such data.⁴

INFRASTRUCTURE AND GOVERNANCE

Agencies establish policies for data governance that dictate access controls. For example, while NIH wants genomic data to be made accessible to the public, it requires a legitimate scientific reason for the use of data derived from humans. Thus, to access data at dbGaP, investigators need to submit a request for access because dbGaP only serves as a repository of the data. NIH did not stand up an analytic workbench to allow people to interact with the hundreds of datasets that were submitted. Moreover, since dbGaP is a federal repository, the Data Access Committee that determined who got access based on their requests was composed solely of federal employees. This was not well received by the research community broadly because it takes the academic community (and commercial communities, if they contributed data) out of the decision-making equation. This was a particular concern because requests for data came from all over the world, and it was unclear whether investigators wanted data on U.S. residents to be shared with foreign entities.

By contrast, the NIH's All of Us Research Program has a different access framework. Rather than serve as only a repository, the program aimed to create a cloud-backed analytic workbench. In doing so, the All of Us Research Program chose to sandbox the data, wherein researchers would be able to access and investigate the data using various programming languages, such as Python, R, and SAS, but would not be permitted to take individual-level data out of the environment. Further, unlike dbGaP, the All of Us Research Program took a different approach than the traditional Data Access Committee model for dbGaP, which required a review for every request of data—including when two different requests came from the same investigator. Specifically, the All of Us Research Program adopted the passport model (Dyke et al., 2018), which worked on a trust model based more on the credentials of the user than on how the data were used. Thus, the All of Us Research Program requires

- institutions associated with where a user originates to enter into a contractual agreement that takes responsibility for their actions (as of December 2024, there were 1,038⁵);
- users to take a series of courses or trainings regarding what they are expected to do (and not do) with the data they are provided access to and pass an examination;⁶
- users to accept a data user code of conduct⁷ in which they agree not to perform stigmatizing research or attempt to re-identify the participants⁸ in the program; and
- users to create a new workspace and a public description of the work that they are doing for every study that they do (to comply with the 21st Century Cures Act⁹), unlike the access request, which only occurs once.

In addition, the All of Us Research Program data is structured in a tiered manner, such that data deemed less risky in regard to privacy or sensitivity are made available in the Registered Tier,¹⁰ while whole genome sequencing data and potentially more private and sensitive data are in the Controlled Tier.

⁴ See <https://commonfund.nih.gov/humanvirome>.

⁵ See <https://www.researchallofus.org/institutional-agreements/> (accessed January 8, 2025).

⁶ See <https://support.researchallofus.org/hc/en-us/articles/6088597693716-All-of-Us-Onboarding-Tutorial-Video> (accessed January 8, 2025).

⁷ See <https://www.researchallofus.org/faq/data-user-code-of-conduct/> (accessed January 8, 2025).

⁸ "Participants" in this case indicates consented research volunteers.

⁹ 21st Century Cures Act of 2016, Pub. L. No. 114-255, 130 Stat. 1033.

¹⁰ See <https://support.researchallofus.org/hc/en-us/articles/4552681983764-How-All-of-Us-protects-participant-privacy> (accessed January 8, 2025).

Xia and colleagues (2023) provide details on how re-identification risk is assessed and managed. Further documentation on how All of Us provisions access is provided by Mayo et al. (2023).

Still the All of Us Research Program data access model creates some limitations with respect to training and developing AI models. At present, the All of Us Research Program does not allow its data to be used to fine-tune or train external foundation models or machine learning models more generally. This restriction stems from concerns of losing control over the data. As such, while a researcher can build a machine learning model within the All of Us Research Program sandboxed workbench, the model cannot be taken out of the workbench. This may change in the future, but it demonstrates why the All of Us Research Program model may not be a suitable example for data governance in considering the various types of biological datasets used to train AI models. Nevertheless, the program is an appropriate case study for how the entire workflow, from the data to the AI-enabled tool to all user interactions with the tool, can be sandboxed and managed accordingly if required. In many respects, this would be analogous to an access-controlled version of AlphaFold.

THE DATA LIFE CYCLE AND NATIONAL REPOSITORIES

Data have a life cycle that is generally composed of the following stages: plan and design, collect and create, analyze and collaborate, evaluate and archive, share and disseminate, and publish and reuse (see Figure 5-1). The various stages inform each other and future experiments. How data are captured and preserved has a critical impact on their integrity and the subsequent validity of AI. Once data and the corresponding contextual information (labels) are available in machine-readable formats, many powerful computational and statistical techniques can be utilized to identify patterns, make predictions, and advance scientific understanding.

The core of the data life cycle is storage and data management, as a long-term home and effective management policies are critical to maximizing the value of a dataset. The data life cycle connects to the AI layer and its associated software life cycle wherever data are used (reuse) or models or data are generated (collect and create). The experimental design or plan impacts the data that are collected and made available for training or building AI because at this stage, the scientist must gather enough contextual information to inform future data reuse. The contextual information links data collected in one experiment to another and builds the necessary connective tissue for large-scale statistical analyses.

National repositories are a valuable resource for scientific progress and AI-driven innovation due to federal investments in curated public databases. The creation, curation, and preservation of national repositories require significant long-term investments in laboratory and computing infrastructure capacity as well as subject matter expertise.

Beyond data storage and management systems with strong data provenance, U.S. federal research funding agencies such as NIH, Advanced Research Projects Agency for Health, Defense Advanced Research Projects Agency (DARPA), National Science Foundation (NSF), U.S. Department of Energy (DOE), and others should fund AI hardware infrastructure so the research community can competitively develop next-generation AI-enabled biological tools that are specifically trained on biological data and focused on biological capabilities. Many researchers lack access to the necessary data, computing, and educational resources to fully leverage or develop AI for biological R&D, including medical countermeasure development.

In order to build AI applications, it is essential to have infrastructure connecting data repositories with the high-performance computing hardware needed to train models, compute inference, and then store those outputs. Many researchers lack access to the necessary data, computing, and educational resources to leverage or develop AI biological models or tools. The National Artificial Intelligence Research Resource (NAIRR) pilot seeks to bring together computational, data, software, model, training, and user support resources as foundational infrastructure for AI research. NAIRR is led by NSF, in partnership with 12 other federal agencies and 26 nongovernmental entities.¹¹ Similarly, DOE is leading the Frontiers

¹¹ See <https://nairrpilot.org/> (accessed November 17, 2024).

in Artificial Intelligence for Science, Security and Technology initiative, offering the connected computing, experimental data generation, and educational infrastructure of the DOE national laboratories to the scientific community to advance AI research. NIH has partnered with cloud vendors to host large repositories such as the Sequence Read Archive (SRA). More than 50 petabytes of raw sequence data are co-located with cloud computing resources that can be used for the development of new tools (Serratus, a cloud computing environment¹²) or new data resources (Logan, a set of compressed sequencing files for SRA¹³), provided that the researcher has the budget to conduct the computations. The United States has led research and development (R&D) in high-performance computing for decades, according to the Supercomputing Top 500 List,¹⁴ and national investments in AI infrastructure will be essential to maintain U.S. competitiveness in AI.



FIGURE 5-1 Various stages of the research data life cycle. Storage and management are core to data generation, characterization, and reuse. National data repositories, such as PDB, GenBank, SRA, and the Global Natural Products Social Molecular Networking (GNPS), are foundational for data science and AI research. SOURCE: Longwood Medical Area Research Data Management Working Group, 2024, under Creative Commons Attribution-NonCommercial 4.0 International License.

¹² See <https://serratus.io/about> (accessed November 17, 2024).

¹³ See <https://registry.opendata.aws/pasteur-logan/> (accessed November 17, 2024).

¹⁴ See <https://top500.org/lists/top500/> (accessed November 17, 2024).

CONSIDERATIONS FOR BUILDING DATA INFRASTRUCTURE

The recognition of AI and enabling data as national R&D, security, and economic assets has been articulated in various policies and strategic plans, including the September 2024 launch of the interagency Task Force on AI Datacenter Infrastructure¹⁵ to promote the development of data center projects and the 2024 National Security Memorandum on AI.

Recommendation: To maximize the benefits of artificial intelligence (AI) in biology, U.S. federal research funding agencies such as the National Institutes of Health, Advanced Research Projects Agency for Health, Defense Advanced Research Projects Agency, National Science Foundation, U.S. Department of Energy, and others should fund efforts and infrastructures to do the following:

- Create and steward AI-compatible data and training sets for biological applications that are made publicly available;
- Initiate and support public–private partnerships for the generation of training datasets; and
- Develop frontier AI-enabled biological tools that are specifically trained on biological data and focused on biological capabilities.

Recommendation: Federal research agencies that house or fund biological databases should consider these to be strategic assets and concordantly implement robust data provenance mechanisms to maintain the highest data quality. They should provide strong incentives and infrastructure for the standardization, curation, integration, and continuous maintenance of high-quality, publicly accessible biological data at scale.

Recommendation: While a public–private partnership is a reasonable approach to pilot the creation of an artificial intelligence (AI)–ready data resource on government-owned infrastructure, a dedicated organization is central to a sustainable national data stewardship resource. The National Artificial Intelligence Research Resource (NAIRR) pilot should explore approaches for establishing central repositories for training AI models.

PROTECTING DATA ASSETS FOR AI-ENABLED BIOLOGICAL R&D

To maintain high-quality and secure national data assets for AI-enabled biological R&D, it is crucial to understand how data flow into AI software and how AI software is used across the design, build, and test stages. Figure 5-2 summarizes the relationships between these layers and the locations of potential vulnerabilities at the data, AI software, and application layers.

- **The Data Layer** shows key datasets used to train AI applied to biology (and beyond). This layer is highly heterogeneous, is decentralized, and includes general text and images mined from the broader internet, creating opportunities to manipulate the biological R&D process by manipulating the data used to train AI (Shan et al., 2024).
- **The AI Layer** highlights many sophisticated forms of AI that are relevant for biological R&D, and these are implemented using a large corpus of open-source software (at least 55% of PubMedCentral–Open Access articles have links to open-source software [Istrate et al., 2022]). This exposes the biological R&D ecosystem to software supply chain attacks, including attacks identified by the Cybersecurity and Infrastructure Security Agency (2024) as major national security risks, which could be used to manipulate AI software used in biomedical R&D.

¹⁵ See <https://bidenwhitehouse.archives.gov/briefing-room/statements-releases/2024/09/12/readout-of-white-house-roundtable-on-u-s-leadership-in-ai-infrastructure/> (accessed November 17, 2024).

- **The Application Layer** emphasizes that AI technologies are being and will be used across the design, build, and test stages, including biological design tools, automation agents to control equipment or order biological materials and reagents (Bose, 2024), and AI analysts to help interpret clinical data to accelerate clinical trials and improve post-market surveillance (e.g., AI for real-world evidence [Subbiah, 2023]). For example, the drug program was approved to prevent lung transplant rejection based on real-world observational data from its off-label use (Food and Drug Administration, 2021). Vulnerabilities introduced at the lower layers could manifest across the biomedical R&D stages and include difficult-to-mitigate vulnerabilities specific to generative AI (Short, La Pay, and Gandhi, 2019; Gupta et al., 2023).

A final challenge is that both noisy data and intentional corruption are likely to be self-amplifying, because corrupted data generated at the application layer will be added back into the data layer, further corrupting AI that is later trained on them and potentially leading to phenomena such as *model collapse* (Dohmatob et al., 2024; Shumailov et al., 2024). Maintaining data provenance (i.e., tracking by whom and where in the application layer data were created, recording all the AI tools and software dependencies used to assist in generating novel data, and linking AI tools to the data used to train their models) creates a complete picture of the history of the data. This allows low-quality data to be rejected, enables rapid identification of AI systems that may have been compromised by noisy or intentionally manipulated data, and ensures high-quality data producers can determine how their data were used to train AI. Boxes 5-1, 5-2, and 5-3 give examples of enabling technologies, which are already utilized in adjacent regulated industries, for creating high-quality and secure data assets with strong data provenance.

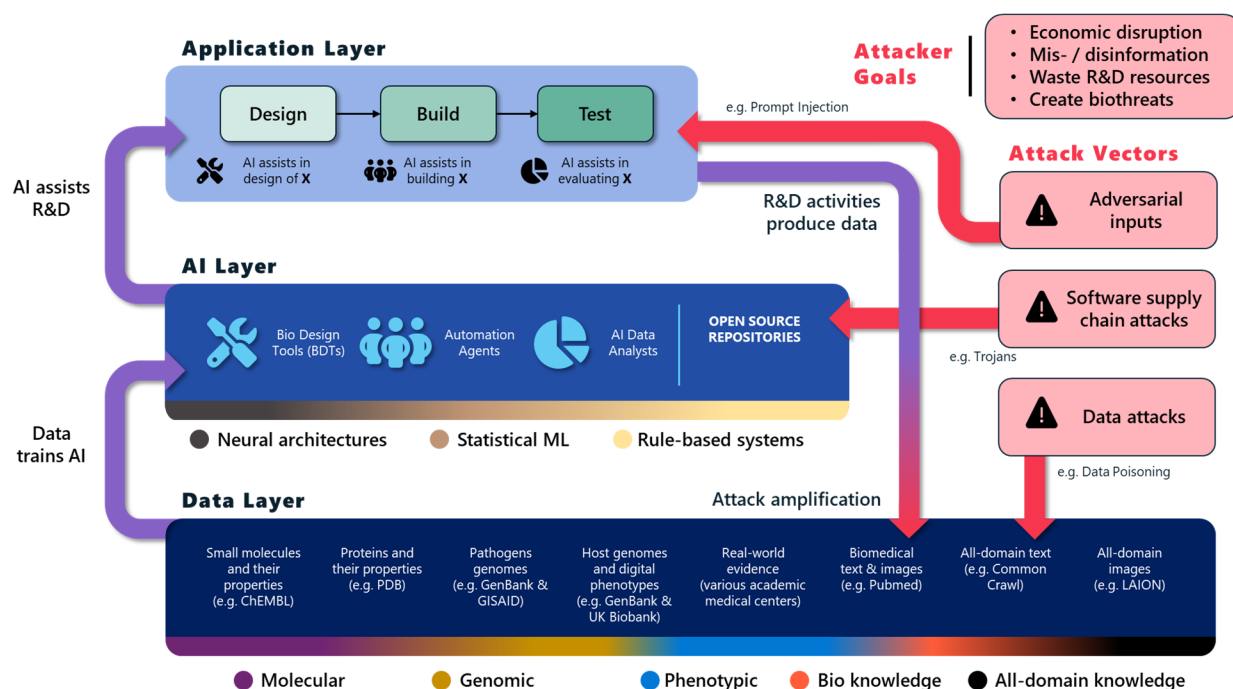


FIGURE 5-2 A summary of the relationships between the application, AI, and data layers and the locations of potential vulnerabilities at those layers.

NOTE: ML = machine learning.

BOX 5-1 Protecting the Data Layer

Major biological data platforms, such as the National Center for Biotechnology Information's GenBank, currently assume data may be (non-maliciously) noisy, and they apply quality assurance (QA) checks to reject bad data,^a although QA remains difficult to achieve (Steinegger and Salzberg, 2020). However, other regulated industries, such as finance and manufacturing, assume actors may intentionally manipulate data to launder money^b or circumvent import controls.^c To mitigate risks posed by malicious actors, additional concepts and techniques are used:

Blockchains^d: Foundational technologies, such as blockchains (Tama et al., 2017), implement distributed ledgers that provide decentralized data provenance with strong guarantees against attacks. Decentralized data provenance is especially appealing when centralized tracking is infeasible or undesirable, as may be the case with production of biological data and is already being trialed in clinical settings on sensitive medical data (Ng, 2021).

Fine-grained access control: Fine-grained access control restricts creation, reading, and manipulation of data using principled algorithms. These approaches can be tailored to create data systems where access is neither too restrictive nor too permissive and security properties can be guaranteed (Cybersecurity and Infrastructure Security Agency, 2019), enabling researchers to share biological data while balancing national security concerns and avoiding coarse-grained treatment of data as classified information.

Privacy-preserving machine learning (Mohassel and Zhang, 2017): For highly sensitive data, a vibrant range of techniques has been and is being developed to perform machine learning on data without revealing the underlying datasets, including methods applicable to viral genomes (Kim and Lauter, 2015; Marcolla et al., 2022).

^a See <https://www.ncbi.nlm.nih.gov/sra/docs/sequence-data-processing/> (accessed October 20, 2024)

^b See <https://www.fdic.gov/banker-resource-center/bank-secrecy-act-anti-money-laundering-bsaaml> (accessed October 20, 2024).

^c See <https://www.dhs.gov/uflpa> (accessed October 20, 2024).

^d Blockchain is defined as “a distributed digital ledger of cryptographically-signed transactions that are grouped into blocks. Each block is cryptographically linked to the previous one (making it tamper evident) after validation and undergoing a consensus decision. As new blocks are added, older blocks become more difficult to modify (creating tamper resistance). New blocks are replicated across copies of the ledger within the network, and any conflicts are resolved automatically using established rules.” See <https://csrc.nist.gov/glossary/term/blockchain> (accessed December 30, 2024).

BOX 5-2 Protecting the AI Software Layer

Biological R&D activities that use AI software or create derivative AI software for redistribution to the broader community can leverage best practices, tools, and advanced research to mitigate risks in their software supply chain. For example, DARPA's AI Cyber Challenge^a is evaluating state-of-the-art approaches to broadly defend the open-source software supply chain, and its results and recommended tools are likely applicable to securing software supply chains in AI-enabled biological R&D. Examples of enabling technologies include the following:

Program analysis: These methods automatically analyze source code for vulnerabilities (Ponta, Plate, and Sabetta, 2018) with automated reasoning and heuristic search methods, including static and dynamic analysis (de Moura and Bjørner, 2008), and are applicable regardless of whether vulnerabilities are intentional or accidental.

continued

BOX 5-2 continued

Automated testing: These methods automatically construct test inputs that can be used to check for spurious results from packages, and they have been applied to detect bugs in critical and widely used machine learning packages (Wei et al., 2022).

Signature-based malicious package detection: These methods build machine learning classifiers based on examples of known malicious packages to automatically detect the signatures of potentially malicious packages in the wild (Sejfić and Schäfer, 2022).

^a See <https://bidenwhitehouse.archives.gov/briefing-room/statements-releases/2023/08/09/biden-harris-administration-launches-artificial-intelligence-cyber-challenge-to-protect-americas-critical-software/> (accessed October 20, 2024).

BOX 5-3**Protecting the Application Layer**

Methods to protect the application layer must consider the specific uses of AI at the design, build, and test stages. Examples include the following:

Generative AI red teaming: Techniques to characterize up-front the potentially dangerous output of AI models (e.g., Urbina et al., 2022), so security risks can be mitigated prior to model deployment and intentional attempts to misuse or jailbreak models can be detected in the wild.

Digital twins: This approach is especially applicable to autonomous laboratories and robotic systems and uses end-to-end digital simulations of autonomous systems to test AI models for dangerous behaviors without the risk of creating dangerous biology (“The increasing potential and challenges of digital twins,” 2024).

Anomaly detection: AI can be applied to learn typical usage patterns at the application level and detect deviations from these patterns for further inspection or terminate actions automatically (Elmrabit et al., 2020).

DATA-DRIVEN VULNERABILITIES AND MITIGATIONS

AI-enabled biological tools bring unique opportunities to biomedical research but may be vulnerable to data corruption. Adversarial attacks exploit vulnerabilities in AI models by introducing small, often imperceptible, perturbations in the input data that can cause the model to make incorrect predictions. We explore potential attack vectors and possible mitigation strategies to ensure data integrity for use in training models.

Potential Data Vulnerabilities*Data Poisoning and Corruption*

Data poisoning is the modification of training data to manipulate the AI model to learn certain patterns or relationships (Jagielski et al., 2018; Yerlikaya and Bahtiyar, 2022). These patterns may be designed to be factually incorrect, to be biased, or to obscure a signal of interest. Data poisoning can be dangerous because small undetected changes in large datasets can lead to significant errors in model predictions and downstream applications.

In the biomedical domain, AI models often rely on large publicly available datasets, such as gene expression profiles or drug efficacy studies. However, one of the concerns of such a reliance is that there is no guarantee that the data contributed to the public domain have been subject to peer review, and those

that have may lack rigorous validation. It is possible that an attacker could publicly disseminate falsified research papers or data, which in turn are mined by AI systems for training data. For instance, an attacker could submit a paper containing slightly altered gene–drug interaction data to a preprint server, where the efficacy of a particular drug is exaggerated.

A more targeted attack could involve altering training data for models that predict protein folding or drug interactions. For instance, a malicious actor could corrupt these data with false protein structures or misleading chemical interaction information, leading an AI system to generate flawed outputs, such as ineffective molecules or incorrect biological pathway predictions. These corrupted outputs could, in turn, derail efforts aimed at MCMs. While these would be caught during the test stage (i.e., preclinical testing and clinical trials), valuable time might be wasted in the context of a rapidly progressing epidemic or pandemic.

Backdoor Attacks

In a backdoor attack (Salem et al., 2022), the adversary introduces a hidden trigger during the training or development of the AI model, such that when a specific input pattern is provided the model produces incorrect or unsafe outputs. A backdoor attack is difficult to detect because it could remain dormant and undetected during normal usage and only be activated under specific conditions.

In this scenario, a public, open-source AI model that is designed to support vaccine discovery could be altered by embedding a backdoor that is activated when the model encounters a specific sequence in a DNA or protein structure.

Jailbreaking Attacks

Jailbreaking attacks involve bypassing the safety mechanisms built into AI models, allowing the system (or user of the system) to perform tasks it was otherwise restricted from doing (Xu, Usuyama, et al., 2024). An actor may also try to exploit the AI's capabilities by rephrasing requests or breaking down the task into smaller, seemingly benign steps. Instead of directly asking the AI to design a lethal substance, the attacker might ask it to suggest molecules with certain properties (e.g., specific toxicity levels) or find ways to bypass the safety filters by describing the task in vague or scientific terms. Over time, these carefully constructed prompts could lead the AI to generate prohibited outputs, despite the model's safeguards. Jailbreaking attacks are particularly concerning because they do not require altering the underlying model; rather, they exploit the model's interpretive processes to circumvent its safety protocols.

Model Fine-Tuning and Instruction Manipulation

In another type of attack, a system user could perform model fine-tuning by injecting malicious data or introducing specific instructions to subtly manipulate the behavior of an AI system. The fine-tuning attack enables an adversary to adjust a pretrained model by retraining it with additional adversarial data, steering it toward incorrect or dangerous outputs. For example, an attacker with access to the model's training process could fine-tune it to produce misleading information about disease progression, biosurveillance, or genetic and molecular interactions. A model fine-tuned in this manner might falsely predict the effectiveness of a particular drug or incorrectly diagnose the health status of an individual, which could lead to improper—or a lack of—identification of or treatments for an individual who has been exposed to some agent. By subtly altering how the AI interprets biological data, an attacker could push the model to design ineffective therapies that would waste time in clinical trials, misinform researchers in critical research, or manipulate public trust.

Attackers could fine-tune biomedical models to align with more harmful objectives, such as providing guidance for the creation of dangerous biological agents. For instance, by feeding the AI malicious training data, an adversary could reorient the model to generate synthetic pathways for

producing toxic compounds or lethal pathogens (Urbina et al., 2022). This kind of attack could enable the development of bioweapons or hazardous chemicals under the guise of legitimate drug discovery efforts. To be successful in such an attack, the adversary would need access to the model’s training pipeline as well as knowledge of how to craft malicious data that will specifically skew the AI’s responses. This would require a nontrivial level of technical expertise in both machine learning and the relevant biological domain. Moreover, this type of attack would require the capability either to inject harmful data during training or to deceive the AI’s deployment environment. Finally, this type of attack will still encounter significant barriers during the build and test phases.

Mitigation Approaches

Continuous Penetration Testing (Pen Testing)

Regular penetration (or pen) testing involves probing AI models for weaknesses by simulating various attack scenarios. This allows researchers to assess the security of their models continuously and identify potential vulnerabilities before they can be exploited by malicious actors. For biomedical AI, this could involve testing how the model responds to harmful prompts, encoded malicious data, or attempts to bypass its safety mechanisms. Robust pen testing ensures that the AI system remains secure even as new threats emerge.

Robust Training Techniques

AI models need to be trained using well-curated and trusted datasets with strong data provenance to minimize the risk of data poisoning. This involves implementing more stringent data filtering and augmentation processes. Additionally, models should be trained to recognize and reject manipulated inputs that could lead to harmful outputs. Training AI models in adversarial environments—where they are exposed to deliberately corrupted or misleading data—can improve their robustness and resistance to attacks.

Benchmarking for Robustness and Risk Evaluation

Developing standardized benchmarks for evaluating the robustness of biomedical AI models is essential. These benchmarks should assess how well a model can resist various forms of attack, including data corruption, jailbreaking, and backdoor manipulation. Consistent benchmarking ensures that models meet safety standards and can be trusted in high-stakes applications, such as pathogen discovery or drug design.

Regular Model Audits

Auditing AI models at regular intervals can help in identifying backdoors or hidden vulnerabilities that may not be immediately apparent. These audits should involve both automated techniques and manual reviews by experts in AI security and biomedical research. Audits can ensure that any updates or changes to the model do not introduce new vulnerabilities.

Assessment of Large Language Models

General-purpose AI models (e.g., large language models [LLMs] like ChatGPT, Gemini, or Claude) are actively assessed to identify different types of vulnerabilities (e.g., cybersecurity), capabilities (e.g., factuality, persuasion), or unwanted or harmful behaviors (e.g., model responses including personal information, or promoting harm or bias). Different approaches can be used to assess or evaluate AI models like LLMs, including the following, among others:

- Static single-turn, open-ended prompts where human raters rate the responses as problematic or not;
- Red teaming where humans take on the role or goals of malicious actors and simulate adversarial “attacks” or questions to the model to see how the model will respond;
- Multiple choice questions addressing a specific scientific domain with automatic grading;
- Uplift studies where the performance of groups of people on a task using an AI model is compared to the performance of an equivalent group that must achieve the same task but without the AI model; and
- Some combinations and variations on these listed approaches.

Each approach brings with it different strengths and weaknesses; for example, automated multiple-choice-question benchmarks or “tests,” which are historically favored by computer engineers, can be performed rapidly and lend themselves easily to comparing the “grades” between models and for different versions of the same model (i.e., one could follow how the grades rise or fall between different models and see if a model is “getting better” at answering biology questions). However, the types of questions posed tend to be simplistic, and the inclusion of reasoning questions has limitations. Closed-ended responses, by definition, also imply that this approach could not be used to “discover” new model capabilities (e.g., trying to find out if an LLM could design good primers for polymerase chain reaction). Moreover, the meaning attributed to any one grade given to a set of multiple-choice-question assessments is unknown—for example, when could we say that an AI model is good at math? What grade would be needed?

Similarly, there are challenges in probing LLMs to see whether they may provide information that could be used by malicious actors to cause harm. To date, most assessments of LLMs related to biosecurity risks are utilized to see how well LLMs can answer scientific questions rather than providing accurate and actionable information that would significantly help a malicious actor to cause harm. If a model performs poorly on biology questions, for example, the assumption is that the model would be unlikely to provide accurate information to a malicious actor in any significant way. There are limitations in extrapolating these assessments, which is indicative of how nascent the science of model assessments and evaluations currently is. More research is needed to map out the approaches and corresponding methodologies to obtain meaningful results.

In addition, disagreement remains on the nomenclature used to describe the different approaches for LLM assessments. Although red teaming, a term used in military applications and predating AI, is one of the best-known approaches, it is not the only approach. Furthermore, the distinction between general LLMs and domain-specific LLMs, such as biological LLMs, should be considered in model assessments and evaluations.

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Appendix A

Mapping the Landscape of AI-Enabled Biological Design¹

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ABSTRACT

This review provides an overview of artificial intelligence-enabled biological models, focusing on their applications in designing biological molecules and systems. We review foundation models that learn data distributions, generative models that sample from these distributions, predictive models that map between biological modalities, and design models that produce desirable biological outputs. The review highlights substantial advances in protein engineering while noting emerging areas in genomic and transcriptomic modeling. We explore these models' potential to accelerate biological design and reveal new fundamental insights, emphasizing their role in augmenting traditional experimental approaches. We also survey diverse biological datasets underlying these models, emphasizing their crucial role in model development. We also discuss the integration of large language models with biology-specific tools as a promising frontier. Finally, we speculate on future developments, envisioning a synergy between computational simulation and experimental validation in biological discovery and design.

GLOSSARY

Important technical terms, which are italicized throughout the text, are also collected with brief definitions in this glossary, in the order in which they appear in the text:

- *Supervised machine learning*: A type of machine learning in which models are trained on labeled data to predict specific outputs from given inputs.
- *Unsupervised machine learning*: A type of machine learning in which models learn patterns and structures from unlabeled data without predefined outputs.
- *Scaling hypothesis*: An empirical finding that language modeling performance improves predictably as the size of the model, dataset, and computational resources increases.
- *Foundation models*: Large-scale unsupervised models trained at scale on vast datasets that exhibit broad capabilities across various downstream tasks.
- *Multilayer perceptron (MLP)*: A basic type of neural network with fully connected layers of artificial neurons.
- *Convolutional neural networks (CNNs)*: Neural networks designed to process data dominated by local interactions among features using convolution operations.
- *Graph convolutional neural networks (GNNs)*: Neural networks that operate on graph-structured data, capturing relationships between nodes in the graph.
- *Transformer architecture*: A neural network architecture that uses self-attention mechanisms to process sequential data, allowing for efficient modeling of long-range dependencies.

¹ The author is solely responsible for the content of this paper, which does not necessarily represent the views of the National Academies of Sciences, Engineering, and Medicine.

- *State space models (SSMs)*: A class of sequence modeling architectures inspired by techniques from signal processing that can be configured to learn local or long-range interactions while maintaining efficient computational scaling.
- *Hybrid architectures*: Neural network designs that combine different layer types, such as transformer layers or SSM layers, to leverage their respective strengths.
- *Autoregressive language modeling*: A training objective in which models predict the next token in a sequence given the previous tokens.
- *Masked language modeling*: A training objective in which models predict masked or corrupted tokens in an input sequence.
- *Discrete diffusion modeling*: A generative modeling approach that progressively removes and reconstructs artificially corrupted discrete tokens.
- *Mean squared error (MSE)*: A common loss function that measures the average squared difference between predicted and actual values.
- *Variational autoencoders (VAEs)*: Generative models that learn to encode inputs into a latent space and decode them back, often used for generating new data.
- *Continuous diffusion modeling*: A generative modeling approach that gradually adds and removes noise from continuous data.
- *Protein language models*: Foundation models trained on large datasets of protein sequences to learn patterns of protein structure and function.
- *Single-cell foundation models*: Unsupervised models trained on single-cell RNA sequencing data to learn cell-type-specific gene expression patterns.
- *Genomic language models*: Foundation models trained on DNA sequences to capture patterns of regulatory DNA, RNA, and proteins, as well as their interactions that create systems and organisms.
- *One-hot encoding*: A simple but effective method of representing categorical data as binary vectors. For a protein sequence, each amino acid is assigned a unique 20-dimensional binary vector in which one element corresponding to that amino acid is 1 and all others are 0.
- *Neural sequence embedding*: A representation based on using a large pretrained neural network to generate dense vector representations of a sequence, potentially capturing complex patterns within the sequence data learned during pretraining.

INTRODUCTION

From molecular interactions to multicellular organisms, biological complexity has required researchers to develop increasingly sophisticated computational approaches both to understand fundamental biological principles and to engineer new biological systems. Rapid progress in artificial intelligence (AI) also promises to improve our ability to do biological discovery and design and has led to a proliferation of tools with applications that span diverse areas of biological research and engineering. The diversity of these tools, alongside an equally diverse set of applications, creates the need for a comprehensive review that catalogues these innovations and provides a structured framework for understanding their relationships. This paper aims to present a systematic overview of AI-enabled biological models as well as the biological datasets that enable these models. By organizing these tools and tasks coherently, we aim to provide researchers, policymakers, and industry professionals with a clear overview of the current landscape, facilitating more effective utilization of these technologies and identifying promising areas for future development.

In this review, we begin with an overview and taxonomy of (1) biological foundation models, (2) predictive models, and (3) biological design tools. Our discussion of foundation models will cover biological sequence models within the central dogma, foundation models of biological representations beyond sequences, multimodal models, and generative approaches. We will then explore prediction tools, with special attention to protein structure and function prediction from sequence, as well as multi-omic

prediction models. We will then examine biological design tools that generate new sequences with desired properties, which are often aided by property prediction tools. Here, we highlight problems in protein design, structure-based sequence generation, and DNA promoter sequence design. Throughout, we will emphasize the role of generative models and modern deep learning techniques.

We will then consider the datasets used to train these models, assessing their quality and information richness. These include comprehensive sequence databases; structural repositories; large datasets that measure function; and multi-omic datasets spanning genomics, transcriptomics, and proteomics. Finally, we will explore how developments in large language models (LLMs) can be integrated with biology-specific tools. While this integration is still in its early stages, we will speculate on both the potential synergies and challenges in combining general-purpose language models with specialized biological tools, considering applications in scientific literature analysis, experimental design, and biological knowledge integration.

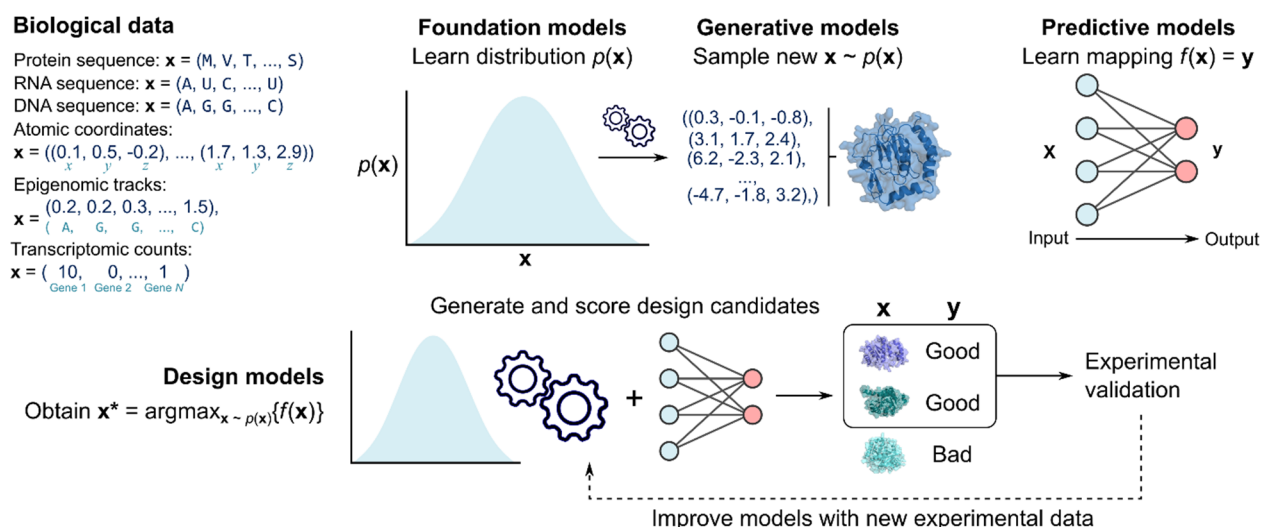


FIGURE A-1 Machine learning models and data for biological design.

NOTES: Data types include discrete sequences (protein, RNA, and DNA), continuous atomic coordinates (molecules represented as a set of three-dimensional atomic coordinates), continuous epigenomic tracks (epigenomic markers at each genomic position), and discrete transcriptomic count vectors (dimensions corresponding to different genes). Foundation models learn complex data distributions at scale (image illustrates a one-dimensional probability density, but biological distributions are much higher-dimensional), while generative models enable efficient sampling from these distributions (image illustrates a generated set of atomic coordinates that forms a protein structure). Predictive models map between different biological data types, potentially with complex structured outputs. Finally, design models often leverage both generative and predictive models to generate and score design candidates for experimental validation, with new experimental data potentially being used to subsequently improve the models. The image illustrates three protein designs produced by a generative model, two of which are labeled as “good” by a predictive model, which are then selected for experimental validation.

BIOLOGICAL MODELS I: FOUNDATION MODELS

In this review, we define foundation models (Bommasani et al., 2021) using two criteria: (1) the model is trained with an unsupervised objective that attempts to reconstruct the underlying data distribution; and (2) when trained on a sufficiently large and complex dataset, the model demonstrates generalist capabilities across a broad set of downstream tasks, with performance improving as both model and data scale increase (see Figure A-1).

Unlike traditional *supervised machine learning*, which maps inputs to different types of outputs (e.g., predicting a caption text given an image), *unsupervised machine learning* models aim to reconstruct their input as accurately as possible (Hastie et al., 2009). This approach encourages the model to learn fundamental patterns and structures within vast amounts of unlabeled data. When trained on sufficiently large and complex datasets (e.g., internet-scale text or images), modern deep learning architectures (Goodfellow, Bengio, and Courville, 2016; Vaswani et al., 2017) have shown the ability to acquire generalist capabilities applicable to a wide range of downstream tasks, which often differ substantially from the original training objective (Brown et al., 2020). As these models increase in size, they demonstrate improved performance on both the unsupervised reconstruction task and specific downstream tasks. This relationship between model scale and performance is often referred to as the *scaling hypothesis* or scaling laws (Kaplan et al., 2020; Hoffmann et al., 2022).

Unsupervised models that exhibit these scaling properties are often referred to as *foundation models* (Bommasani et al., 2021). The ability to learn generalist information across various tasks from a single large-scale unsupervised model distinguishes foundation models from smaller task-specific unsupervised models. The term “foundation” refers to how the information learned through a single unsupervised training objective can transfer to many task-specific downstream applications. Foundation models for text, code, images, and speech have made significant progress in their respective domains. In this review, we examine the current landscape of foundation models for biological data, discussing their progress and potential opportunities.

Preliminaries: Model Architectures and Objectives

Foundation models in biology use neural networks based on a common set of architectures and training objectives. Architectures define how neural connections are arranged within the model, while training objectives are loss functions that guide the optimization of the model. Understanding the distinction between a model’s architecture (e.g., a transformer) and its training objective (e.g., a diffusion model) is crucial for understanding the model’s capabilities and limitations.

Architectures

Several key architectures are fundamental to modern deep learning models and to foundation models in particular. The most standard deep learning architecture is the fully connected neural network, sometimes referred to as a deep neural network or as a *multilayer perceptron* (MLP) (Rumelhart, Hinton, and Williams, 1986; Goodfellow, Bengio, and Courville, 2016), which consists of an input layer of artificial neurons that receives data, one or more fully connected hidden layers to process the data, and an output layer to generate the final predictions. *Convolutional neural networks* (CNNs) (LeCun et al., 1989; Goodfellow, Bengio, and Courville, 2016) process information organized by position (e.g., one-dimensional text or two-dimensional images), focusing on interactions between nearby elements. *Graph convolutional neural networks* (GNNs) (Scarselli et al., 2009; Wu et al., 2019), a subvariant of CNNs, model elements as nodes and consider interactions only between connected nodes, making them useful for graph-representable data. These architectures excel when the data distribution is dominated by local interactions. The *transformer architecture* (Vaswani et al., 2017), based on an operation called self-attention (Bahdanau, Cho, and Bengio, 2014), allows each element in an input sequence to interact with all others, regardless of distance. This enables the capture of both local and long-range dependencies, proving remarkably effective for many tasks. However, its all-by-all element comparison leads to quadratic computational scaling with sequence length. *State space models* (SSMs) (Gu and Dao, 2023; Poli et al., 2023) are an emerging class of architectures that can be configured to prefer learning local or long-range interactions while maintaining near-linear computational scaling with sequence length. *Hybrid architectures* combine different layer types (e.g., a mixture of CNN layers, transformer layers, or SSM layers) within a single deep neural network, potentially benefiting from the strengths of each constituent architecture (Poli et al., 2024).

Objectives

A foundation model can be trained with different objectives that all attempt, in some way, to reconstruct the input. Different objectives are used depending on whether the underlying data are discrete or continuous. When data samples are sequences of discrete tokens (e.g., a sequence of DNA base pairs), a typical language model is trained via *autoregressive language modeling* in which the model is asked to predict the next token given a sequence prefix, enforcing a left-to-right order (Radford et al., 2019). An alternative training objective for discrete sequence models is *masked language modeling* in which a model is asked to predict the true values of positions that are masked or corrupted in the input (Devlin et al., 2018). A generalization of masked language modeling is *discrete diffusion modeling* in which a model is trained to progressively remove artificially masked or corrupted tokens in its input (Lou, Meng, and Ermon, 2023). When the input takes the form of continuous data (e.g., an image containing pixels along a color spectrum), a common objective is to simply minimize the *mean squared error* (MSE) separating the prediction value from the ground truth value (Hastie et al., 2009). For example, MSE loss is frequently used in a class of models called *variational autoencoders* (VAEs) (Kingma and Welling, 2014), which encode the input as a Gaussian latent variable encouraging a smooth representation of the data, combined with an MSE loss that compares the output of a decoder with the original sample. While VAEs can successfully represent some simple data distributions, more powerful recent models have been trained with *continuous diffusion modeling* (Ho, Jain, and Abbeel, 2020) in which, as in the discrete case, the model is trained to iteratively remove artificial noise from an input sample. Importantly, any architecture can be trained with any objective, though certain architectures are commonly paired with certain objectives (e.g., autoregressive transformer models).

Generative Models

Some objectives are particularly suited for efficient sampling from learned probability distributions. Autoregressive models, for example, enable left-to-right sampling based on the model's next-token predictions. A VAE enables decoding of Gaussian noise into new samples. Both continuous and discrete diffusion models facilitate iterative denoising or reconstruction of purely noised or corrupted samples. Models trained with these sampling-friendly objectives are often referred to as *generative models*.

Modeling the Distribution of Sequences

We first consider a major class of biological foundation models in which the underlying data are a sequence of discrete tokens, where each token corresponds to a DNA base pair, an RNA base pair, or a protein amino acid. We describe notable models, along with their corresponding model architecture and objective, in the sections below and in Table A-1.

General Protein Sequence

Many biological sequence models have been trained on large corpora of protein amino acid sequences across diverse protein families and have been shown to learn properties of both structure and function (Bepler and Berger, 2021). By learning which positions in a sequence input are useful when predicting values at other positions, *protein language models* identify covarying positions across protein evolution, which typically reflects structural proximity. By learning which amino acids are more likely (or unlikely) at given positions, they also learn which mutations are better (or worse) tolerated. A notable suite of protein language models is the ESM family of models, including the first-generation ESM-1b model (released as a general-purpose foundation model) (Rives et al., 2021) and ESM-1v models (specialized to the task of variant effect prediction) (Meier et al., 2021) and the second-generation ESM-2 models (released at different scales from 6 million to 15 billion parameters) (Lin et al., 2023); we discuss

TABLE A-1 Foundation Models of Biological Data

Category	Model name	Architecture	Training objective	Reference
General protein sequence	ESM-1b	Transformer	Masked language modeling	Rives et al., 2021
General protein sequence	ESM-1v	Transformer	Masked language modeling	Meier et al., 2021
General protein sequence	ESM-2	Transformer	Masked language modeling	Lin et al., 2023
General protein sequence	ESM3	Transformer	Discrete diffusion modeling	Hayes et al., 2024
General protein sequence	ProGen	Transformer	Autoregressive language modeling	Madani et al., 2021
General protein sequence	ProGen2	Transformer	Autoregressive language modeling	Nijkamp et al., 2023
General protein sequence	CARP	CNN	Masked language modeling	Yang, Fusi, and Lu, 2024
General protein sequence	ProteinBERT	Transformer	Masked language modeling	Brandes et al., 2023
General protein sequence	TAPE	Transformer	Masked language modeling	Rao et al., 2019
General protein sequence	ProtTrans	Transformer	Autoregressive language modeling	Elnaggar et al., 2022
General protein sequence	ProtGPT2	Transformer	Autoregressive language modeling	Ferruz, Schmidt, and Höcker, 2022
General protein sequence	RITA	Transformer	Autoregressive language modeling	Hesslow et al., 2022
Antibody protein sequence	AbLang	Transformer	Masked language modeling	Olsen, Moal, and Deane, 2022
Antibody protein sequence	IgBERT	Transformer	Masked language modeling	Kenlay et al., 2024
Antibody protein sequence	IgT5	Transformer	Autoregressive language modeling	Kenlay et al., 2024
Antibody protein sequence	AntiBERTa	Transformer	Autoregressive language modeling	Leem et al., 2022
Antibody protein sequence	AntiBERTy	Transformer	Autoregressive language modeling	Ruffolo, Gray, and Sulam, 2021
Antibody protein sequence	Sapiens	Transformer	Masked language modeling	Prihoda et al., 2022
Coding RNA sequence	CaLM	Transformer	Masked language modeling	Outeiral and Deane, 2024
Coding RNA sequence	CodonBERT	Transformer	Masked language modeling	Ren et al., 2024
Noncoding RNA sequence	RiNALMo	Transformer	Masked language modeling	Penić et al., 2024
Noncoding RNA sequence	RNA-FM	Transformer	Masked language modeling	Chen et al., 2022
Noncoding RNA sequence	Uni-RNA	Transformer	Masked language modeling	Wang et al., 2023
Noncoding RNA sequence	RNAErnie	Transformer	Masked language modeling	Wang et al., 2024
Genomic DNA sequence	GenSLM	Transformer	Autoregressive language modeling	Zvyagin et al., 2023
Genomic DNA sequence	Nucleotide Transformer	Transformer	Masked language modeling	Dalla-Torre et al., 2023
Genomic DNA sequence	GPN	CNN	Masked language modeling	Benegas, Batra, and Song, 2023
Genomic DNA sequence	regLM	Transformer	Autoregressive language modeling	Lal et al., 2024
Genomic DNA sequence	HyenaDNA	SSM	Autoregressive language modeling	Nguyen, Poli, Faizi, et al., 2024

continued

TABLE A-1 *continued*

Category	Model name	Architecture	Training objective	Reference
Genomic DNA sequence	Caduceus	SSM	Autoregressive language modeling	Schiff et al., 2024
Genomic DNA sequence	Evo	Hybrid (SSM + Transformer)	Autoregressive language modeling	Nguyen, Poli, Durrant, et al., 2024
Molecular structure	RFdiffusion	Hybrid (GNN + Transformer)	Continuous diffusion	Watson et al., 2023
Molecular structure	Chroma	GNN	Continuous diffusion	Ingraham et al., 2023
Molecular structure	Protpardelle	Transformer	Continuous diffusion	Chu et al., 2024
Cellular transcriptomes	Geneformer	Transformer	Masked language modeling	Theodoris et al., 2023
Cellular transcriptomes	scBERT	Transformer	Masked language modeling	Yang et al., 2022
Cellular transcriptomes	scFoundation	Transformer	Masked language modeling	Hao et al., 2024
Cellular transcriptomes	scGPT	Transformer	Specialized autoregressive language modeling	Cui et al., 2024

NOTE: Summary of various foundation models across different categories of biological data, including protein sequences, antibody sequences, RNA sequences, genomic DNA sequences, molecular structures, and cellular transcriptomes. For each model, the architecture, training objective, and citation are also provided.

ESM3, the third-generation ESM model (Hayes et al., 2024), under the subsection on multimodal models below. ESM-1 and ESM-2 models are trained with masked language modeling and use a standard transformer architecture. Another notable class of models is the ProGen family of models, including the first ProGen model (Madani et al., 2021) and ProGen2 (released at different scales from 151 million to 6 billion parameters) (Nijkamp et al., 2023). ProGen models are also transformers trained on diverse protein families using an autoregressive language modeling objective and have been used to sample diverse sequences from enzyme families while retaining catalytic activity (Madani et al., 2023). Several other notable masked language models, which are evaluated and used as general-purpose protein language models, include CARP (Yang, Fusi, and Lu, 2024), ProteinBERT (Brandes et al., 2023), and TAPE (Rao et al., 2019). Other notable autoregressive language models include ProtTrans (Elnaggar et al., 2022), ProtGPT (Ferruz, Schmidt, and Höcker, 2022), and RITA (Hesslow et al., 2022).

Antibody Protein Sequence

Several protein language models have been developed specifically for antibody sequences. These include AbLang (Olsen, Moal, and Deane, 2022), IgBert and IgT5 (Kenlay et al., 2024), AntiBERTa (Leem et al., 2022), AntiBERTy (Leem et al., 2022), and Sapiens (Prihoda et al., 2022). The ProGen2 suite also contains a version that is fine-tuned on antibody sequences. These models typically train on sequence datasets of the heavy chain variable region (VH) and light chain variable region (VL), which form the antigen-binding part of antibodies and are highly diverse. The largest antibody sequence database is the Observed Antibody Space (OAS), containing ~2.4 billion VH and VL sequences from animal immune cells. Interestingly, specializing models on antibody datasets like OAS often results in degraded performance on tasks like mutational effect prediction compared to general protein language models, suggesting that some antibody-specific models may lose general structural and functional information learned by training on many protein families (Nijkamp et al., 2023; Hie et al., 2024).

Coding RNA Sequence

Each protein amino acid is encoded by an RNA codon with three base pairs, where 64 codons encode the 20 canonical amino acids plus three stop codons. Rather than train on sequences with an amino acid vocabulary, some models train on sequences with a codon vocabulary. Codon models can capture structural and functional information similar to protein language models while also incorporating codon biases that contain species-related information. In turn, these codon biases can have functional effects on biological processes like translation rate or fidelity. Codon language models include CaLM (Outeiral and Deane, 2024) and CodonBERT (Ren et al., 2024), which can be competitive with or even outperform protein language models on function prediction tasks and transcript abundance datasets.

Noncoding RNA Sequence

Sequence models are also trained on large databases of noncoding RNA (ncRNA) sequences, such as sequences of transfer RNAs (tRNAs), ribosomal RNAs (rRNAs), or regulatory ncRNAs. Many ncRNAs rely on functionally relevant secondary or tertiary structure formation, and mutations can substantially affect their functional activity. Inspired by the success of protein language models at learning analogous aspects of structure and function, ncRNA language models are evaluated on similar tasks. Notable ncRNA models include RiNALMo (Penić et al., 2024), RNA-FM (Chen et al., 2022), UniRNA (Wang et al., 2023), and RNAErnie (Wang et al., 2024). These models are smaller than typical protein language models (the largest is RiNALMo at 650 million parameters) and have made some early-stage progress on structure and function prediction tasks.

Genomic DNA Sequence

DNA sequence, the fundamental layer of information in molecular biology, encodes proteins, RNA, and regulatory elements. The vast scale of genomic data presents substantial challenges for sequence modeling, leading to the development of various genomic sequence models with different architectures. These range from models that combine nucleotides into larger tokens, such as GenSLM (Zvyagin et al., 2023) or Nucleotide Transformer (Dalla-Torre et al., 2023), to those maintaining single nucleotide resolution that operate at shorter sequence lengths, such as GPN (Benegas, Batra, and Song, 2023) or regLM (Lal et al., 2024), with recent architectures like Hyena or Mamba offering longer context windows (Nguyen, Poli, Faizi, et al., 2023; Schiff et al., 2024). Genomic sequence models have diverse applications, including fitness prediction of genetic variants using models like GPN-MSA (Benegas et al., 2024) (similar to how protein sequence models are used for fitness prediction). Another major application is generative, with models like regLM or DNA-Diffusion (DaSilva et al., 2024) emphasizing novel sequence design via autoregressive sampling or discrete diffusion. Models like GenSLM and Nucleotide Transformer emphasize transfer learning for tasks such as gene annotation and chromatin accessibility prediction using the embeddings learned by the DNA language model. More capable language models, trained with billions of parameters on hundreds of billions of base pairs with long context, such as Evo (Nguyen, Poli, Durrant, et al., 2024), have the capacity to make progress on both predictive and generative tasks and at multiple levels of biological complexity.

Modeling the Distribution of Other Biological Modalities

While sequence models have dominated much of the recent progress in biological foundation models, there are notable efforts to model distributions of other biological data modalities that also leverage modern architectures, training objectives, and scale. These non-sequence modalities provide complementary insights into biological systems beyond the genetic information contained in DNA, RNA, and protein sequences.

Molecular Structure

Recent advancements in generative models have expanded to encompass atomic-level protein structures, most notably in the realm of de novo protein design, in which the goal is to design proteins with structures that are unconstrained by the natural repertoire of protein structures and folds. Fundamentally, these molecules are represented as a set of atoms, where an atom is defined by its (x, y, z) coordinates. Early generative models of protein structure based on VAEs were proposed to generate the backbone atomic coordinates of proteins (Lin et al., 2021). Diffusion modeling, which led to more powerful generative models of images, has also shown particular promise in protein structure generation. An early protein diffusion model demonstrating the feasibility of this approach was proposed by Anand and Achim (2022); more advanced protein backbone diffusion models, such as RFdiffusion (Watson et al., 2023) and Chroma (Ingraham et al., 2023), apply progressive noise to known protein backbone coordinates during training and learn to reverse this process, generating diverse and realistic protein backbones. Recent models such as Protpardelle (Chu et al., 2024) have extended this approach to generating all of the atoms in a protein structure (both the backbone atoms and side-chain atoms). These generative models are either applied unconditionally (simply generating a complete structure from scratch) or conditionally (fixing a part of the structure and generating a compatible completion of the structure). Conditional generation is particularly useful for generating a protein that binds to another target protein (Bennett et al., 2024), where the user conditions on the target and the binder is generated by, for example, a diffusion model. Beyond proteins, conditional diffusion models have also been used to design small molecules or peptides that bind to a given protein, which has applications in small molecule drug design (Peng et al., 2022; Huang et al., 2024).

Cellular Transcriptomes

Ideas from sequence modeling have influenced models of transcriptomic data collected by single-cell RNA sequencing (scRNA-seq) technologies. In this formulation, each sample represents a single cell described by a vector of integer RNA molecule counts, corresponding to expression levels of different genes. *Single-cell foundation models* adapt concepts from autoregressive and masked language modeling, applying analogous objectives to these gene expression vectors. Several of these single-cell foundation models have been proposed, including Geneformer (Theodoris et al., 2023), scBERT (Yang et al., 2022), scFoundation (Hao et al., 2024), and scGPT (Cui et al., 2024), which all learn patterns in scRNA-seq data by predicting gene expression levels of some genes based on the gene expression levels of other genes. Consequently, these models can generate realistic expression profiles, impute missing data, and provide insights into cell types and states. While these models could theoretically also learn complex genetic interactions and regulatory networks, the extent to which these higher-order concepts are learned directly by the current models during pretraining appears to be more limited (Kedzierska et al., 2023). An important challenge in this domain is the lack of pretraining data available compared to protein or *genomic language models*. While there are 50–100 million publicly available single-cell transcriptomes, these represent around several thousand underlying cell types; and while a transcriptome is theoretically high-dimensional (~20,000 genes in the human transcriptome), many of these dimensions are highly correlated and can be well represented by tens of principal components. Progress in single-cell transcriptomic modeling is most likely data limited compared to other domains.

Multimodal Foundation Models

While we have primarily discussed single-modality foundation models, a growing trend is the development of multimodal foundation models in biology. These models are trained with unsupervised objectives to learn information about multiple biological modalities simultaneously. Some models, like

the genomic sequence model Evo (Nguyen, Poli, Durrant, et al., 2024), use a single fundamental data type (DNA) to capture information across multiple modalities (RNA and protein), enabling multimodal generative tasks across the central dogma. Others, such as ESM3 (Hayes et al., 2024), explicitly combine data from disparate modalities, taking protein sequence, structure, and function as input and training with a masked reconstruction objective across all three.

The motivation for multimodal learning extends beyond capturing different facets of biological processes. Integrating multiple data modalities allows for the incorporation of additional constraints in the learning process, potentially reducing the number of learned parameters and leading to more compact yet equally capable models. This approach holds promise for democratizing access to powerful models, making them more accessible to academic laboratories and individual researchers.

Furthermore, multimodal models offer an opportunity to incorporate biological priors that encode physics and expert knowledge. This is crucial because data alone may never be sufficient to fully capture complex biological processes occurring at various timescales. By integrating these priors, models could learn more robust and biologically relevant representations. For instance, recent work has demonstrated that while language models trained solely on sequence data can capture some structural information, they still struggle at remote homology prediction, highlighting that sequence-only approaches could benefit from additional data modalities (Kabir et al., 2024).

However, significant challenges remain in multimodal biological modeling. Current models often do not fully account for the disparate processes, timescales, granularities, and varying fidelities of different biological data types. Resolving these discrepancies and effectively integrating diverse data sources remains an open area of research.

Despite these challenges, the development of multimodal foundation models represents a promising direction for complex biological modeling. The ability of these models to learn joint data distributions, incorporate diverse constraints, and potentially yield more efficient models aligns well with biology's inherently multimodal and multiscale nature. As the field progresses, we anticipate further advancements in integrating multiple data modalities and biological priors, leading to more comprehensive and robust models of biological systems.

BIOLOGICAL MODELS II: PREDICTIVE MODELS

Predictive models in computational biology serve a crucial role by mapping one biological modality to another (see Figure A-1), often aimed at simulating complex biological processes or experimental outcomes. Unlike foundation models, which are unsupervised and learn general patterns, prediction models are typically supervised and designed for specific tasks. These models aim to transform input data, such as biological sequences, into output predictions of properties, structures, or functions that are often experimentally challenging or resource-intensive to determine. This approach has led to advances in various areas of biology, from protein structure prediction to functional annotation and epigenomic profiling.

There are many supervised or predictive modeling efforts in biology, often using lightweight models either with simple input features or input features derived from a foundation model. In this review, rather than attempting to document all examples of supervised learning in biology, we instead focus on challenging prediction tasks that are widely used as part of biological design workflows. These prediction tasks are largely focused on biomolecular applications given the large amount of research focused on protein design, though we also review prediction of epigenomic features that is used in DNA regulatory design as well. We also focus mostly on prediction tasks involving complex structured outputs that often require specialized neural network architectures and information-rich training datasets. We also provide a summary of these predictive models in Table A-2.

TABLE A-2 Predictive Models

Category	Model name	Input data type	Output data type	Citation(s)
Sequence to structure	AlphaFold	Protein sequence	3D protein structure	Senior et al., 2020
Sequence to structure	AlphaFold2	Protein sequence	3D protein structure	Jumper et al., 2021
Sequence to structure	AlphaFold3	Biomolecular sequence	3D biomolecular structure	Abramson et al., 2024
Sequence to structure	ESMFold	Protein sequence	3D protein structure	Lin et al., 2023
Sequence to structure	OmegaFold	Protein sequence	3D protein structure	Wu et al., 2022
Sequence to structure	RoseTTAFold All-Atom	Biomolecular sequence	3D biomolecular structure	Krishna et al., 2024
Image to structure	CryoDRGN	Cryo-EM images	3D protein volume	Zhong et al., 2021
Image to structure	cryoSPARC	Cryo-EM images	3D protein volume	Punjani et al., 2017
Image to structure	tomoDRGN	Cryo-ET images	3D protein volume	Powell and Davis, 2024
Image to structure	CryoDRGN-ET	Cryo-ET images	3D protein volume	Rangan et al., 2024
Protein fitness	Low-N supervision	Protein sequence	Fitness score	Biswas et al., 2021; Hsu, Nisonoff, et al., 2022
Protein fitness	General language models	Protein sequence	Fitness score	Livesey and Marsh, 2020, 2023; Meier et al., 2021
Viral protein fitness	Constrained Semantic Change Search (CSCS)	Viral protein sequence	Escape score	Hie et al., 2021
Viral protein fitness	EVEscape	Viral protein sequence	Escape score	Thadani et al., 2023
Viral protein fitness	Early-Warning System	Viral protein sequence	Escape score	Beguir et al., 2023
Viral protein fitness	CoVFit	Viral protein sequence	Epidemiological fitness	Ito et al., 2024
Multi-omic prediction	Enformer	DNA sequence	Epigenomic tracks	Avsec et al., 2021
Multi-omic prediction	Borzoi	DNA sequence	Gene expression tracks	Linder et al., 2023

NOTE: This table summarizes various predictive models across different categories, including sequence-to-structure prediction, image-to-structure reconstruction, protein fitness prediction, viral fitness prediction, and multi-omic prediction. For each model, the input and output data types are provided, illustrating the diverse range of prediction tasks using modern machine learning methods.

Notable Prediction Task 1: Molecular Structure Prediction

Sequence to Structure

Protein structure prediction, exemplified by AlphaFold, represents a notable breakthrough in computational biology. This task involves determining a protein's three-dimensional (3D) structure from its amino acid sequence. Notable modeling efforts that advanced this field incorporated physical information, coevolutionary statistical data, and deep learning architectures. Building on these ideas and the preliminary success of an initial AlphaFold model (Senior et al., 2020), AlphaFold2 achieved prediction performance that approaches or matches experimental accuracy for many important proteins, reducing reliance on time-consuming experimental methods like X-ray crystallography (Jumper et al., 2021). The latest iteration, AlphaFold3, extends molecular structure prediction beyond proteins to include other modalities such as small molecules, nucleic acids, and lipids from their sequence descriptions (Abramson et al., 2024). Complementing these advancements, language models like ESM have also

contributed to structure prediction models. Trained on vast protein sequence datasets, these models capture deep evolutionary relationships and form the basis for single-sequence structure prediction tools such as ESMFold (Lin et al., 2023) and OmegaFold (Wu et al., 2022). By leveraging the rich information encoded in protein language models, these tools can predict structures accurately without requiring multiple sequence alignments of evolutionarily related proteins that are critical to AlphaFold's performance. However, language model-based approaches to protein structure prediction have not yet matched the level of accuracy achieved by AlphaFold2 or AlphaFold3. Aside from AlphaFold3, other notable models attempt to predict multimodal biomolecular structures. RoseTTAFold All-Atom (Krishna et al., 2024) models the structures of proteins alongside nucleic acids, metal ions, small molecules, and post-translational modifications, though with lower accuracy than AlphaFold3. Diffusion modeling approaches have been applied to predicting the structural pose of small molecules that bind to proteins (Corso et al., 2023), which is an important problem in small molecule drug design.

Image to Structure

Complementing sequence-based structure prediction, an important area of research focuses on inferring protein volumes from cryo-electron microscopy (cryo-EM) and cryo-electron tomography (cryo-ET) images (Benjin and Ling, 2020). A protein volume is a 3D representation of the electron density distribution within a protein, depicting the spatial arrangement of its mass. Notable work in this field includes CryoDRGN (Zhong et al., 2021) and cryoSPARC (Punjani et al., 2017; Punjani and Fleet, 2021), which use deep learning techniques to reconstruct (potentially heterogeneous) 3D protein volumes from cryo-EM data, and tomoDRGN (Powell and Davis, 2024) and CryoDRGN-ET (Rangan et al., 2024) for 3D reconstruction from cryo-ET data. Although these volumes do not directly provide atomic-level details, they serve as crucial intermediates in structure determination; to derive the actual protein structure from a volume, researchers employ computational methods that fit atomic models into the density map, leveraging prior knowledge of the protein's sequence and secondary structure predictions (Benjin and Ling, 2020). CryoDRGN and similar approaches are particularly valuable for capturing protein dynamics, as they can reveal conformational heterogeneity and structural flexibility. By leveraging advanced machine learning algorithms to generate and interpret these volumes, these methods enhance the determination of structures for challenging protein targets and improve understanding of protein behavior in near-native environments.

Notable Prediction Task 2: Protein Fitness Prediction

Many supervised machine learning models predict protein fitness as a scalar value, trained on existing datasets that map protein sequences to fitness values. These models input protein sequences, represented either through simple *one-hot encoding* (where each amino acid is assigned a unique binary vector) or more advanced *neural sequence embeddings* from foundation models. The choice of input representation can substantially impact model performance. Most supervised models use straightforward approaches such as linear regression, Gaussian process regression for predictions with uncertainty estimates, random forest regressors, or small MLPs or CNNs. The choice of features and the regression model often depends on factors like dataset size, fitness landscape complexity, and requirements for interpretability or uncertainty quantification (Biswas et al., 2021; Hsu, Nisonoff, et al., 2022).

Unsupervised models of protein sequences, trained on large datasets of naturally occurring proteins, have demonstrated remarkable success in zero-shot prediction of mutational effects on protein fitness (Livesey and Marsh, 2020, 2023). These models leverage sequence likelihoods to predict the impact of mutations, with lower likelihoods generally indicating more deleterious effects. This computational approach can be thought of as an *in silico* version of deep mutational scanning (DMS), an experimental technique that measures the functional effects of thousands of small mutations in a protein (e.g., all possible single-amino-acid substitutions) (Fowler and Fields, 2014). DMS data are used to evaluate these unsupervised models and, in some cases, to train supervised models for comparison.

Interestingly, unsupervised models often outperform supervised models at mutational effect prediction tasks, particularly in zero-shot scenarios. For example, the likelihoods from language models like ESM-1v and ESM-2 are competitive mutational effect predictors across many DMS datasets (Meier et al., 2021) and across datasets of clinically relevant human disease variants (Brandes et al., 2023), though there is still room for improvement in variant effect prediction (Bromberg et al., 2024). This advantage likely stems from the unsupervised models capturing broad evolutionary information across protein families, providing a strong prior for fitness effects (Hie et al., 2024). In contrast, supervised models may overfit to assay-specific nuances, limiting their generalizability. To assess the effectiveness of these approaches, benchmark suites such as ProteinGym (Notin et al., 2024) and FLIP (Fitness Landscape Inference for Proteins; Dallago et al., 2021) compile diverse DMS datasets, allowing for comprehensive evaluation of predictive models across various proteins and experimental conditions.

Predicting Viral Infectivity or Escape

Building on the broader fitness-prediction applications of protein sequence models, these approaches have found critical relevance in virology. Unsupervised protein language models have shown promise in predicting viral infectivity and escape, utilizing high-throughput DMS data that measure how amino acid changes affect these properties (Haddox, Dingens, and Bloom, 2016; Haddox et al., 2018; Lee et al., 2018; Starr et al., 2020; Wu et al., 2020; Starr et al., 2022). Viral escape refers to mutations allowing viruses to evade immune responses, while infection is mediated by specific proteins interacting with host cell receptors. Initial work demonstrated that statistical properties learned by protein sequence models correspond to viral escape characteristics (Riesselman, Ingraham, and Marks, 2018; Hie et al., 2021; Thadani et al., 2023). Subsequent work showed that larger protein language models like ESM, either zero-shot or when fine-tuned on viral proteins, can competitively model various aspects of viral evolution (Hie, Yang, and Kim, 2022; Beguir et al., 2023; Ito et al., 2024; Lamb et al., 2024). These diverse methods offer promising tools for studying mutational effects on near-term infectivity or escape, but their ability to predict long-term viral evolution remains limited. Viral evolution across multiple generations and long evolutionary trajectories inherently has substantial levels of uncertainty. Better models are needed to forecast viral evolution accurately over the long term, highlighting the ongoing challenges in translating these insights into reliable vaccine design and pandemic preparedness strategies.

Notable Prediction Task 3: Multi-omic Sequence Prediction

Another notable direction in biological prediction tasks relevant to design workflows is the development of tools that infer complex genomic and epigenomic features directly from DNA sequences. Enformer (Avsec et al., 2021) predicts multiple epigenomic tracks from raw DNA sequence, capturing regulatory interactions across long genomic distances. These tracks typically include data on chromatin accessibility, histone modifications, and transcription factor binding, often measured experimentally through techniques like ATAC-seq (Assay for Transposase-Accessible Chromatin using sequencing). Extending these models, Borzoi (Linder et al., 2023) focuses on predicting gene expression tracks, which reflect the activity levels of genes across different cell types or conditions, usually measured by RNA-seq. Both models use CNN deep learning architectures to learn the intricate relationships between DNA sequence and functional genomic readouts. These approaches are valuable as they provide insights into gene regulation and expression without requiring extensive experimental data when given new DNA sequences, and are therefore used as part of DNA regulatory element design pipelines (which we review below) to evaluate if a given design produces the desired impact on the downstream epigenomic sequence. However, like other predictive models in computational biology, the predictions of these models, and subsequent designs, should be interpreted cautiously and validated experimentally when possible.

BIOLOGICAL MODELS III: DESIGN MODELS

We have so far considered foundation models that learn unsupervised information (e.g., learning the distribution of sequences) and predictive models that map sequences to functions. We now consider biological design, which benefits from combining both generative and predictive models (see Figure A-1). This section examines the landscape of machine learning–guided biological design tools; their underlying principles; and their potential to transform key processes in drug discovery, enzyme engineering, and synthetic biology. We begin with a general discussion on how advancements in both generative models and predictive models can be combined to improve biological design pipelines. As protein design has been a primary focus of biological design efforts, we then review two major categories in this field: function-guided adaptive protein design, which relies on iterative rounds of experimental testing and evaluation, and de novo protein design, which emphasizes the creation of structurally novel proteins. We conclude by exploring machine learning–guided biological design beyond protein molecules.

Generative Models, Predictive Models, and Biological Design

In the earlier section on foundation models, we described how generative models enable a user to efficiently sample from a complex data distribution. While generative models play a crucial role in biological design tasks, they are often insufficient on their own. These models learn to capture the distribution of naturally occurring biological sequences or structures, providing a foundation for generating plausible new designs. However, generating designs with specific desired functions requires additional steps. Typically, generative models are coupled with predictive models in a rejection sampling framework (Hastie et al., 2009); in this approach, the generative model proposes candidate sequences, which are then evaluated by a predictive model. Only sequences predicted to have the desired function are selected for experimental testing, while the remainder are “rejected.” This combination allows for efficient exploration of the vast sequence space while focusing on promising candidates. This strategy not only increases the likelihood of discovering sequences with desired properties but also significantly reduces the experimental burden by prioritizing the most promising candidates for laboratory validation (Hie and Yang, 2022).

Beyond rejection sampling, a predictive scoring function could be used to directly bias the generative model’s samples using algorithms like supervised fine-tuning (SFT), reinforcement learning techniques, or direct preference optimization (DPO) (Rafailov et al., 2024). SFT involves further training the generative model on a curated dataset of high-scoring sequences to shift its distribution. Reinforcement learning techniques, such as proximal policy optimization (Schulman et al., 2017; Ouyang et al., 2022), iteratively update the model to maximize a reward signal derived from the scoring function. DPO aligns the model’s outputs with desired preferences without explicit reward modeling, often using pairwise comparisons between positively and negatively labeled samples (Rafailov et al., 2024). These approaches aim to align the generative model’s output more closely with the desired functional properties, potentially increasing the efficiency of the design process compared to simple rejection sampling.

Notable Design Task 1: Function-Guided Adaptive Protein Design

Function-guided biological design specifies a desirable biological function and aims to produce a sequence that exhibits that desirable function. In recent years, adaptive machine learning approaches have emerged as powerful tools for function-guided biological design, particularly in the realm of protein engineering to navigate the vast and complex landscape of possible sequences. These methods combine computational predictions with iterative experimental validation to search efficiently for proteins with desired properties (Hie and Yang, 2022).

At the heart of many of these approaches is Bayesian optimization, a strategy for finding the global optimum of an unknown function. In the context of protein engineering, this function might map

protein sequences to a desired property like enzyme activity or stability. Bayesian optimization typically uses a probabilistic surrogate model, often a Gaussian process, to approximate this unknown function (Rasmussen and Williams, 2005). Gaussian processes are flexible, nonparametric models that can capture complex relationships and provide uncertainty estimates for their predictions.

The optimization process iterates between using the surrogate model to select promising candidates and experimentally testing these candidates to refine the model. This selection is guided by an acquisition function, which balances the exploitation of designs predicted to be good with the exploration of uncertain regions of the sequence space. A common acquisition function is the upper confidence bound, which combines the predicted value with the uncertainty of that prediction (Rasmussen and Williams, 2005). Several studies have used Gaussian process predictions and uncertainty estimates to guide iterative rounds of experiments (Romero, Krause, and Arnold, 2013; Hie, Bryson, and Berger, 2020; Greenhalgh et al., 2021).

Generative models, such as VAEs or protein language models, are often used to propose new candidate sequences. These models can learn the distribution of valid protein sequences and generate novel sequence designs that can then be tested for functional activity. For example, Madani and colleagues (2023) use an autoregressive protein language model to generate lysozyme proteins with active catalytic activity and as low as 40 percent sequence identity to natural lysozymes. Similarly, Hayes and colleagues (2024) use a masked language model combined with rounds of experimental data collection to design a green fluorescent protein (GFP) homolog with less than 60 percent sequence identity to natural GFPs.

Adaptive sampling techniques can further refine these generative models based on the results of the surrogate model, steering the generation process toward more promising regions of the sequence space. A key advantage of these methods is their ability to learn and improve over multiple rounds of experimentation. This iterative process, sometimes called active learning, allows the models to become increasingly accurate and efficient at identifying promising designs (Hie and Yang, 2022).

Because active learning often requires multiple rounds of experimental data collection, a recent trend has been to augment these processes with automated experimental platforms. For example, Rapp, Bremer, and Romero (2024) use multiple rounds of Bayesian optimization with a Gaussian process combined with an experimental pipeline with a heavy use of automation to design a suite of more active enzymes. These approaches could accelerate the design-build-test cycle in biological engineering, but laboratory automation pipelines are still difficult to set up and are less reliable for more complex biological assays. While relatively robust binding measurements or enzymatic activity assays are amenable to laboratory automation in the immediate term, improvements in laboratory automation technologies could improve the generalizability of this approach.

Other challenges to function-guided adaptive learning strategies remain. Protein sequences are discrete and high-dimensional, which can pose difficulties for traditional optimization techniques. Additionally, biological experiments often involve large batch sizes and limited rounds of testing, which require careful consideration in the design of optimization strategies. Despite these challenges, adaptive machine learning approaches are becoming a more regular part of a protein engineering pipeline.

Notable Design Task 2: Structure-Guided De Novo Protein Design

While adaptive machine learning techniques have been developed mainly to enhance existing proteins through iterative rounds of experimentation, structure-guided “de novo design” aims to create entirely new proteins using computational methods with structures that are substantially different from the structures of natural proteins (Chu, Lu, and Huang, 2024). The sophistication of de novo design has increased by replacing early approaches based on physical principles with recent advances in machine learning.

The modern protein design workflow often uses a two-step process. First, structure generation models like RFdiffusion (Watson et al., 2023) create protein backbones, allowing designers to specify constraints such as binding interfaces or symmetry. Then, inverse folding methods like ProteinMPNN

(Dauparas et al., 2022) or ESM-IF1 (Hsu, Verkuil, et al., 2022) determine amino acid sequences likely to fold into the generated structures. Designs are typically evaluated using a “self-consistency” metric, which assesses whether a designed sequence is likely to fold into the intended structure under a structure prediction model like AlphaFold or ESMFold.

De novo protein design to engineer proteins with useful functions, in addition to novel structures, typically still uses parts of natural proteins that are involved in a specific function (e.g., an enzyme active site or a binding interface) but replaces other components of the protein with a nonnatural protein scaffold. These developments have enabled the creation of de novo proteins with complex functions, including enzymes, protein binders, and designed proteins that can form large assemblies or span cell membranes (Courbet et al., 2022; Wang et al., 2022; Watson et al., 2023).

While progress has been substantial, challenges remain, particularly in areas where fine control over chemical interactions and conformational dynamics is crucial. The field is moving toward more integrated approaches that consider multiple states, dynamics, and the co-design of sequence and structure (Chu, Lu, and Huang, 2024). Future advancements likely will come from a combination of data-driven approaches and first-principles reasoning (e.g., directly incorporating biophysical principles or experimental data into protein design tools), potentially leading to greater versatility of de novo designed proteins.

Other Design Tasks

Machine learning-guided biological design has extended beyond proteins to other biological modalities, albeit to a more limited extent. One notable example is the work by Nguyen and colleagues, who developed a genomic language model, Evo, that learns sequence information underlying DNA, RNA, and proteins from raw genome sequences (Nguyen, Poli, Durrant, et al., 2024). This model enables the design of multimodal biological systems, including CRISPR-Cas systems (involving protein-RNA co-design) and transposon systems (focusing on protein-DNA co-design). Another notable area is machine learning-guided promoter design, which is often used in synthetic biology applications to control levels of gene expression in *E. coli* or to control cell-type-specific gene expression in mouse or human cells (LaFleur, Hossain, and Salis, 2022; Zhang et al., 2023; DaSilva et al., 2024; Reddy et al., 2024). The approach here parallels that of protein design: either a generative model produces promoter sequences, or a large collection of promoters is utilized as a starting point. These candidate promoters are then evaluated using a promoter activity prediction model. The designs are subsequently tested in the laboratory, with the results feeding back into the system to refine future predictions and designs. This iterative process allows for continuous improvement in the design of promoters with desired activities. As generative and predictive machine learning models improve, we can expect to see further integration of machine learning across various aspects of biological design, potentially leading to more efficient approaches to engineering biological systems.

BIOLOGICAL DATASETS

Biological datasets are crucial for training foundation models, as they provide the raw material from which these models learn complex patterns. When training foundation models by leveraging the scaling hypothesis, the scale, quality, and complexity of data are as important as the model size and the amount of compute used to train the model (Kaplan et al., 2020; Hoffmann et al., 2022). This review focuses on large-scale public datasets representing a diverse range of biological data types, including genetic sequence data (spanning proteins, RNA, DNA, and whole genomes), molecular structures, molecular properties, epigenomic information, and transcriptomic data, which we also describe in Table A-3. We also summarize ways to link information across multiple modalities when training multimodal language models.

TABLE A-3 Biological Databases

Category	Database name	Type of data stored	Citation or URL
Genome sequence	GenBank	Genomic sequences and annotations	Sayers et al., 2019
Genome sequence	European Nucleotide Archive (ENA)	Genomic sequences and annotations	https://www.ebi.ac.uk/ena/
Genome sequence	RefSeq	Genomic sequences and annotations	O’Leary et al., 2016
Genome sequence	Ensembl	Genomic sequences and annotations	Martin et al., 2023
Genome sequence	Ensembl Genomes	Genomic sequences and annotations	Yates et al., 2022
Genome sequence	Genome Taxonomy Database (GTDB)	Prokaryotic genome sequences	Parks et al., 2022
Genome sequence	Unified Human Gastrointestinal Genome (UHGG)	Metagenomic sequences	Almeida et al., 2021
Genome sequence	JGI IMG	Microbial genome and metagenome sequences	Markowitz et al., 2006
Genome sequence	Human Gastrointestinal Bacteria Genome Collection	Gut microbiome metagenomic sequences	Forster et al., 2019
Genome sequence	MGnify	Metagenomic sequences	Richardson et al., 2023
Genome sequence	Youngblut et al.	Animal gut metagenomic sequences	Youngblut et al., 2020
Genome sequence	Tara Oceans Project	Marine metagenomic sequences	Pesant et al., 2015
Protein sequence	UniProt	Protein sequences and annotations	UniProt Consortium, 2019
Protein sequence	UniRef50/UniRef90	Clustered protein sequences	Suzek et al., 2007
Protein sequence	Pfam	Protein domain sequences and annotations	El-Gebali et al., 2019
Protein sequence	Observed Antibody Space (OAS)	Antibody variable region sequences	Olsen, Boyles, and Deane, 2022
Viral sequence	GISAID	Influenza and SARS-CoV-2 sequences	Shu and McCauley, 2017
Viral sequence	BV-BRC	Bacterial and viral sequences	Olson et al., 2023
Viral sequence	Influenza Research Database	Influenza virus sequences	Zhang et al., 2017
Viral sequence	LANL HIV database	HIV sequences	Foley et al., 2018
Noncoding RNA sequence	Rfam	RNA family sequences	Kalvari et al., 2021
Noncoding RNA sequence	RNAcentral	Noncoding RNA sequences	Sweeney et al., 2019
Biomolecular structure	Protein Data Bank (PDB)	3D structures of biological macromolecules	Berman et al., 2000
Biomolecular structure	CATH	Protein domain structures and classifications	Knudsen and Wiuf, 2010
Biomolecular structure	CASP	Protein structure prediction targets	https://predictioncenter.org/
Biomolecular structure	AlphaFoldDB	Predicted protein structures	Varadi et al., 2022
Biomolecular structure	ESM Metagenomic Atlas	Predicted metagenomic protein structures	Lin et al., 2023

continued

TABLE A-3 *continued*

Category	Database name	Type of data stored	Citation or URL
Protein function	ProteinGym	Protein mutational effect datasets	Notin et al., 2024
Protein function	FLIP	Protein fitness landscape datasets	Dallago et al., 2021
Protein function	ClinVar	Genetic disease variants and phenotypes	Landrum et al., 2018
Protein function	SKEMPI	Protein-protein interaction affinity changes	Jankauskaitė et al., 2019
Protein function	PDBbind	Biomolecular binding affinity data	Wang et al., 2005
Protein function	BindingDB	Protein-small molecule interaction data	Liu et al., 2007
Protein function	BRENDA	Enzyme and enzyme-ligand data	Chang et al., 2021
Protein function	Binding MOAD	Protein-ligand binding data	Wagle et al., 2023
Protein function	2P2Idb	Protein-protein interactions targetable by small molecules	Basse et al., 2016
Protein function	ProThermDB	Protein stability data	Nikam et al., 2021
Protein function	STITCH	Protein-chemical interactions	Szklarczyk et al., 2016
Protein function	DrugBank	Drug-target interactions	Knox et al., 2011
Protein function	Gene Ontology (GO)	Gene and protein function annotations	Aleksander et al., 2023
Epigenomic datasets	ENCODE	Epigenomic data for human and mouse	ENCODE Project Consortium, 2012
Epigenomic datasets	Roadmap Epigenomics Project	Human epigenomic data	Kundaje et al., 2015
Epigenomic datasets	PsychENCODE	Brain epigenome data	Akbarian et al., 2015
Epigenomic datasets	4D Nucleome Project	3D genome organization data	Dekker et al., 2017
Epigenomic datasets	FANTOM	Transcription start sites and enhancers	Andersson et al., 2014
Transcriptomic datasets	Human Cell Atlas (HCA)	Human single-cell transcriptomic data	Regev et al., 2017
Transcriptomic datasets	Tabula Sapiens	Human single-cell transcriptomic data	Jones et al., 2022
Transcriptomic datasets	Tabula Muris	Mouse single-cell transcriptomic data	The Tabula Muris Consortium, 2018
Transcriptomic datasets	Fly Cell Atlas	Fruit fly single-cell transcriptomic data	Li et al., 2022
Transcriptomic datasets	CELLxGENE	Aggregated single-cell data	CZI Single-Cell Biology Program et al., 2023
Transcriptomic datasets	Broad Single Cell Portal	Aggregated single-cell data	https://singlecell.broadinstitute.org

NOTE: This table provides an overview of major biological databases related to biological model training, organized by category of biological data. It includes databases for genome sequences, protein sequences, coding and noncoding RNA sequences, biomolecular structures, protein functions, epigenomic data, and transcriptomic data. For each database, we provide a brief description of the type of data stored. Datasets of sufficient diversity and quality to train foundation models are highlighted in light blue.

TABLE A-4 Example Training Datasets and Associated Models

Model	Model type	Training datasets
ESM-2	Protein sequence model	UniRef90
Evo	Genomic sequence model	GTDB, IMG/PR, IMG/VR
AlphaFold2	Structure prediction model	PDB
ProGen2	Protein sequence model	UniRef90, metagenomic sequences, Observed Antibody Space (OAS)
AbLang	Antibody protein sequence model	Observed Antibody Space (OAS)
CaLM	Codon sequence model	Coding sequences from European Nucleotide Archive (ENA)
RiNALMo	Noncoding RNA sequence model	Rfam, RNACentral, NCBI
GenSLM	Genomic sequence model	GenBank
Enformer	Multi-omic prediction model	ENCODE datasets
scGPT	Transcriptomic count model	CELLxGENE

NOTE: Example models and their corresponding training datasets. These datasets are typical for each of the types of models in the table. Some models draw from multiple publicly available data sources for training.

Genetic Sequence Datasets

Genome Sequence

Genomic sequencing data form the foundation for databases of proteins, RNAs, and regulatory DNA sequences. Most genome sequences are obtained through shotgun sequencing, where short DNA fragments are sequenced, quality-filtered, and assembled into longer contiguous sequences called contigs. These contigs can be derived from a single species' genomic DNA (or RNA for some viruses) or from mixed species samples (metagenomic data) (Giani et al., 2020). Bioinformatic tools then identify potential genes, protein-coding sequences, and other functional elements, though annotation quality can vary by species (Yandell and Ence, 2012). The primary repositories for these genomic sequences and annotations are GenBank from the National Center for Biotechnology Information (NCBI) (Sayers et al., 2019) in the United States and the European Nucleotide Archive (ENA) from EMBL-EBI.² Specialized databases often curate subsets of NCBI and ENA data and offer additional metadata. For instance, RefSeq is a high-quality subset of NCBI genome data with annotations (O'Leary et al., 2016); the Ensembl and EnsemblGenomes databases likewise curate ENA genomes along with annotations (Yates et al., 2022; Martin et al., 2023), while the Genome Taxonomy Database (GTDB) focuses on high-quality prokaryotic genomes (Parks et al., 2022). Specialized protein and RNA sequence databases also heavily leverage the annotated sequences in these larger genomic sequence databases. Metagenomics databases, which are often also deposited into NCBI or ENA, include the following: Unified Human Gastrointestinal Genome (UHGG) (Almeida et al., 2021), Joint Genome Institute Integrated Microbial Genomes (JGI IMG) (Markowitz et al., 2006), Human Gastrointestinal Bacteria Genome Collection (Forster et al., 2019), MGnify (Richardson et al., 2023), Youngblut and colleagues' (2020) animal gut metagenomes, and the Tara Oceans Project (Pesant et al., 2015). Genomic sequence models often train on a much smaller subset of the terabytes of sequences found in these databases. Many genomic sequence models have only been applied to the human reference genome. Evo, a genomic language model trained on prokaryotic genomes, was mostly trained on GTDB.

² See <https://www.ebi.ac.uk/ena/> (accessed November 17, 2024).

Protein Sequence

Protein sequence datasets are primarily sourced from UniProt (Universal Protein Resource), which is a high-quality database that includes many manually annotated and experimentally validated sequences (UniProt Consortium, 2019). UniRef50 and UniRef90 provide clustered versions of UniProt at 50 percent and 90 percent sequence identity thresholds, respectively, which removes redundant proteins to ensure that protein language models are trained on diverse sequences (Suzek et al., 2007). Another highly curated protein database is Pfam, which uses hidden Markov models to identify and classify recurring structural and functional units within proteins; all protein sequences in Pfam have an associated Pfam domain categorization (El-Gebali et al., 2019). Metagenomics databases such as MGnify (Richardson et al., 2023) and databases maintained by the Joint Genome Institute (JGI) (Markowitz et al., 2006) identify potential protein sequences from prokaryotic genomes by locating open reading frames, which are DNA sequences between start and stop codons that potentially encode proteins. While potentially lower quality than UniProt, metagenomic databases help to expand the known protein sequence space greatly. Antibody language models primarily make use of the Observed Antibody Space (OAS) dataset (Olsen, Boyles, and Deane, 2022), containing ~2 billion antibody variable domain sequences. Viral protein language models use protein sequences identified during viral surveillance (primarily directed at the highly mutable pandemic viruses such as influenza A, HIV, and severe acute respiratory syndrome coronavirus 2 [SARS-CoV-2]) and deposited into databases such as NCBI RefSeq (O’Leary et al., 2016), GISAID (Global Initiative on Sharing All Influenza Data) (Shu and McCauley, 2017), the Bacterial and Viral Bioinformatics Resource Center (BV-BRC) database (Olson et al., 2023), Influenza Research Database (Zhang et al., 2017), and the Los Alamos National Laboratory (LANL) HIV database (Foley et al., 2018).

Coding RNA Sequence

Codon language models are trained on coding sequences extracted from natural genomes, utilizing the coding sequence annotations found in NCBI or ENA databases. In prokaryotic genomes, these coding sequences are contiguous. However, in eukaryotic genomes, coding sequences are noncontiguous due to the presence of introns. Eukaryotic gene expression involves splicing, a process in which introns are removed and exons are joined to form the mature mRNA. Thus, eukaryotic coding sequence datasets require annotations specifying which genomic regions are retained in the final transcript. Notable codon language models include CaLM, which is trained on coding sequences from ENA, and CodonBERT, which utilizes coding sequences from NCBI (Outeiral and Deane, 2024; Ren et al., 2024).

Noncoding RNA Sequence

Several language models are specifically trained on ncRNA sequences. As with protein coding sequences, many ncRNA sequences are derived from annotated ncRNA data in GenBank and Ensembl, and eukaryotic ncRNA sequences must account for splicing. Additional ncRNA-specific databases include Rfam (Kalvari et al., 2021), which organizes ncRNA sequences into RNA families (analogous to Pfam for proteins), and RNAcentral (Sweeney et al., 2019), a comprehensive collection of ncRNAs that incorporates sequences from various sources, including GenBank and Ensembl.

Biomolecular Structure Datasets

Decades of experimental work in structural biology, resulting in large curated datasets of molecular structures, have been instrumental in driving the progress in biomolecular structure prediction represented by AlphaFold. The Protein Data Bank (PDB) serves as the primary repository for experimentally determined 3D structures of biological macromolecules, including proteins and nucleic

acids (Berman et al., 2000). The structures in the PDB are primarily determined through experimental methods such as X-ray crystallography, nuclear magnetic resonance (NMR) spectroscopy, and cryo-EM. X-ray crystallography provides high-resolution structures by analyzing X-ray diffraction patterns from protein crystals. NMR spectroscopy determines structures in solution, offering insights into protein dynamics. Cryo-EM, which has seen significant recent advancements, images flash-frozen proteins using electron microscopy, allowing for the determination of large complex structures and conformational states.

Many additional datasets are built off of data that have been deposited in the PDB. CATH (Class, Architecture, Topology, Homology) offers a hierarchical classification of protein domains based on their 3D structures, organizing PDB entries into structurally related groups (Knudsen and Wiuf, 2010). CASP (Critical Assessment of protein Structure Prediction)³ is a biannual competition that assesses the state of the art in protein structure prediction method; as part of CASP, new protein structures are experimentally determined and used as a test set for structure prediction models. Both CATH and CASP are particularly valuable for developing machine learning models for structural biology, as they provide splits of protein structures based on structural similarity and, in the case of CASP data, temporally held-out data with respect to different versions of the PDB. Ensuring proper held-out structures during model development helps mitigate data leakage, a common issue in machine learning where information from the test set inadvertently influences the training process, leading to overly optimistic performance estimates. By using structural similarity-based splits, researchers can more accurately assess generalization capabilities to new protein folds.

A more recent development in structural biology is the emergence of large databases of AI-generated protein structure predictions. AlphaFoldDB contains AlphaFold2-predicted structures for nearly all proteins in UniProt, representing ~214 million structure predictions (Varadi et al., 2022). Another database, the ESM Metagenomic Atlas, provides predicted structures for more than 617 million metagenomic proteins using the ESMFold structure prediction method (Lin et al., 2023). This atlas extends structural coverage to diverse microbial proteins from various environments. While these databases can be useful for providing structural hypotheses for many natural proteins and potentially for augmenting machine learning models that require large amounts of structural training data, they should also be treated with increased caution. The best-performing models that incorporate protein structure during training still mainly utilize experimental structure data, with some mixture of predicted structure data.

Protein Function Datasets

Large datasets that map many protein sequences to an associated fitness value (where fitness is defined as the strength of some function-of-interest and which is typically tied to some kind of experimental assay) are useful for both model evaluation and development. Here, we use definitions common in the directed evolution literature in which a “function” is defined as a particular biological property or process (e.g., stability, catalysis, binding) and “fitness” is defined as how well or strongly the protein achieves that function (e.g., scoring a variant by melting temperature, Michaelis-Menten kinetics, or binding affinity). As described in the section above on biological foundation models, many biological sequence models are evaluated based on their ability to predict mutational effects on fitness, which is done by correlating the fitness score of a given mutation with the language model likelihood assigned to that mutation. Protein fitness datasets are also used to develop models that predict functional activity from the protein sequence; these models can then be used as part of an adaptive machine learning loop for machine learning-guided directed evolution and iterative protein design. More recently, these datasets are also being used to align protein generative models so that sampling from these models results in more desirable designs.

³ See <https://predictioncenter.org/> (accessed November 17, 2024).

High-throughput protein measurement techniques often use cellular expression systems with sorting or selection methods (Fowler et al., 2010; Jacquier et al., 2013). Cell surface display technologies, such as yeast display, express proteins on cell surfaces for easy measurement, often using flow cytometry to quantify expression and binding. Likewise, intracellular expression coupled with fluorescent reporters enables fluorescence-activated cell sorting to isolate desired variants. Some techniques use cell viability or fitness as a readout, linking protein function to cell survival or growth rate for selection through competition or survival experiments. Other high-throughput methods for collecting many protein fitness measurements are based on microfluidics; for example, the HT-MEK platform uses droplet-based microfluidics to perform thousands of individual enzyme assays in parallel at nanoliter scale (Markin et al., 2021). When a study measures an exhaustive set of mutations to a protein—for example, all single-residue substitutions—the study is referred to as a deep mutational scan (DMS) (Fowler and Fields, 2014).

Phage and yeast display apply these principles to study antibody–antigen interactions, expressing antibody variants on bacteriophage or yeast surfaces, followed by incubation with a solution containing the antigen (Hunter and Cochran, 2016). These systems enable rapid screening of large collections of antibody sequences, isolating high-affinity binders through multiple selection rounds. High-throughput methods based on cell-based assays followed by nucleotide sequencing are also used to identify viral protein variants with increased infectivity or antibody escape. This involves creating mutated viral or pseudoviral particles (often modified so that they cannot replicate), optionally exposing them to antibody selection pressure, infecting cells, and sequencing the more successful viral variants. This approach allows for high-throughput assessment of viral mutations that impact infectivity or immune evasion (Haddox, Dingens, and Bloom, 2016; Haddox et al., 2018; Lee et al., 2018; Starr et al., 2020; Wu et al., 2020; Starr et al., 2022).

Other large-scale protein fitness datasets increase the scope of data collected via laboratory automation, large-scale manual effort, or a combination of both. For example, Cao and colleagues (2022) were able to collect neutralization data representing 2,096 mutations to the receptor binding domain of the SARS-CoV-2 spike protein in the presence of 1,548 antibodies, representing a total of 173,384 neutralization measurements. While impressive in scope, fitness data collection at this scale still requires substantial human resources beyond what can be accomplished by laboratory automation alone.

Unlike protein structure data, there is no single comprehensive database that systematically curates large-scale protein fitness data. Often, these data are simply included as supplementary information in research publications. Some initiatives have aimed to consolidate DMS datasets into standardized formats with consistent preprocessing, primarily to serve as benchmark evaluation sets for comparing protein language models. Notable examples include ProteinGym (Notin et al., 2024), a collection curated by Livesey and Marsh (2020), and the FLIP dataset collection (Dallago et al., 2021). In addition to DMS data, ProteinGym incorporates human genetic variants commonly associated with disease from the ClinVar database (Landrum et al., 2018), further enhancing its utility for model evaluation and development. However, additional effort is most likely needed to curate protein fitness datasets more consistently across many high-throughput studies.

Beyond DMS datasets, there are some databases specific to certain kinds of protein function. In particular, several databases have been developed to curate and provide access to protein binding data, serving as valuable resources for understanding protein–protein and protein–ligand interactions. SKEMPI (Structural Kinetic and Energetic database of Mutant Protein Interactions) focuses on experimentally determined binding affinity changes upon mutation for protein–protein interactions (Jankauskaitė et al., 2019). PDBbind offers a comprehensive collection of experimentally measured binding affinity data for biomolecular complexes found in the PDB (Wang et al., 2005). BindingDB provides data on protein interactions with small drug-like molecules (Liu et al., 2007), while BRENDA, primarily an enzyme database, includes enzyme–ligand binding information (Chang et al., 2021). Other notable resources include Binding MOAD (Mother of All Databases) (Wagle et al., 2023), 2P2Idb for protein–protein interactions targetable by small molecules (Basse et al., 2016), and ProThermDB for protein stability data (Nikam et al., 2021). Additionally, STITCH integrates various sources of protein–chemical interactions

(Szkarczyk et al., 2016), while DrugBank, primarily a drug database, contains extensive information on drug-target interactions (Knox et al., 2011).

Finally, there are some databases that map protein sequences to standardized functional descriptions in natural language. Gene Ontology (GO) provides a structured vocabulary categorizing gene and protein functions into molecular function, biological process, and cellular component, with a hierarchical structure allowing varied detail (Aleksander et al., 2023). UniProt contains comprehensive protein information, including function descriptions, catalytic activities, cofactors, regulation, structure, and pathway involvement, while also incorporating GO annotations. These systems enable translation of sequence data into functional descriptions, or vice versa, and can be used to connect protein generative models with LLMs.

Epigenomic Datasets

Many predictive models have attempted to predict epigenomic tracks from DNA sequence, leveraging the wealth of data generated by various epigenomic sequencing technologies. These technologies provide insights into chromatin structure, DNA modifications, and 3D genome organization, offering a comprehensive view of the functional state of the genome beyond its primary sequence.

ATAC-seq identifies open chromatin regions by inserting sequencing adapters into accessible areas of the genome (Buenrostro et al., 2013). This method has become increasingly popular due to its simplicity and effectiveness in mapping regulatory regions. DNA methylation, another crucial epigenetic modification, can be measured through various techniques such as bisulfite sequencing, which provides single-base resolution methylation maps, or MeDIP-seq, which uses immunoprecipitation to capture methylated DNA fragments (Greenberg and Bourc'his, 2019). Hi-C (High-throughput chromosome conformation capture) offers insights into the 3D organization of the genome by capturing long-range chromatin interactions (Lieberman-Aiden et al., 2009).

The ENCODE (Encyclopedia of DNA Elements) project is a large source of epigenomic data for human and mouse genomes (ENCODE Project Consortium, 2012). This resource includes data from various assays, such as chromatin accessibility (ATAC-seq, DNase-seq), histone modifications (ChIP-seq for various histone marks), transcription factor binding sites, DNA methylation, RNA-seq for gene expression, and 3D chromatin interactions. The standardized high-quality datasets provided by ENCODE across multiple cell types and tissues serve as the primary current resource for developing and benchmarking predictive models of epigenomic features.

Beyond ENCODE, other major epigenomic datasets and projects complement its efforts. The Roadmap Epigenomics Project has mapped epigenomic landscapes across a wide range of human cell types and tissues (Kundaje et al., 2015). More specialized projects include PsychENCODE (Akbarian et al., 2015), which maps the brain epigenome with emphasis on neuropsychiatric disorders, and the 4D Nucleome Project (Dekker et al., 2017), which explores the 3D organization of the nucleus. The FANTOM project provides extensive data on transcription start sites and enhancers (Andersson et al., 2014). While these diverse projects collectively provide a rich landscape of epigenomic data across various biological contexts, cell types, and organisms, many of these datasets have yet to be fully utilized in machine learning modeling efforts.

Single-Cell Transcriptomic Datasets

scRNA-seq aims to obtain a more accurate representation of cellular heterogeneity by enabling gene expression profiling at individual cell resolution. scRNA-seq is often compared to bulk RNA sequencing, which measures averaged expression levels over a sample containing multiple cell populations, potentially masking important cell-type-specific variation. scRNA-seq data are typically collected by isolating individual cells using microfluidic devices, followed by sequencing of cellular mRNA. Large-scale “cell atlas” efforts, such as the Human Cell Atlas (HCA) (Regev et al., 2017) and Tabula Sapiens (Jones et al., 2022), aim to create comprehensive reference maps of all human cell types

across multiple tissues and organs. Other large-scale scRNA-seq atlases have also been created for other model organisms include the Tabula Muris (The Tabula Muris Consortium, 2018) in mice and the Fly Cell Atlas (Li et al., 2022) in fruit flies. To facilitate access to these diverse datasets, centralized resources like the CELLxGENE database (CZI Single-Cell Biology Program et al., 2023) and the Broad Single Cell Portal⁴ have been developed. These platforms allow researchers to explore, analyze, and download single-cell data from multiple studies, providing interactive visualization tools and basic analysis capabilities. These atlas-level datasets make up the bulk of the training data for single-cell foundation models.

LARGE LANGUAGE MODELS AND BIOLOGICAL INTEGRATION

Integrating LLMs like GPT and Claude with biological design tools represents an emerging area of research. This integration could leverage the rich representations and reasoning capabilities of LLMs alongside the specialized knowledge encoded in biological data-specific models. Early efforts in this field show promise, though many technological challenges remain.

Some efforts focus on creating AI co-pilots for bioinformatics or drug discovery, integrating LLMs with tools like BLAST for sequence similarity searches or genome annotation software. These systems aim to enhance the efficiency and accuracy of bioinformatics workflows by providing intelligent assistance to researchers. Most of the efforts are carried out in the private sector; for example, LOWE by Valence Labs is a scientific co-pilot for bioinformatic tool usage in pharmaceutical companies.⁵ More general-purpose software co-pilots like Devin by Cognition⁶ have also been demonstrated for their biology-specific applications.

Another avenue of exploration involves generating protein sequences from natural language prompts. The aspiration is to enable users to specify functional proteins purely through natural language descriptions. For example, ESM3 contains a protein function track that also enables a user to, for example, condition sequence and structure generation given a natural language function-of-interest (Hayes et al., 2024). While potentially promising, this approach may be limited by the number of proteins with comprehensive natural language labels in datasets like UniProt.

Perhaps the most ambitious efforts involve directly connecting LLMs with laboratory experimental feedback. Proposals for “cloud labs” envision remote users (or potentially LLMs themselves) writing programs to specify experimental protocols, which are then executed in automated laboratory environments.⁷ The resulting data would be fed back to the user or LLM for analysis, new hypothesis generation, and experimental planning. While this concept is tantalizing, it does face substantial near-term challenges. Current cloud labs struggle with complex biological workflows and are currently limited mainly to inorganic chemistry applications (Abolhasani and Kumacheva, 2023) or basic liquid handling. Furthermore, LLMs would likely require substantial improvements in scientific knowledge, reasoning capabilities, and long-term planning to effectively manage such systems.

Despite these challenges, the integration of LLMs with biological design tools holds substantial potential for accelerating scientific discovery and innovation in the life sciences. As both LLM capabilities and biological automation continue to advance, we can anticipate increasingly sophisticated and powerful systems that can be integrated with human scientific input to advance biological discovery and design.

DISCUSSION

This work presents a comprehensive overview of the current landscape of biological machine learning models and datasets, highlighting several key themes and trends in the field. The past four years

⁴ See https://singlecell.broadinstitute.org/single_cell (accessed November 17, 2024).

⁵ See <https://www.valencelabs.com/lowe> (accessed November 17, 2024).

⁶ See <https://www.cognition.ai/blog/introducing-devin> (accessed November 17, 2024).

⁷ See <https://www.futurehouse.org/> (accessed November 17, 2024).

have witnessed an extraordinary acceleration in the development of AI models for biological applications. This period has seen the emergence of numerous foundation models across protein sequences, genomic sequences, and cellular transcriptomes, alongside significant advancements in predictive models such as AlphaFold2. The pace of development has been remarkable, with new models often surpassing their predecessors within months (see Figure A-2). This rapid progress, driven by increasing computational resources, larger datasets, and architectural innovations, demonstrates the field's dynamic nature and the potential for AI to affect many areas of biology. However, this pace also presents challenges in model evaluation, reproducibility, and the continuous adaptation of downstream applications. The taxonomy of models discussed can be broadly categorized into four interconnected types: foundation models, generative models, predictive models, and design models.

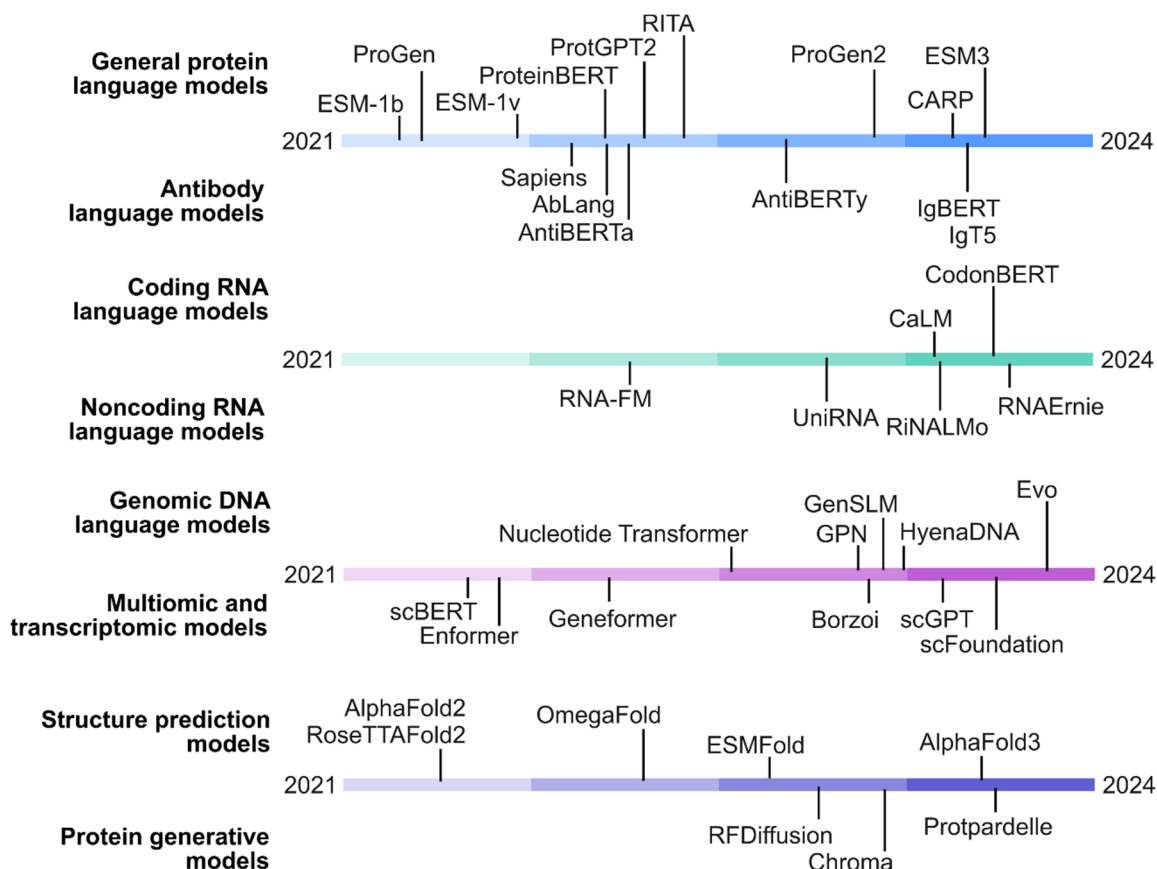


FIGURE A-2 Timeline of model development since 2021.

NOTES: Notable models (see Tables A-1 and A-2) arranged according to their broad categories and the time of the last archival work describing the model (either the preprint date if no journal publication followed, or the date of the final journal publication). On this timeline, a burst of model development occurred at the level of protein sequence and structure prediction in 2021 and 2022. More recent modeling has focused on models of genome sequence, transcriptome counts, or generative protein models.

Foundation models, trained on large-scale biological datasets using unsupervised objectives, learn to capture the underlying distributions of biological data. These models, exemplified by protein language models like ESM and genomic language models like Evo, demonstrate remarkable generalist capabilities across various downstream tasks. Generative models, often built upon or closely related to foundation models, enable efficient sampling from learned biological distributions and have shown particular

promise in generating novel protein sequences and structures. Predictive models, on the other hand, focus on mapping between different biological modalities, such as predicting protein structure from sequence (e.g., AlphaFold) or inferring epigenomic features from DNA sequence (e.g., Enformer). Finally, design models leverage both generative and predictive capabilities to produce biological entities with desired properties, as seen in protein engineering and promoter design tasks.

Throughout this review, protein tasks are a primary focus of modern deep learning applications in biology. This emphasis is likely due to the availability of large high-quality datasets and the immediate practical applications in areas such as drug discovery and enzyme engineering. However, the field is expanding to encompass other biological modalities. Genomic sequence modeling, while currently less mature, shows promise for advancing our understanding of gene regulation and expression and enabling multimodal and multiscale design tasks. Similarly, single-cell transcriptomics modeling is an emerging area with potential for revealing insights into cellular heterogeneity and developmental processes. As more comprehensive datasets become available and model architectures continue to evolve, we can expect significant advancements in these areas.

The integration of LLMs with biology-specific tools represents an exciting frontier in the field. While still in its early stages, this integration holds the potential to enhance bioinformatics workflows, facilitate natural language-guided protein design, and potentially automate aspects of the experimental process through cloud laboratories. However, significant challenges remain in terms of model capabilities, data integration, and experimental automation.

Looking forward, the continued development of AI-enabled biological models holds immense promise for both biological design and scientific discovery. We can anticipate more sophisticated approaches to engineering proteins, genetic circuits, and other biological systems, potentially accelerating drug discovery, improving enzyme engineering, and facilitating the development of novel synthetic biological systems. These models also have the potential to reveal new biological insights by capturing complex patterns within large-scale data, as demonstrated by protein language models' ability to provide insights into protein folding and function. However, it is crucial to view these models as tools to augment and accelerate biological research that work alongside experimental approaches. The complexity of biological systems necessitates continued experimental validation and rigorous scrutiny of AI-generated hypotheses. As the field evolves, integrating more sophisticated models with experimental workflows, we should expect substantial advancements in both basic biological science and applied biotechnology.

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Appendix B

Public Meeting Agendas

**Assessing and Navigating Biosecurity Concerns and Benefits of Artificial Intelligence
Use in the Life Sciences—August Information-Gathering Meeting
The Keck Center, 500 Fifth Street, NW
Washington, DC 20001
August 13–14, 2024**

FRIDAY, JUNE 21, 2024

- 1:00 Welcome
Michael Imperiale & Lynda Stuart, Co-Chairs
- 1:05 Sponsor Perspective on Charge to Committee
Shankar Sundaram, National Security Council
Richard C. Johnson, Deputy Assistant Secretary of Defense, Nuclear and CWMD Policy
Michael Horowitz, Deputy Assistant Secretary of Defense for Force Development and Emerging Capabilities
- 1:30 Discussion with Committee
- 2:00 Adjourn

TUESDAY, AUGUST 13, 2024

- 9:30–12:00 The Convergence of AI and Biology
John Jumper, Google DeepMind
Eric Horvitz, Microsoft
Yana Bromberg, Emory University
Sean Ekins, Collaborations Pharmaceuticals
David Baker, Institute for Protein Design, University of Washington
- 12:00–1:00 Lunch
- 1:00–2:15 Panel: GenAI Tools and Techniques for Biology
Kathy Wei, 310 AI
Mengdi Wang, Princeton University
Dan Hendrycks, Center for AI Safety
- 2:30–4:00 Panel: Understanding Pathogenesis of Transmissible Biological Agents and AI
Craig Wilen, Yale University
Sarah Richardson, Industrial Biotechnologist
Andrew Pekosz, Johns Hopkins University
Pardis Sabeti, Harvard University

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The Age of AI in the Life Sciences: Benefits and Biosecurity Considerations

4:00 Adjourn

WEDNESDAY, AUGUST 14, 2024

11:00–12:00 Data Security Implications for Biological Data
Bo Li, University of Chicago
Jennifer Roberts, ARPA-H

12:00–12:30 Opportunities to Use AI in Pandemic Countermeasures Development
Jimmy Gollihar, Houston Methodist Research Institute

12:30 Adjourn

**Assessing and Navigating Biosecurity Concerns and Benefits of Artificial Intelligence
Use in the Life Sciences—September Information-Gathering Meeting
Virtual**

SEPTEMBER 27, 2024

12:00 Welcome
Michael Imperiale & Lynda Stuart, Co-Chairs

12:05 Panel: High-Performance Computing in Life Sciences and Biological Data
Resources and Infrastructure
Dan Stanzione, Texas Advanced Computing Center, University of Texas at
Austin
Kim Pruitt, National Library of Medicine

12:30 Panel Discussion with Committee

1:00 Adjourn

**Assessing and Navigating Biosecurity Concerns and Benefits of Artificial Intelligence
Use in the Life Sciences—October Information-Gathering Meeting
The Keck Center, 500 Fifth Street, NW
Washington, DC 20001**

OCTOBER 1, 2024

9:00 Welcome
Michael Imperiale & Lynda Stuart, Co-Chairs

9:05–9:30 Understanding Pathogenesis of Transmissible Biological Agents and Impact of AI
Arturo Casadevall, Johns Hopkins University

9:30–10:30 Committee Discussion

10:30–11:15 Opportunities to Use AI in Medical Countermeasures Development
Jason McLellan, University of Texas at Austin
Emanuele Andreano, Biotechnopolo di Siena Foundation

Prepublication Copy

11:15–11:30	Break
11:30–12:00	High-Performance Computing in the Life Sciences <i>Benjamin Brown</i> , Advanced Scientific Computing Research, Department of Energy
12:00	Adjourn

Appendix C

Committee Biographies

Michael J. Imperiale (*Co-chair*) is the Arthur F. Thurnau Professor Emeritus in the Department of Microbiology and Immunology at the University of Michigan. He previously served as associate vice president for research–research policy and compliance from 2017 to 2022 and as an editor of the *Journal of Virology*, *PLoS Pathogens*, and *mBio*. He is founding editor-in-chief of *mSphere*. Imperiale’s research interests focus on the study of DNA tumor viruses. He was an inaugural member of the National Science Advisory Board for Biosecurity from 2005 to 2014 and served on the Planetary Protection Subcommittee at the National Aeronautics and Space Administration, and he was a member of the board of directors of the Van Andel Institute Graduate School. In 2009, he was the recipient of the University of Michigan Distinguished Faculty Achievement Award, and in 2016 he received the Rackham Distinguished Graduate Mentor Award. He was elected as a fellow of the American Academy of Microbiology in 2010 and as a fellow of the American Association for the Advancement of Science in 2011. Imperiale received a Ph.D. in biological sciences from Columbia University. He previously served as the chair of the National Academies of Sciences, Engineering, and Medicine’s Committee on Strategies for Identifying and Addressing Potential Biodefense Vulnerabilities Posed by Synthetic Biology and as a member of the Committee on Science, Technology, and Law. Imperiale published an article in *Organisation for Economic Co-operation and Development Library* in 2022, titled “Integrity and Security in the Global Research Ecosystem,” in which he and his co-authors describe policy initiatives to safeguard national and economic security and briefly summarize initiatives relevant to artificial intelligence.

Lynda M. Stuart (*Co-chair*) is the executive director of the Institute for Protein Design and acting professor in biochemistry at the University of Washington School of Medicine. As a physician scientist with more than 20 years of experience in immunology, global health, and product development, she directs translational research and corporate and foundation collaborations, transforming laboratory breakthroughs into real-world impact. As deputy director of the Gates Foundation from 2013 to 2022, Stuart oversaw the preclinical development of vaccines, including mRNA vaccines, and antibody therapies to address urgent global health challenges. In the wake of Ebola, she contributed to the conceptualization of “just-in-time” platform approaches for pandemic responses. From 2020 to 2022 she led the Foundation’s COVID-19 discovery and translational vaccine response efforts. After leaving the Gates Foundation, Stuart spent one year as the vice president of infectious disease at the mRNA company BioNTech. Stuart is a fellow of the American Society for Clinical Investigation. She also currently serves on the Scientific Advisory Committee of CEPI (the Coalition for Epidemic Preparedness Innovations) and is a member of the Scientific Technical Expert Group for the International Pandemic Preparedness Secretariat supporting the 100 Days Mission. Stuart received a Ph.D. in immunology from the University of Edinburgh and an M.D. from the University of London. Stuart published a 2024 article, titled “Community Values, Guiding Principles, and Commitments for the Responsible Development of AI for Protein Design,” in which she and her co-authors propose a set of values and principles to guide the responsible development of artificial intelligence in the field of protein design.

Patrick M. Boyle is the chief scientific officer at Ginkgo Bioworks, the leading horizontal platform for cell programming, providing flexible end-to-end bioengineering services for organizations across diverse markets—from food and agriculture to pharmaceuticals to industrial and specialty chemicals. Over the past 12 years, Boyle has worked on Ginkgo’s core microbial technologies at the intersection of biology, computation, and data. Ginkgo helps hundreds of companies to leverage software, robotics, and novel biological assets to bring new products to market. Dr. Boyle serves as a technical advisor to Social

Impact, which is a small venture capital firm investing in biotechnology and deep tech companies. Boyle was a 2014 fellow in the Johns Hopkins University Center for Health Security's Emerging Leaders in Biosecurity Program, a member of Schmidt Futures' Task Force on Synthetic Biology and the Bioeconomy, and a member of the American Society for Microbiology's Climate Change Task Force, and he is an advisor to Social Impact Capital. Boyle currently serves on the Board on Life Sciences for the National Academies of Sciences, Engineering, and Medicine. Boyle previously worked on two consensus studies for the National Academies: *Biodefense in the Age of Synthetic Biology* (2018) and *Safeguarding the Bioeconomy* (2020). Prior to Ginkgo, Boyle received a Ph.D. from Harvard Medical School in 2012, where he developed synthetic biology applications in bacteria, yeast, and plants in the laboratory of Dr. Pamela Silver. He received an S.B. in biology from the Massachusetts Institute of Technology in 2006. Boyle has presented at several conferences on the use of automation to generate biological data for the training of artificial intelligence (AI) models for biology and the implications of AI in biology.

Kjiersten M. Fagnan is currently the chief informatics officer of the U.S. Department of Energy (DOE) Joint Genome Institute (JGI), a national user facility responsible for generating petabytes of environmental genomics data deposited into national repositories. She contributed to the development and establishment of JGI's data management system that has been in production for more than 10 years. She also supported the team that built the biosecurity screening software for the first version of JGI's DNA synthesis platform that ensures JGI does not construct any pathogens on behalf of users. Additionally, she was part of the Association for Computing Machinery's Gordon Bell-winning team in 2018 for the paper "Attacking the Opioid Epidemic: Determining the Epistatic and Pleiotropic Genetic Architectures for Chronic Pain and Opioid Addiction," which used high-performance computing infrastructure to synthesize and aggregate large omics data volumes. Fagnan received a Ph.D. in applied mathematics from the University of Washington. Fagnan published a report for the DOE Office of Scientific and Technical Information in 2019, titled *A Framework for Advancing AI in Science*, in which she and her co-authors outlined discussions at a roundtable on enhancing access to research data and computing resources for artificial intelligence research and development.

Gigi Kwik Gronvall is an associate professor in the Department of Environmental Health and Engineering and a senior scholar at the Center for Health Security in the Johns Hopkins Bloomberg School of Public Health. An immunologist by training, she leads work to advance biosecurity, biosafety, pandemic preparedness, and the bioeconomy. Gronvall is the author of *Synthetic Biology: Safety, Security, and Promise* (2016) and *Preparing for Bioterrorism: The Alfred P. Sloan Foundation's Leadership in Biosecurity* (2012). She is a member of the U.S. Department of State's International Security Advisory Board and the Novel and Exceptional Technology and Research Advisory Committee, and she is co-chair of the BioMADE Safety, Security, Sustainability and Social Responsibility Committee. Gronvall received a Ph.D. from Johns Hopkins University and was a National Research Council Postdoctoral Associate at the U.S. Army Medical Research Institute of Infectious Diseases in Fort Detrick, Maryland. She currently serves on the National Academies of Sciences, Engineering, and Medicine's Forum on Microbial Threats and was previously a member of the Committee on Strategies for Identifying and Addressing Potential Biodefense Vulnerabilities Posed by Synthetic Biology. Gronvall served as an expert research participant to OpenAI from March 24, 2024, through April 2, 2024.

Heidi C. Howard is currently an associate professor at University College Cork, Ireland, and a bioethicist at Google DeepMind; she heads the chemical, biological, and radiological/nuclear evaluations for large language models. She is also an active member of the European Society of Human Genetics Policy and Ethics Committee. Since 2008 she has been studying the ethical, legal, and social aspects of emerging technologies in biomedicine. A large focus of Howard's work has been on genetics/genomics and data, including the model of provision direct-to-consumer, public health genomics, personalized medicine, and new technologies such as next-generation sequencing, gene editing, data-driven life sciences, and AI.

Howard received a Ph.D. in molecular genetics from McGill University and an M.Sc. in bioethics from KU Leuven.

Ethan Jackson is senior director and senior principal researcher at Microsoft where he leads Microsoft Premonition, Microsoft's artificial intelligence (AI) platform for biosecurity and bio-intelligence, and manages an interdisciplinary team of researchers, engineers, and scientists. Dr. Jackson previously was lead author of Microsoft's response to the AI Executive Order regarding topics on AI and biothreats, and he regularly engages with the Executive Branch and Congress as a subject matter expert. He is the chief science officer for the Computing the Biome program focusing on domestic biosecurity—one of the top programs at the National Science Foundation's Convergence Accelerator. In addition, his work extends to broader robustness of cyber-physical systems, which has been applied to harden essential parts of the Windows Operating System, develop aspects of the HoloLens augmented reality platform, and implement key protocols in Microsoft's Cloud platform. He works at the intersection of technology, biology, and security, having managed programs with GSK, the Gates Foundation, and Wellcome Trust, and directed the introduction of biosensing into Microsoft's own data centers. Dr. Jackson's program won Fast Company's World Changing Ideas Award in 2021. He holds a Ph.D. in computer science from Vanderbilt University and a B.S. in computer engineering from University of Pittsburgh.

Suman Jana is an associate professor in the Department of Computer Science and the Data Science Institute at Columbia University. His primary research interest is at the intersection of computer security and machine learning. Jana works on building automated tools for fixing and finding privacy and security vulnerabilities in large real-world systems. His research has led to fixing and reporting about 250 high-impact security vulnerabilities across a diverse range of popular software. Several open-source software including Apache Cordova, Mozilla Firefox, and curl have incorporated his research software. His research has received seven best paper awards, a *Communications of the ACM* research highlight, a Google faculty fellowship, a JPMorgan Chase Faculty Research Award, a National Science Foundation CAREER award, and an Army Research Office young investigator award. Jana received an M.S. in computer science from the University of Utah and a Ph.D. in computer science from the University of Texas at Austin.

Jens H. Kuhn is a principal at Tunnell Government Services, tasked as one of two principal scientists, and the director of virology (contractor) at the National Institutes of Health/National Institute of Allergy and Infectious Diseases/Division of Clinical Research/Integrated Research Facility at Fort Detrick, a biosafety Level 4 facility. He was the first western scientist permitted to work in a former Soviet biological warfare facility, the State Research Center of Virology and Biotechnology "Vektor" in Siberia, Russia, within the U.S. Department of Defense's Cooperative Threat Reduction Program. Kuhn specializes in pathogenesis of and medical countermeasures against highly virulent viral pathogens. He contributed to the Center for International and Security Studies at Maryland's Controlling Dangerous Pathogens Project and recently was a taskforce member of the Bulletin of the Atomic Scientists' Pathogens Project—Creating the Framework for Tomorrow's Pathogen Research. He has authored *Filoviruses: A Compendium of 40 Years of Epidemiological, Clinical, and Laboratory Studies* (2008) and co-authored *The Soviet Biological Weapons Program—A History* (2012). Kuhn received a Ph.D. in biochemistry and a Ph.D. in medical sciences from Freie Universität Berlin and an M.D. in human medicine from Universitätsmedizin Berlin, Germany. He previously served on the National Academies of Sciences, Engineering, and Medicine Committee on Animal Models for Assessing Countermeasures to Bioterrorism Agents.

Bradley A. Malin is the Accenture Professor of Biomedical Informatics, Biostatistics, and Computer Science at Vanderbilt University and the vice chair for research affairs in the Department of Biomedical Informatics at Vanderbilt University Medical Center. He is the co-founder and co-director of the AI Discovery and Vigilance to Accelerate Innovation and Clinical Excellence Center at Vanderbilt. He is

also one of the principal investigators of the National Institutes of Health's (NIH) Bridge2AI Center and the Infrastructure Core of NIH's AIM-AHEAD program. His research is in the development and implementation of computational technologies to facilitate the collection, sharing, and automated analysis of biomedical data in a socially responsible manner. Malin is also a consultant to Microsoft for the development of data privacy technologies. He is an elected fellow of the National Academy of Medicine, the Institute of Electrical and Electronics Engineers, the American Institute for Medical and Biological Engineering, the American College of Medical Informatics, and the International Academy of Health Sciences Informatics. Malin received an M.Phil. in public policy and management from Carnegie Mellon University, where he also received a Ph.D. in computer science. He previously served on eight National Academies of Sciences, Engineering, and Medicine committees as a member and one as a chair, notably AI for Scientific Discovery: Toward an International Grand Challenge—A Workshop, which investigated the future role of artificial intelligence in scientific discovery.

Amarda Shehu is a professor of computer science in the College of Engineering and Computing at George Mason University, where she is also an associate dean for artificial intelligence (AI) innovation, associate vice provost for research for the Institute for Digital Innovation, and Mason representative in the National Institute of Standards and Technology AI Safety Institute Consortium. Shehu served as a National Science Foundation (NSF) program director in the Directorate for Computer and Information Science and Engineering (CISE) from 2019 to 2022, where she led the NSF-wide working group on the Harnessing the Data Revolution: Data Science Corps program and led CISE efforts in the COVID-19 RAPID Program and the Molecular Foundations in Biotechnology Program. Previously a National Institutes of Health Predoctoral Fellow in the Nanobiology Program at Rice University, Shehu's research advances foundational and use-inspired AI research to enable and advance scientific discovery. Her laboratory has made significant advances at the intersection of AI, generative AI, and molecular biology. She is a 2022 fellow of the American Institute for Medical and Biological Engineering and has received several awards, including the 2022 Outstanding Faculty Award from the State Council of Higher Education for Virginia, the 2012 NSF CAREER Award, and two Director Recognition awards for her work at NSF. Shehu received a Ph.D. and an M.S. in computer science from Rice University.

Alexander Titus is vice president of artificial intelligence (AI) and machine learning (ML) at Avidity Biosciences, Inc., as well as a principal scientist with the University of Southern California's (USC's) Information Sciences Institute and research faculty at USC's Iovine and Young Academy. Previously, he was the vice president of strategy and computational sciences at Colossal Biosciences, where he led strategic projects across the organization, and his team worked cross-functionally to provide AI/ML and genotype-to-phenotype analyses to make de-extinction a reality. Titus joined Colossal from Google Cloud, where his work as a strategic business executive included leading AI/ML and life sciences solution work for the global public sector. Previously, he was the inaugural assistant director (now called principal director) for biotechnology within the Office of the Chief Technology Officer at the U.S. Department of Defense (DoD), where he led the team developing DoD's roadmap toward biotechnology modernization. Titus received a Ph.D. in quantitative biomedical sciences from Dartmouth College and a B.S. and B.A. in biochemistry and biology, respectively, from the University of Puget Sound. Titus is a member of the National Security Commission on Emerging Biotechnologies. He has also provided a written statement to the U.S. Senate AI Insight Forum in 2023, titled "Risk, Alignment, & Guarding Against Doomsday Scenarios." He published a preprint article on *arXiv* in 2023, titled "The Promise and Peril of Artificial Intelligence—Violet Teaming Offers a Balanced Path Forward," in which he and his co-author review emerging issues with opaque and uncontrollable AI systems and propose an integrative framework called violet teaming to develop reliable and responsible AI.

